

Project Smores

Toyota Sienna Modular Camper Conversion — Full Build Plan

Vehicle measurements surveyed and verified Aug 1, 2026. Dimensions in this document drive real cuts — re-measure your own vehicle before building.

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- **Cooking, heat, and carbon monoxide.** Cooking or running any fuel-burning appliance in or near an enclosed vehicle risks fire, burns, and carbon monoxide poisoning, which can be fatal while you sleep. Ventilate, and fit working CO and smoke alarms.
- **Weight, handling and load limits.** Added weight affects braking, handling, tire load, and your vehicle's GVWR and axle ratings. Verify all figures against your own vehicle's door-jamb placard.
- **Tools and materials.** Power tools, sharp edges, dust, adhesives, and finishes cause injury. Follow every manufacturer's instructions and use appropriate protective equipment.
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Verify every dimension on your own vehicle before cutting anything. The measurements here were taken on one specific van and may not match yours, and errors — in measurement, in transcription, or in the drawings — are possible throughout.

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Project Smores

Full Build Plan — Modular Lift-Out Design

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Project Smores — Full Build Plan

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Two-person setup, 2nd row seats **removed entirely** (3rd row folded flat; seat removal/reinstall procedure incl. the SRS airbag emulators: Section 9). Layout, tailgate to front seats: **Kitchen (Panel C's fridge/kitchen void) → Rear pantry (prefab drawers on Panel C's deck) → Bed (Panel B + Panel A) → front seats**. The rear pantry is a **prefab drawer cluster** (owner, July 2026): a 2×2 array of IRIS USA 12"W stackable drawers sitting on the tailgate end of Panel C's deck, held by a cleat pocket + a cam strap, with a pot/pan crate in the ~21.8" of open deck beside it — nothing built, ~27 lb lighter, and each drawer unit lifts straight out (Component 1). Panel A and B have no top of their own (capped by the three-piece bed frame — Panel A's section is screwed down, and Panel B's top is two centreline halves that lift out individually), while Panel C keeps its fixed top — **recessed flush into its rail plane** (the deck ply drops between the rails onto cleats; Panels A/B run 16.25" legs vs Panel C's 17", so the platform on their rails lands in the same flush 18.5" plane — ¾" more headroom) — forming one continuous deck with 79¼" of sleeping length for the HEST Dually Long mattress (78" x 50" x 4") (Panel A now sits flush with the front seatbacks, using up the floor space that used to sit empty in front of it). Panel B is a bare-frame deep-storage bay — the side doors don't reach it, so it has no drawers and loads from above by lifting out either half of its **two-piece bed top** (owner, Aug 2026); Panel A holds a right drawer (EcoFlow DELTA 3) plus a left open-storage bay (EcoFlow WAVE 3, no drawer box); Panel C's under-deck void instead houses a real BougeRV compressor fridge and a real JAGAHABA slide-out camp kitchen, both on heavy-duty slides pulling straight out the open tailgate — bought products, not built from plywood — plus a shallow slide-out **kitchen drawer** hung under the deck in the dead air above the kitchen unit. The fridge and its cooling fans run off the EcoFlow DELTA 3 stack (Panel A) via their own DC line, so they keep running when people are away from the van regardless of ignition state; the DELTA 3 itself gets AC-charged from the front console outlet while driving, while the induction cooktop and Power strip 1 plug into the van's REAR AC outlet (back passenger area — verified to exist), right beside them: two short cords, no seam crossings.

Vehicle measurements — the van was surveyed on Aug 1 2026 and every dimension in this plan comes from that survey. The full record (V1–V10 interior/opening measurements, F1–F8 rear-floor survey, what each finding changed, and the few items still open) is **Appendix A**.

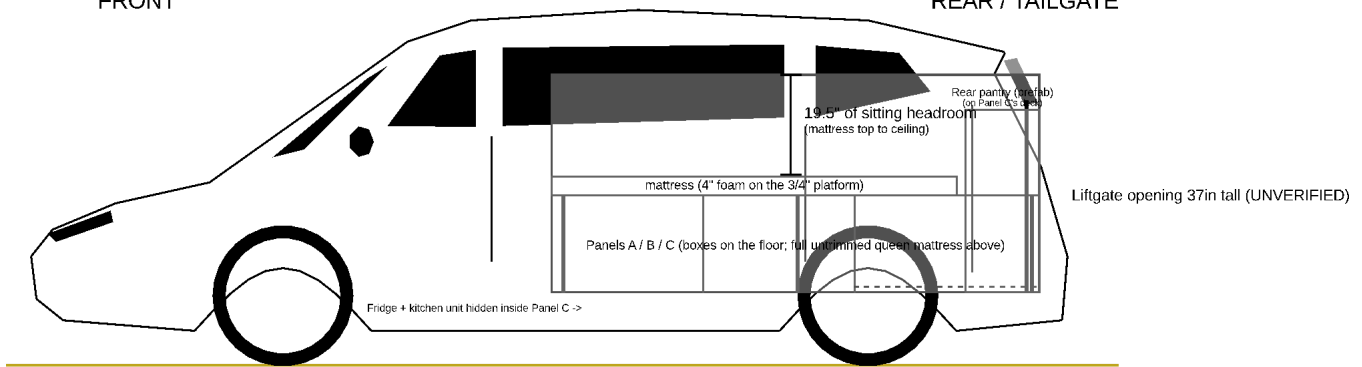
Renders

Parametric 3D model: `platform.scad` (dimensions in `params.scad` — edit one file to regenerate every view via `./render.sh`).

Whole-Vehicle Overview

FRONT

REAR / TAILGATE

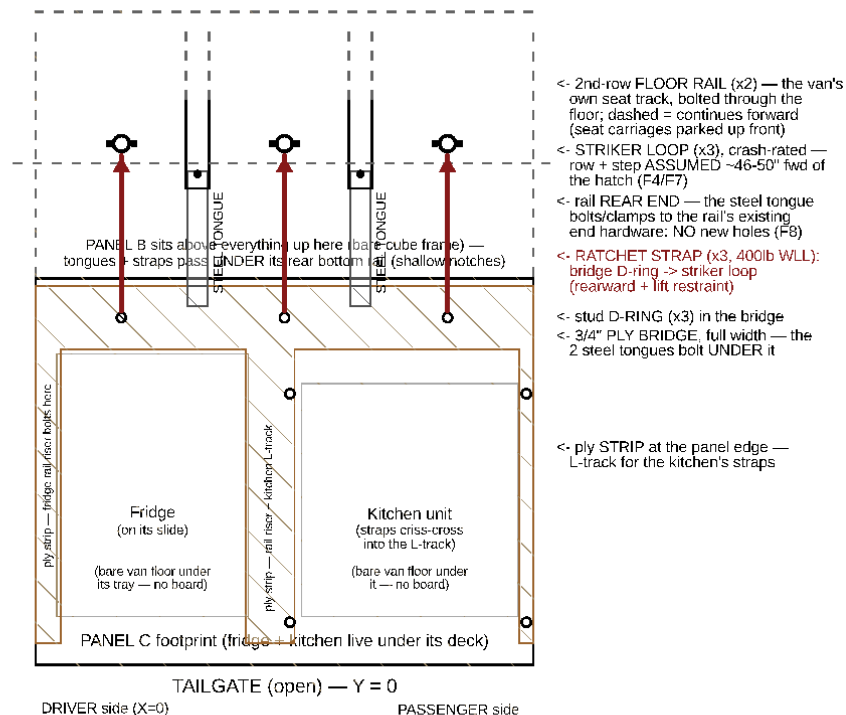


Cargo interior envelope (thin outline): 93.75in x 42in, 2nd row removed — UNVERIFIED length
 Exterior body outline is illustrative only — cargo interior, liftgate opening, and platform dimensions are to scale.

Side profile of the van itself (illustrative exterior body outline — hood, windshield, roofline, tailgate, wheels) with the cargo interior envelope, liftgate opening, and platform assembly drawn to scale inside it. Front at left, tailgate at right.

Overhead Floorplan

NO-DRILL ANCHOR PLATFORM — overhead: Panel C + the factory hardpoints forward of it (Section 8)



READ ME — what this platform is (Section 8):

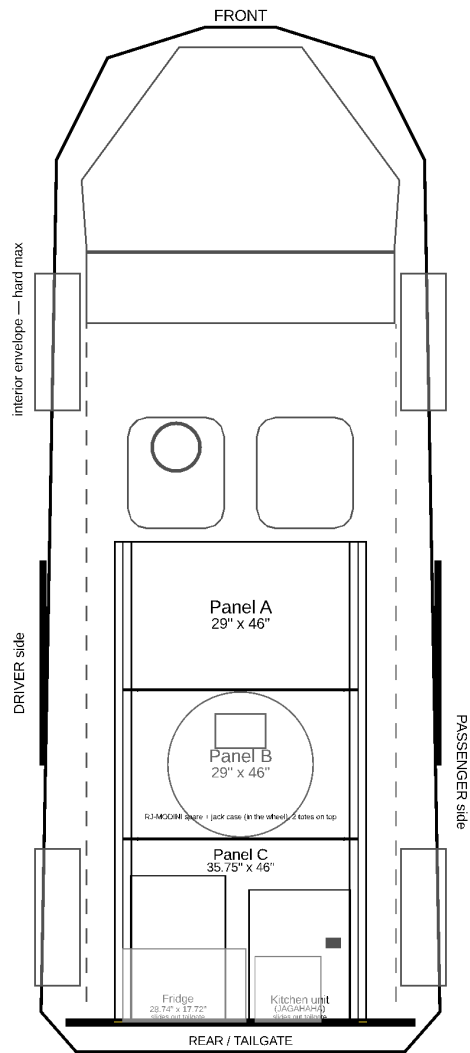
A skeleton of 3/4" plywood on a non-slip rubber mat, laid on the van floor:
 ONE comb-shaped piece — a full-width BRIDGE with 3 narrow STRIPS running back from it (the hatched shape above), cut from a single 46"x33" blank. There are NO joints in it: nothing is glued or screwed to another piece of ply.
 It is NOT a full plywood floor — nothing sits under the fridge tray or the kitchen unit; the van floor there stays bare.

The APPLIANCES anchor to the BOARD: the fridge side's steel riser angles bolt to the two rail-line strips (1/4-20 T-nuts), and the kitchen's 4 ratchet straps criss-cross into L-track D-rings on its two flanking strips.

The BOARD anchors to the VAN at two factory hardpoints — zero new holes:
 -> RAILS: 2 steel tongues (2"x3/16" flat bar) bolt under the bridge and bolt/clamp to the 2nd-row floor rails' rear ends -> FORWARD crash load.
 -> STRIKERS: 3 ratchet straps run from bridge D-rings into the 3rd-row striker loops -> REARWARD + LIFT loads (and they pin the board down).

Rail-end + striker positions are ASSUMED until the Section 0 F1-F8 survey.

The no-drill rail platform (Section 8) in plan view, before the whole-van floorplan below: the mat + ply anchor board under Panel C, its 2 steel tongues bolted to the rear ends of the 2nd-row long-slide floor rails, and its 3 ratchet straps dropping into the crash-rated 3rd-row striker loops. Tailgate at the bottom, forward at the top. Rail-end and striker positions are ASSUMED until the Section 0 F1-F8 survey.



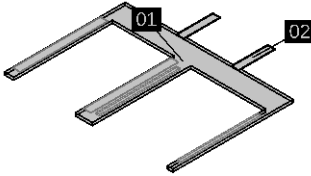
Vent intrusion (3.5" per side, floor level only) shown along both edges — legs must stay clear
 Usable boundary line = 0" hatch-curvature reserve, nothing built below it toward the tailgate
 Fridge + kitchen unit live inside Panel C (not a separate row) — see rear-view diagram for cooling/control detail
 Rear pantry (dark gray, tailgate end of Panel C): prefab 2x2 drawer cluster + pot bin ON the deck above the fridge/kitchen void,
 not a separate footprint — food side faces the kitchen, personal shelf (power strip 1) faces the mattress

*Top-down view: Panel A/B/C plus the fridge and kitchen unit living inside Panel C's footprint, drawn to scale inside the interior envelope. Front of the vehicle is at the top, tailgate at the bottom.
 Seam bumpers, alignment pins, and grab-handle positions are all marked.*

Anchor Board — Assembly & Connection Details

The IKEA-style header for the platform — the finished piece, its bought hardware (lettered), and its cut parts (numbered), in the same format as the Section 6 component banners:

ANCHOR BOARD — NO-DRILL RAIL PLATFORM (SECTION 8)

FINISHED COMPONENT	ACCESSORY LIST				PART LIST	
	A  L-track punch x2	B  stud D-ring x7	C  ratchet strap x7	D  bolt & T-nut x30	01  beam 46x25 ONE PC 1x	02  stud for gus 2x16 2x

The banner above only *names* accessories A–G. **Where each one goes is the install map below** — the board in plan with every accessory ballooned to its actual position. The what/where/how for each letter, and the build order, follow it as tables.

ANCHOR BOARD — ACCESSORY INSTALL MAP (A-G), Section 8

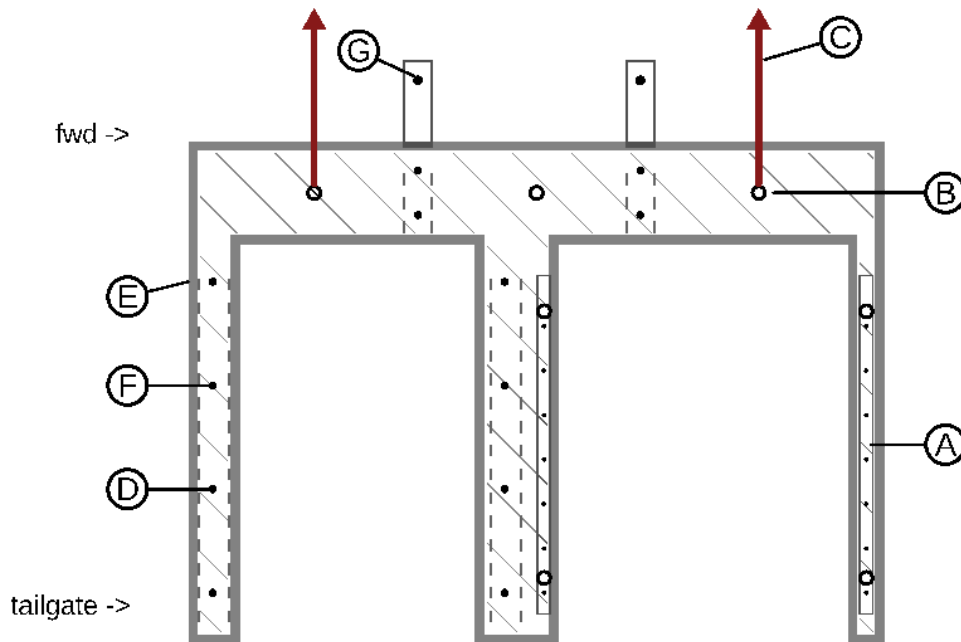
Every letter matches the ACCESSORY LIST in Section 8 and the anchor-board banner icons.

The board is ONE piece of ply: every fastener holds **HARDWARE** to it, never ply to ply.

2 STEEL TONGUES -> the 2nd-row FLOOR RAILS' rear ends, MEASURED 42" (F8a)

how they engage the track: see the tongue-to-rail connection detail

3 RATCHET STRAPS -> the 3rd-row STRIKER LOOPS, MEASURED 44.5" (F4a)



ONE-PIECE 3/4" PLY BOARD, 46" x 33" — assembly sheet V1 for dimensions, V5 for the cut

Anchor board — accessory install map and build order

The map is the drawing; the descriptions are the table below. They used to be two more columns on the same sheet, which doubled its width and drove every label down to about 4pt on paper — a figure is only ever as legible as page-width ÷ sheet-width allows.

Accessory list A–G

	Accessory	Where it goes, and how it fastens
A	L-track, cut down (×2)	Bolted flat onto the two KITCHEN-SIDE strips: the centre strip's kitchen face (X=23) and the 1.5" panel-edge strip (X=44.75), ~23" each. 7 bolts per length, ~3" apart, down into T-nuts. Cut the 48" stock down first; deburr the slots
B	Stud fitting w/ D-ring (×7)	No fasteners — each one drops into an L-track slot and twist-locks. 4 go in the kitchen L-track (2 per side, at Y=6 and Y=24) as the kitchen strap corners; 3 go on the BRIDGE (X=8, 23, 38) as the striker-strap points
C	Ratchet strap, 400 lb WLL (×7)	Never fastened to the board — they hook to B's D-rings. 4 criss-cross over the kitchen unit into its 4 L-track D-rings; 3 run forward from the bridge D-rings into the van's 3rd-row striker loops, measured 44.5" from the hatch (F4a). Tension last; re-tension after one drive
D	½"-20 bolt + T-nut (~30 pairs, incl. spares)	The T-nut goes in the board's underside , counterbored flush so the board still lies flat on its mat; the bolt comes down from above through whatever it holds: riser flanges (4 each = 8), L-track (7 each = 14), steel tongues (2 each = 4). Seat every T-nut before any hardware goes on — you cannot reach them afterward
E	Non-slip rubber mat (×1)	Cut to the board's outline and laid under it — friction, trim protection and rattle control. Not fastened to anything. It goes in only when the finished board is set into the van (step 7), not on the bench
F	Threadlocker, blue 242 (×1)	One drop on every machine screw as you fit it — that is the A, D and G fasteners (B needs none, it twist-locks). They all end up hidden under the appliances, so they must never need re-checking. Cure ~24 h before loading
G	Rail-end engagement (×2)	In-van only. The forward end of each steel tongue engages the 2nd-row floor rail's own track — measured 42" from the hatch (F8a), rails 17.5" apart (F8b), so cut both tongues 16" long . No new holes in the vehicle. The forward load is carried by a butt joint — a downturned lug bearing on the rail's rear end face — with the slot fastener acting only as a hold-down. Full detail in "How the tongues connect to the rails" later in this section, with a top-down plan and a side section

Build order — steps 1–6 on the bench, step 7 in the van

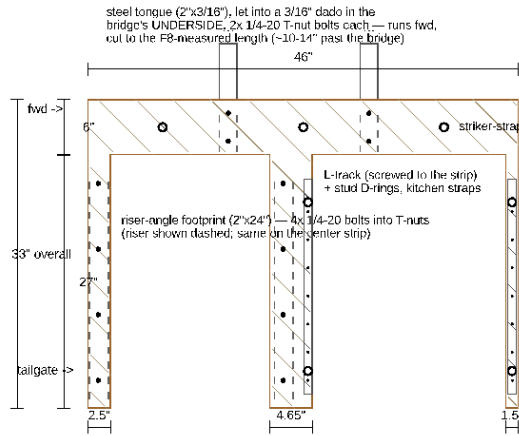
1. **Cut** the comb as ONE piece from the ¾" sheet (grain along the strips, ~½" fillet at every inside corner — assembly sheet V5). Have the store break the sheet down first (Section 3).
2. **D** — Mark and drill every bolt hole, then counterbore the **underside** and seat all ~30 **T-nuts** flush. Do this before anything is in the way.
3. **D** — Bolt the 2 **steel riser angles** onto the rail-line strips (4 bolts each) — the fridge's fixed slide rails land on these.
4. **A + D** — Bolt the two cut-down **L-track** lengths onto the kitchen-side strips (8 bolts each).
5. **D** — Bolt the 2 **steel tongues** into the dado in the bridge's underside (2 bolts each). Leave their forward ends unfixed until step 7.
6. **B + F** — Drop the 7 **stud D-rings** into their slots. Threadlocker on every machine screw as it goes in; let it cure ~24 h before loading.
7. **E + G + C** — **In the van:** lay the **mat**, set the board on it, engage the tongue ends to the floor rails (**G** — see the connection detail), notch Panel B's rear bottom rail 3/16" where the tongues and straps pass under it, then hook and **tension** the straps (**C**) — 3 to the striker loops, 4 over the kitchen unit.

Nothing in this sequence joins plywood to plywood — the board arrives at step 2 already whole.

How the platform actually goes together, in five views: **V1** the bare plywood board from above — ONE 46"x33" comb (full-width bridge + 3 strips, continuous ply, no wood joints), dimensioned, with the steel tongue runs, L-track, D-rings, and riser bolt patterns; **V2** a 4x-scale side section through a rail-line strip (mat → ply → T-nut → riser angle → rail, with the tray hanging over bare floor beside it); **V3** the kitchen's criss-cross straps into the stud D-rings, rear elevation; **V4** the board-to-van side view — the steel tongue let into the bridge's underside dado, bolted, running forward to the floor rail's rear end, plus the striker strap and the step fallback; **V5** the cut plan — how the comb nests in the ¾" sheet's spare, which way the face grain runs, the two offcut rectangles that fall out of it, the cut order and the fillet at each inside corner (there is no joinery view, because the board has no joints) — with a SYMBOLS legend decoding every mark:

ANCHOR BOARD — assembly & connections (Section 8; companion to the overhead platform diagram)

V1 — TOP: the one-piece plywood board, 46"x33" (bench-built, then laid in as ONE piece)



ONE PIECE — cut the whole comb from the 3/4" sheet's spare (V5):
NO ply-to-ply joints, no glue, no lap screws. Rubber mat under it.
Every fastener on this board holds HARDWARE, never wood to wood.
(gaps = bare van floor; the tray hangs there, the kitchen sits there)

RATCHET STRAP (x3): bridge D-ring —> striker loop, tensioned (rearward + lift restraint)

Panel B rear bottom rail sits over the run — its shallow notch lets the tongue + strap pass

^ floor STEP at the striker row = the tongue's bearing FALLBACK (F7)



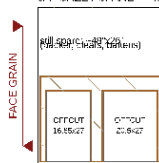
V3 — REAR elevation, kitchen side: straps -> D-rings -> L-track -> BOARD

SYMBOLS

- 3/4" plywood (cut piece, section-hatched)
- non-slip rubber mat (under every piece)
- steel tongue 2x3/16" flat bar
- dashed = hidden or overlaid part (riser footprint, tongue under the bridge, fridge above the tray)
- stud-fitting D-ring (drops into an L-track slot)
- 1/4-20 bolt into a T-nut (through the ply)
- L-track, screwed to the strip (slot dots)
- ratchet strap (400lb WLL)

V4 — SIDE: how the board grabs the VAN — forward at left, tailgate at right (vertical scale exaggerated)

V5 — ONE-PIECE CUT: how the comb nests in the 3/4" sheet's spare



ONE PIECE, NO WOOD JOINTS. The bridge and all three strips are a single continuous 46"x33" comb. Nothing on this board is glued or screwed to another piece of ply — the 12 lap screws and the wood glue are GONE from the BOM.

IT FITS THE STOCK YOU ALREADY BUY. The 3/4" sheet carries only Panel C's duck plus some ripped cleats, leaving a contiguous spare of at least 48"x60"; the comb drops in with ~26" of sheet length still left over for the control-panel backer, the rear-pantry hold-down cleats and the spare-tire skid battens.

FACE GRAIN RUNS ALONG THE STRIPS (the 33" direction = the sheet's 96" length). That is the axis a rearward pull loads in tension, and the strips are the narrow members — the panel-edge one is only 1.5" wide, so it wants its face plies running lengthwise. The bridge takes its load in-plane over a 6" depth, where cross-grain costs nothing that matters.

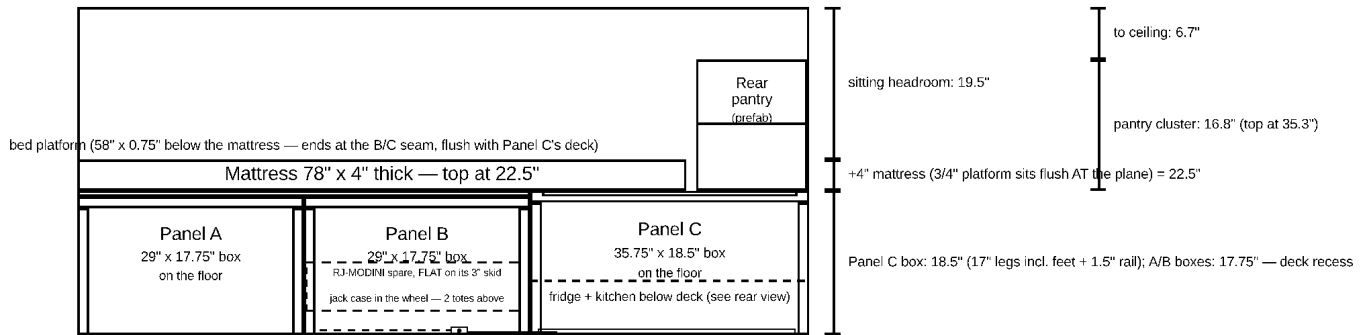
THE FALLOFF IS NOT WASTE. The two gaps come out as clean rectangles (16.85"x27" and 20.5"x27") — keep both as 3/4" offcut stock.

CUT ORDER: rip the two gaps out with a track saw or a circular saw on a straightedge (both cuts run the full 27" and STOP at the bridge line), then crosscut the teeth free. FILLET every inside corner ~1/2" radius — drill a 1/2" hole centered ON the corner first, then saw into it. A square re-entrant corner is exactly where a comb starts a tear, and it is the only place this one-piece board is weaker than three separate strips would be.

Why one piece matters: the striker straps hold the BRIDGE — the strips (and everything bolted to them) hang off it in a rearward pull. As one piece that load path is continuous ply, instead of three glue lines working in tension. (The steel tongues are NOT wood joints — they bolt through the bridge with the 1/4-20 T-nuts, V1.)

Anchor board — assembly and connection details

CEILING — 42" interior height (hard max; curves down toward the walls)



rear-pantry cluster top at 35.3" -> 6.7" of roof clearance | Power strip 1 + ROLL level on the deck edge beside it

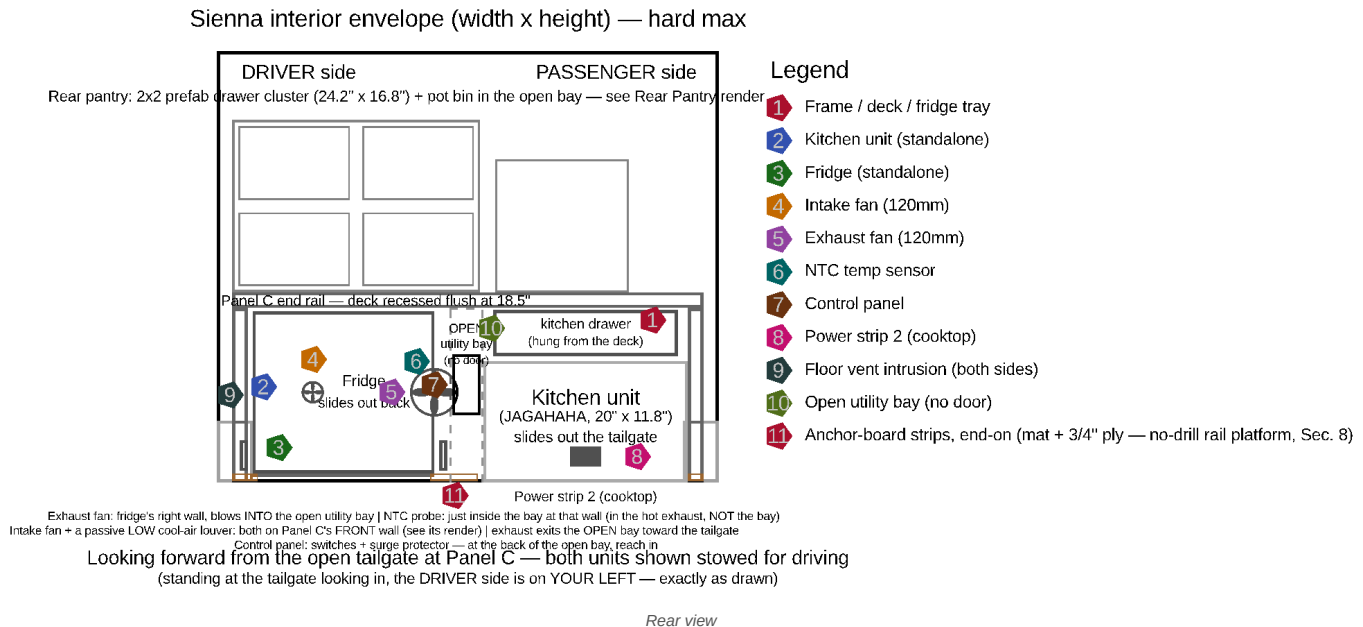
FRONT SEATS at left — TAILGATE at right. Every height above is computed from params.scad, not sketched.

no-drill anchor board (mat + ply, Sec. 8) — its steel tongue bolts to the 2nd-row floor rail's rear end (F8)

Side profile

The interior envelope used for fit checks (Section 1: 93.75" x 49" x 42" — all three MEASURED, owner Aug 1 2026) is a simplified rectangular box — the real hatch curvature, vent shapes, and wall taper aren't modeled beyond the reserves called out in `params.scad`.

Rear view, looking forward from the open tailgate at Panel C with the fridge and kitchen unit both stowed for driving:

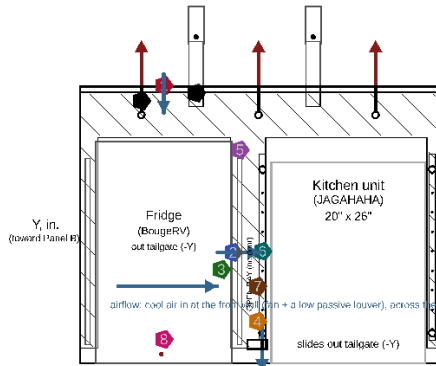


Fridge Installation Detail

The rear view above is too small-scale to label the fridge's 2 cooling fans and temperature sensor clearly. This top-down detail of Panel C zooms in on just that area, with numbered markers and a coordinate list (see Section 2 for the coordinate system and the full component coordinate table):

PANEL C — 46" wide x 35.75" deep, looking down from above

2 steel rail tongues (2"x3/16") BOLT to the rear ends of the 2nd-row FLOOR RAILS (seat anchorage — no new holes; F8; fallback: butt the striker-row step, F7) — forward restraint is steel-to-steel to the van's own rails (Sec. 8)
3 ratchet straps -> 3rd-row STRIKER LOOPS (crash-rated, ~10-14" fwd of Panel C — position UNVERIFIED, F4)



Component Position / fastener

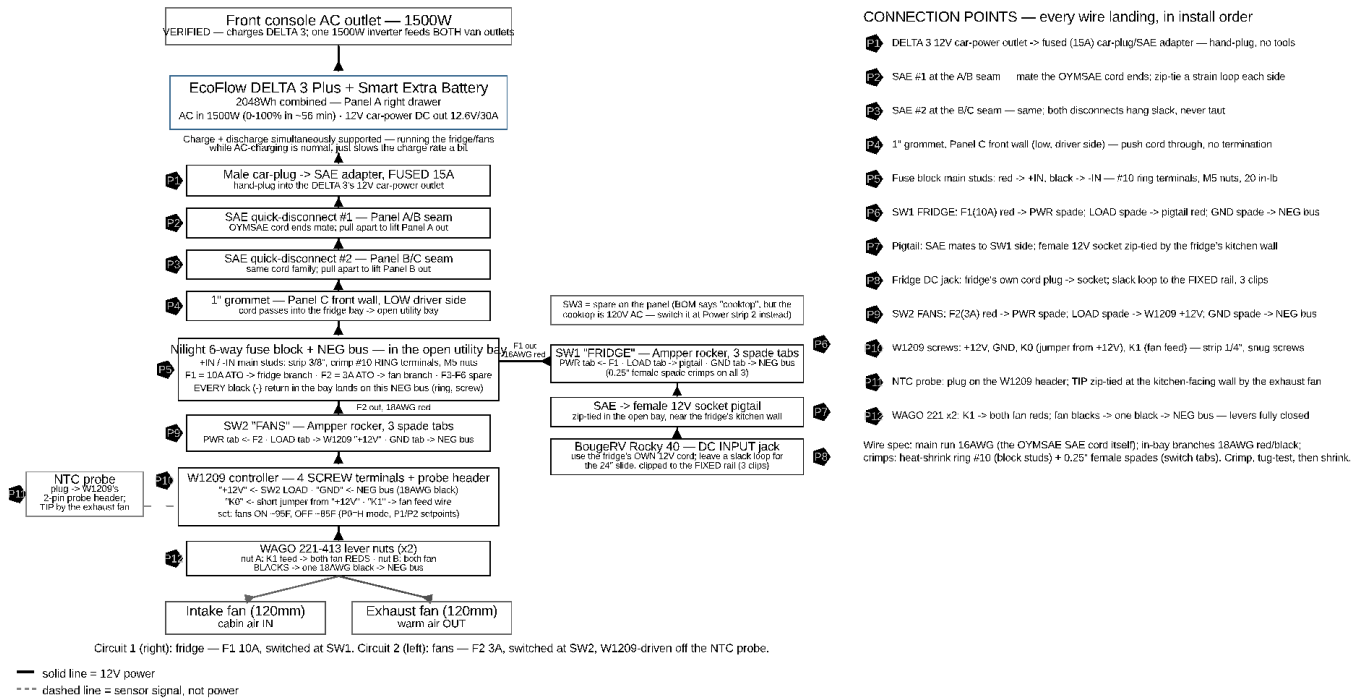
- 1 Intake fan (120mm) — blows IN
X=10.86 Y=35.75 (Panel C's FRONT wall) Z=8.8
4x M4x20 machine screws over the wall's fan hole (Front Wall render)
- 2 Exhaust fan (120mm) — blows INTO the open utility bay
4x M4x20 machine screws, 10mm bolt circle, into a plywood fan ring
- 3 NTC temp sensor
X=18.22 Y=12.2 Z=8.8
adhesive thermal pad or 1x3/4" screw through its bracket tab
- 4 Control panel enclosure
X=21.5-24.3 Y=2 Z=6.5-12.5 — in the OPEN utility bay (no door: reach in)
backer board (3/4" birch) hung from the deck; 4x #8x1" screws
- 5 Anchor-board rail strips (2 of the board's 3, under the slide rails)
mat + 3/4" ply band on each rail line BESIDE the tray (side-mount) — part of the ONE-piece board, not separate strips; the fixed rails' risers bolt to them 1/4-20 machine screws into T-nuts from below — NO holes in the van (Section 8)
- 6 Kitchen tie-down: L-track + 4 stud D-rings
on the board's kitchen-side strips (utility-bay gap + the 1.5" band at the panel edge)
4 ratchet straps (400lb WLL) criss-crossed over the top into the D-rings
- 7 WAVE 3 hose/cord hook
in the open bay — screwed up into the deck underside, kitchen side; bundle the hoses so nothing swings out
1x heavy-duty hook, #8x1.5" screw — stows hoses+cord when not in use
- 8 Fridge hold-down strap D-rings (x2)
tray side apron, near the tailgate end — hooks to the fridge's 2 end handles
cam strap, snug not tight — secures fridge TO its tray (the anchor board secures the tray to the van); side profile in fridge-slide-detail
- 9 Board bridge (same piece) + 2 steel rail tongues
the board's full-width front edge at Y=29.25 — one continuous 48"x33" comb with the strips, NO joints; 2"x3/16" flat bars run fwd and BOLT to the 2nd-row floor rails' rear ends (F8)
forward restraint = steel-to-steel to the van's own seat rails; no new holes (fallback: butt the striker-row step, F7)
- 10 Striker straps (x3)
ridge D-rings -> the 3rd-row seat striker loops, fwd of Panel C (crash-rated; position UNVERIFIED, F4)
400lb WLL ratchet straps — rearward + tilt restraint; re-tension after the first drive

All coordinates measured from Panel C's tailgate-facing left corner, floor level (see file header).
NO DRILL (Section 8): no new holes in the van. The board straps to the 3rd-row striker loops and its tongues bolt to the 2nd-row floor rails' rear ends. Cut the board's outline only AFTER the Section 0 F1 F8 survey

Fridge install detail

And the electrical side — a schematic (not to physical scale) of how the fridge, both fans, and the NTC sensor are powered and wired — now down to **every physical wire landing**: connection points P1–P12 are numbered on the diagram, each naming the exact terminal, connector type, and crimp, with the full list repeated in Section 5's connection-point table:

Fridge System Wiring — every connection point (P1-P12), schematic, not to physical scale

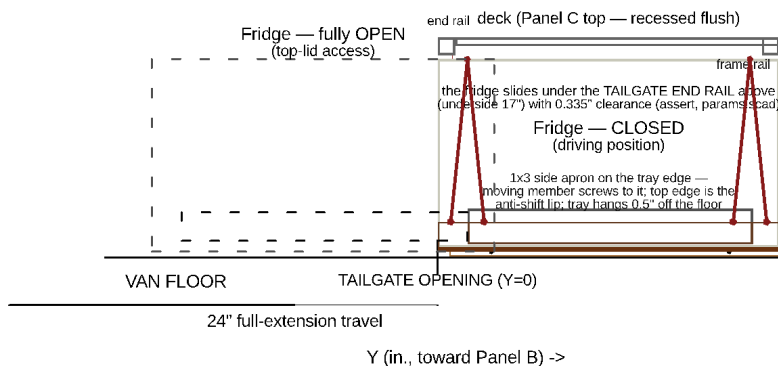


Fridge wiring diagram

The fridge's slide mechanism itself, closed vs. fully extended. The VADANIA rails stand vertically BESIDE the tray (side-mount): each fixed rail on a steel riser angle bolted to the no-drill anchor board's mat + ply rail-line strip (Section 8 — the board raises only the rails; the tray keeps its hang), each moving rail on the tray's 1x3 side apron — nothing under the hanging tray, so the slide hardware adds zero height and the mounted fridge keeps 0.33" of running clearance under Panel C's end rail:

Fridge Slide Mechanism — side view (one of 2 parallel slides shown)

CLOSED (solid) vs fully OPEN (dashed) — pulled out through the tailgate for top-lid access



Fridge slide detail

The kitchen drawer — a shallow slide-out drawer hung under Panel C's deck in the dead air above the kitchen unit (11.8" unit, 17" clear void), riding a 24" full-extension slide pair and pulling out the open tailgate; ~3.5" clear inside for utensils, cutting board, flat dry goods, and the cooktop's griddle plate:

Hold-down strap (x1, cam-buckle): fridge's end handle -> 2 D-rings on the tray's side apron — stops the fridge lifting off the TRAY (the riser/board bolting only pins the TRAY to the van). Snug, not tight — must clear the lid at OPEN.

VADANIA VD2576 industrial pair, 24", 379lb, LOCKS closed (transit) and extended (loading) — loaded fridge can hit 60-90lb, well within rating.

Tray (3/8" ply + two 1x3 side aprons) hangs BETWEEN the rails on the moving members; the aprons' top edges are the fridge's anti-shift lip.

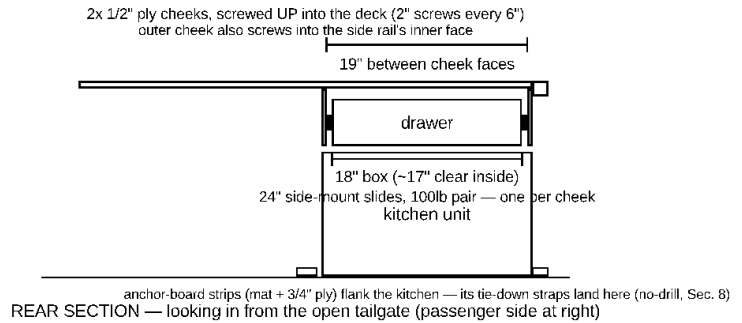
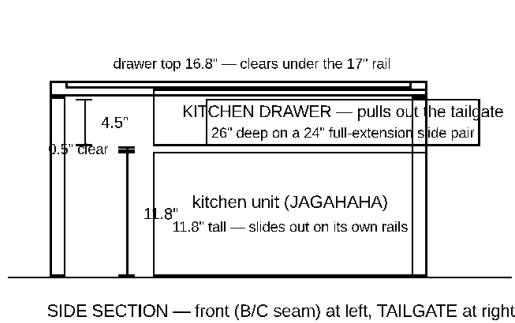
DC line slack clips to THIS fixed rail (3 screw-mount clips, Sec. 5/6) — never to the moving tray, so it can't get pinched between apron and rail when the slide comes home.

FIXED rail (VADANIA VD2576, 3" tall, stands VERTICALLY beside the tray — screwed to a steel riser angle bolted to the ANCHOR BOARD strip; it adds WIDTH beside the tray, ZERO height under it — never mount it underneath: flat under the tray it would add ~1.2" and hit the end rail above.

ANCHOR BOARD strip under this rail line (Sec. 8, no-drill): non-slip mat + 3/4" ply, riser bolted with 1/4-20 T-nuts — NO holes in the van. The board runs fwd (+Y) to the bridge: 2 steel tongues bolt to the 2nd-row floor rails' rear ends (F8) + 3 straps drop into the 3rd-row striker loops.

KITCHEN DRAWER — hung under Panel C's deck, above the kitchen unit (Component 7)

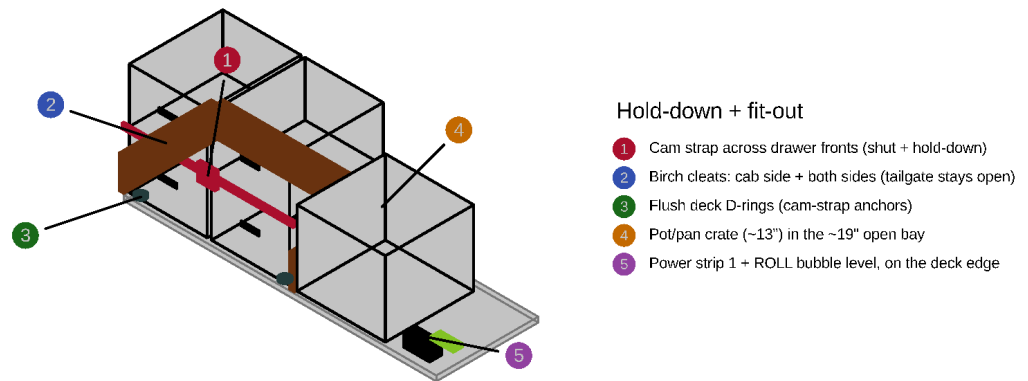
Box: 1/2" ply, glue + 1-1/4" screws, ~3.5" clear inside — utensils / cutting board / flat dry goods / griddle plate.
Face gets a magnetic catch so nothing rattles in transit (the old cabinet door it matched was cut — open bay now).



Kitchen drawer detail

Rear Pantry — Prefab Drawer Cluster + Pot Bay

The custom plywood pantry was **swapped for bought prefab drawers** (owner, July 2026 — Component 1): a 2x2 array of IRIS USA 12"W stackable drawers (Home Depot #500163) on the tailgate end of Panel C's deck, plus a pot/pan bay in the ~21.8" of open deck beside it. Nothing is built — a cleat pocket + one cam strap hold it, and each drawer unit lifts straight out (removable). This deletes the plywood carcass, the cam-lever clamp set, the 2 steel sway braces, and the fiddle-lip retention kit, and drops the pantry from ~42 lb to ~15 lb:



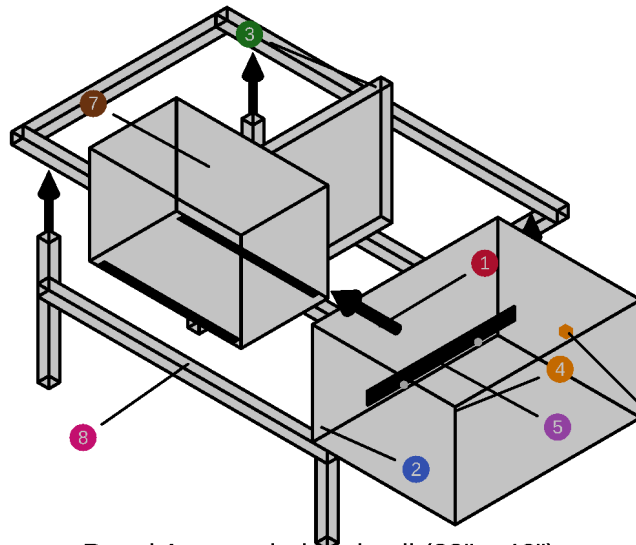
REAR PANTRY — prefab 2x2 IRIS drawer cluster (24.2" x 16.8") + pot crate, on Panel C's deck (Component 1)

Replaces the plywood pantry — ~27 lb lighter, nothing built. Each drawer unit lifts straight out once the cam strap is loosened.
Canned goods LOW, boxed dry goods UP. Cab-side cleat stops the forward slide under braking; the strap doubles as drawer-shut + hold-down.

Rear pantry layout

Floor Panel Detail

Each floor panel, exploded (Panel A: frame + legs + center divider + one right drawer, left bay is WAVE 3 open storage; Panel B: a bare cube frame — no divider or drawers; Panel C: no divider or drawers — the fridge/kitchen units live in its void, shown here as reserved footprint):



Panel A — exploded detail (29" x 46")

Left (driver-side) bay: WAVE 3 open storage, no drawer box or slide — the unit is too wide for a boxed drawer.

Component

Position / fastener / material

- 1 End rails (x2) + side rails (x2)
29" x 46" perimeter, 2x2 pine
corner brackets + 2" screws + glue
- 2 Legs (x4)
16" cut + leveling foot, inset 3.5" from the edge
1/2" x 3/4" insert hole in each bottom (Panel B detail's inset)
- 3 Center divider
26" cut, 2x2 pine
splits the bay — drawer (right) / WAVE 3 (left)
- 4 Drawer box (right, DELTA 3 side)
19.25" x 25" x 13.75", 3/8" ply, 1/2" bottom
5 pieces, glued + biscuited
- 5 Drawer slide (right)
20" full-extension pair
box-to-rail + box-to-divider
- 6 Drawer catch (right)
friction catch or small turn latch
keeps the drawer shut in transit
- 7 WAVE 3 + glide strips (left)
2x glide strip, UHMW/laminate scrap
no box, no slide — rests directly on the bay floor
- 8 Bottom rails (x2, END faces only)
2x2 pine, underside at 1"
the SIDE faces stay open — the drawer + WAVE 3 exit there at floor level

Panel A detail

LEG BOTTOM, zoomed — same hole on all 12 legs
1/2" dia x 3/4" deep, dead center of the
1.5" x 1.5" end grain -> 3/8-16 insert

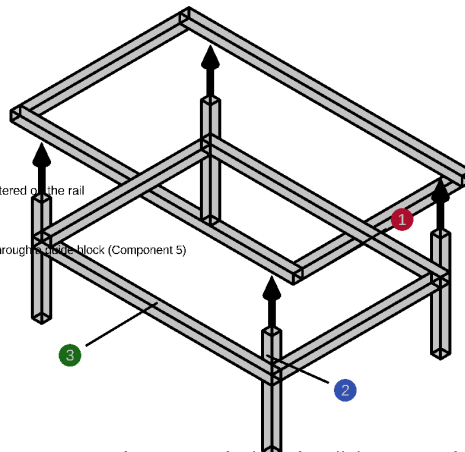


drill BEFORE assembly (Leveling Foot render)

END-RAIL SEAM FACE — both ends
2x 3/8" dia x 3/8" deep pin holes, 3" in from each side edge, centered on the rail



drill the mating panel's face as a matched pair — clamp both, drill through the glue block (Component 5)



Panel B — exploded detail (29" x 46")

NO divider, NO drawers, no skirts — the side doors don't reach Panel B, so its whole bay is deep storage, loaded from above.

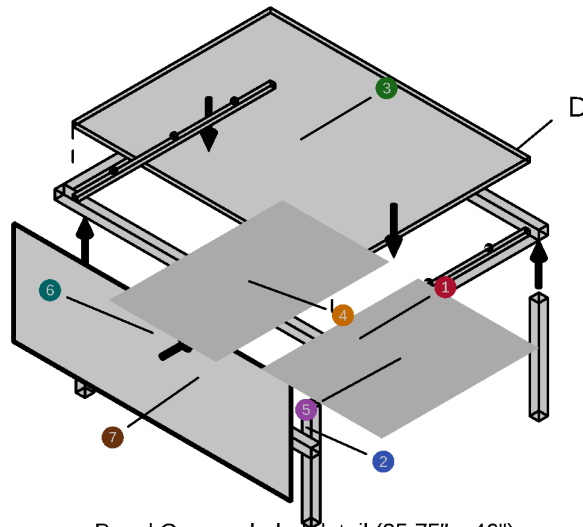
THE FULL CUBE: bottom rails on all 4 faces (nothing exits Panel B sideways). Both hole types dimensioned in the insets at left.

Component

Position / fastener / material

- 1 End rails (x2) + side rails (x2)
29" x 46" perimeter, 2x2 pine
corner brackets + 2" screws + glue; grip these rails to lift the panel
- 2 Legs (x4)
16" cut + leveling foot, inset 3.5" from the edge
1/2" x 3/4" insert hole in each bottom (inset at left)
- 3 Bottom rails (x4 — full cube)
2x2 pine, underside at 1" (clears the feet/knobs)
2x 2" screws + glue into each leg — the frame racks far less as a closed box

Panel B detail



Panel C — exploded detail (35.75" x 46")

No divider, no drawers — the void stays open for the bought fridge + kitchen unit (Component 7). ONE wall: the front (B-facing) face.

Panel C detail

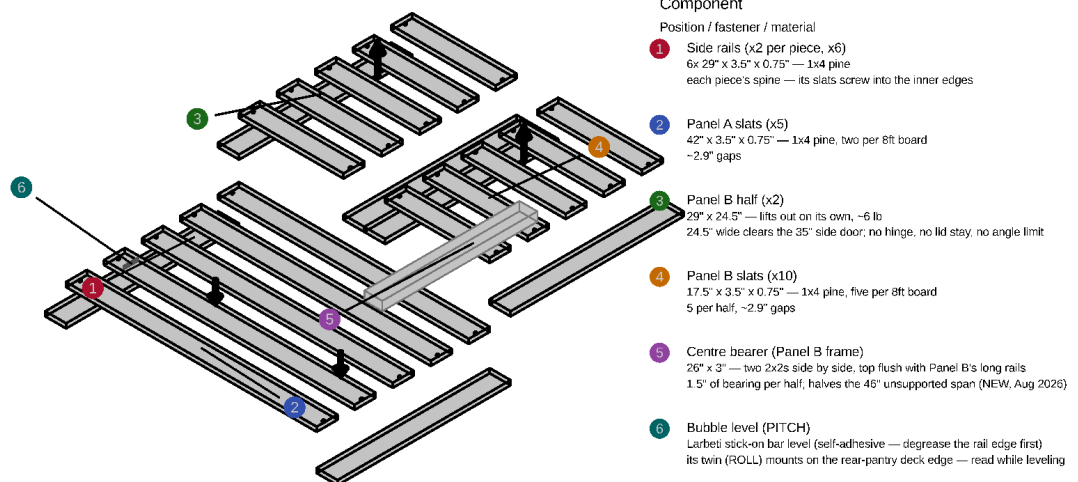
Component

Position / fastener / material

- 1 End rails (x2) + side rails (x2)
35.75" x 46" perimeter, 2x2 pine
corner brackets + 2" screws + glue
- 2 Legs (x4)
16" cut + leveling foot, inset 3.5"
insert hole in each bottom — see the Panel B detail's inset
- 3 Fixed top
43" x 32.75", 3/4" ply — RECESSED between the rails, flush with the rail tops
on 3/4x3/4 bearer cleats; screwed down — NOT a lift-off lid, unlike Panel A/B's bed-frame cap
- 4 Fridge zone (reserved)
17.72" x 28.74", flush LEFT/driver
bought product — see fridge-install-detail
- 5 Kitchen zone (reserved)
20" x 26", flush RIGHT/passenger
bought product — see fridge-install-detail
- 6 Front wall (the ONLY wall)
46" x 17", 3/8" ply, on the B-facing face
fan hole + 2 grommets — Panel C Front Wall render
- 7 Bottom rail (x1, FRONT face only)
2x2 pine, underside at 1", behind the wall
tailgate face stays open (appliances exit there); REAR legs at the TRUE corners

Bed Platform Detail

Component 2's slatted bed platform (58" x 49" overall, 15 slats pocket-screwed between 1x4 side rails — one flush 3/4" plane, cantilevered 1.5" past the boxes on each side; **width cut back from 52" by measurement V7, Aug 2026**), exploded. It is **THREE lift-out pieces (owner, Aug 2026): Panel A's 29" x 49" section, plus Panel B's top as two 29" x 24.5" centreline halves that each lift out on their own**. Per-site leveling is done at the WHEELS with blocks (Block Calculator, Appendix E); the interior leg feet are a one-time set. Both sections rest DIRECTLY on Panel A/B's top rails and the platform ENDS at the B/C seam — Panel C's own fixed deck sits at exactly the same surface height, so the two meet flush and the mattress's last ~20" rides that deck. Leveling happens at the leg feet, down at the floor, and an RV bar bubble level screwed to the driver-side rail's outer edge reads fore-aft pitch while you turn the knobs (its twin on the rear-pantry deck edge reads roll):



BED PLATFORM — 3 lift-out pieces, exploded (Component 2, 58" x 49" overall, 15 slats, 0.75" thick)

SPLIT Aug 2026: Panel A keeps one 29 x 49" section. Panel B's top is TWO 29 x 24.5" halves, split on the centreline, each lifting out on its own.

Panel B is top-load-ONLY — the 35 x 45" side door sits over Panel A and only 29" of it is ever clear. At 24.5" wide a half fits back out through that door; a 49" piece never would.

Lift the mattress clear, take out only the half on the side you are standing at. The centre bearer also halves the deck's unsupported span over Panel B (46" -> ~22").

Bed frame detail

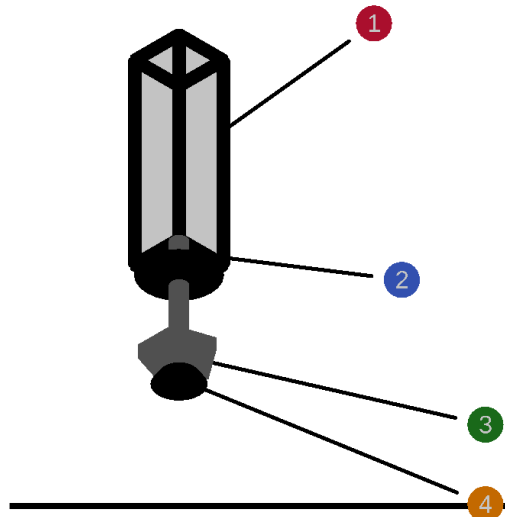
Leg Leveling Foot Detail

Leveling lives at the floor: each leg is cut 1" short and gets a 3/8-16 insert in its bottom end grain, taking a leveling glide bolt with a floor pad and a ~2" star-knob hand grip (12 total: 4 per panel x 3). Effective leg heights are 17" on Panel C / 16.25" on A/B (cuts: 16" and 15.25" — deck recess); dropping the old between-layers adjusters bought 1.25" of headroom, and recessing the deck ply into the rail plane buys another 3/4". To adjust: tip that corner of the box slightly and spin the knob — every leg is exposed at floor level:

Component

Position / fastener / material

- 1 Leg (bottom end shown)
2x2 pine (1.5" x 1.5" actual), cut to 16"
drill centered, 1/2" dia x 7/8" deep for the insert
- 2 3/8-16 threaded insert
driven UP into the leg's end grain
screw-in style, 7/16" OD coarse outer thread
- 3 Star knob, ~2" dia (hand grip)
3/8-16 thru-hole knob + jam nut on the bolt shaft
the adjustment — big enough to turn with the box tipped
- 4 Leveling glide bolt + pad
1.375" floor pad; +/-0.5" travel around 1" nominal
threads into the insert — leveling only, not height changes



LEG LEVELING FOOT — shown 3x actual size (12 total: 4 per panel x 3 panels)

Legs are CUT to 16" (Panel C) / 15.25" (A/B — deck recess) — the foot makes up the last 1".

To level: kneel at the side door, tip that corner of the box slightly, spin the star knob. Every leg is exposed at floor level — nothing boxes it in. The platform above rests DIRECTLY on the box rails (no adjusters up top anymore).

Leveling foot detail

Load: worst case is ~700 lb total (2 people + mattress + platform + boxes + cargo) spread over 12 feet — ~60 lb per foot against the feet's 330 lb rating each, a 5x margin before dynamic factors. The feet are not the weak point. If you want more margin anyway, 1/2"-13 leveling mounts (1,100 lb each) are a drop-in upgrade — same install, just a 5/8" insert hole instead of 1/2".

Access: the practical answer is to stop crawling under the bed at camp entirely — **level the VAN at the wheels** (Andersen/Camco-style curved ramps, standard van-life practice) each site, and set the interior feet ONCE against the van's own floor irregularities. After that first setup you shouldn't need to touch a knob; the two bubble levels tell you at a glance whether the wheels need a ramp.

Blocks + calculator (the no-feet workflow): the two bubble levels read DEGREES (-7 to 7). Degrees plus the wheelbase are all the math needs, so there's a phone calculator for it — [Sienna Block Calculator](#) (also saved in the repo as `leveling_calculator.html`): enter the two readings and which end/side is low, and it returns how many Lynx-style leveling blocks to stack under each tire, the total, and the residual tilt after driving up. One 10-pack of blocks covers ~2.9° of correction at 1.5"/block over the 120.5" wheelbase — beyond ~4 blocks per tire, find a flatter spot. This makes the interior feet a ONE-TIME set (against the van's own floor irregularities), not a per-site chore — and if you'd rather delete the feet entirely, replace them with fixed shims set once and let the blocks do all site leveling (saves ~\$72, loses the fine-trim option).

Installing the calculator on a phone (two ways):

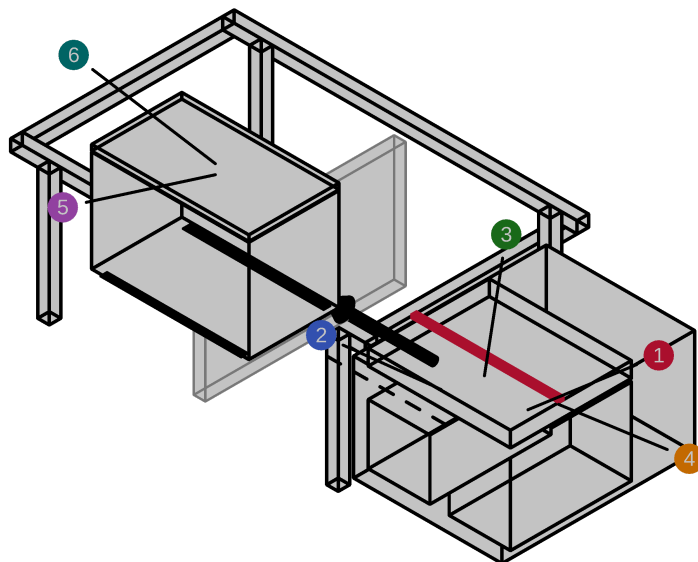
1. **Online:** open the [calculator link](#) in Chrome on the phone (it needs the claude.ai login), then Chrome menu → **Add to Home screen** — it gets an icon and opens full-screen. Needs signal at launch.
2. **Offline — the one to rely on at camp:** `leveling_calculator.html` is a single self-contained file with zero network dependencies. Send the copy from this repo (or `~/Downloads/sienna_block_calculator.html`) to the phone by email/Drive/Nearby Share, open it once via **Files** → **Open with** → **Chrome**, and bookmark it. It runs with no signal forever and remembers the settings below.

Using it at a site:

1. Park in the spot you actually want, engine off, **parking brake on**.
2. Read the two bed-mounted levels: the fore-aft level on the platform's driver-side rail (pitch) and the side-to-side level on the rear-pantry deck edge (roll). Each reads in degrees; the bubble floats toward the HIGH side, so the opposite end/side is the LOW one.
3. Enter both degree readings and tap which end (nose/tail) and side (driver/passenger) are low. One-time settings under "Van & block settings": wheelbase 120.5", track ~68", and the height each block adds — Lynx-style ≈ 1.5", but **measure your own stack once** and enter that.
4. The van diagram shows blocks per tire. Lay each stack just ahead of its low tire, **drive up slowly**, and chock a wheel.
5. Re-read the levels and refine once if needed — suspension squish and block nesting make a second small pass normal. If any tire calls for more than 4 blocks, the calculator says so: re-park (turning the van around often halves the stack) rather than building a tower.

DELTA 3 / WAVE 3 Stowage Detail

Stowage for the EcoFlow gear: the DELTA 3 Plus + Smart Extra Battery, stowed unstacked side by side in Panel A's right drawer, and the WAVE 3, stored as open bay storage (no drawer box, no slide) in Panel A's left bay (its hoses/cord stow separately — item 12 in the fridge installation detail above):



DELTA 3 STACK (right drawer) + WAVE 3 (left bay)

Panel A, exploded — Components 2 & 8

DELTA 3 Plus outboard (the WAVE 3 plugs into it), Extra Battery inboard; ~48 lb in a normal drawer — no E-track. The WAVE 3 rests in the raw bay on 2 glide strips.

Found storage: a ~2.6" lift-out tray tops the DELTA stack, and a shelf @ 13.7" tops the WAVE 3 — it still slides out beneath.

Component / position

- 1 DELTA 3 Plus — 7.95"x15.7"x11.16", ~28 lb outboard (pull wall); cam-strapped over locating cleats
- 2 Smart Extra Battery — 8"x15.7", ~20 lb inboard — extra runtime only, grabbed less often
- 3 Lift-out tray — reclaims ~2.6" dead air cables, the DELTA's own cords, dongles
- 4 Cam strap + D-rings + 1" grommet strap over both units; WAVE 3 charge cable exits the grommet
- 5 WAVE 3 — 20.4"x11.7"x13.2", 33.7 lb raw bay, 0.35" clear — a boxed drawer would NOT fit
- 6 Overhead shelf @ 13.7" on 1x1 cleats flat soft goods + the WAVE 3's own hoses ride up here

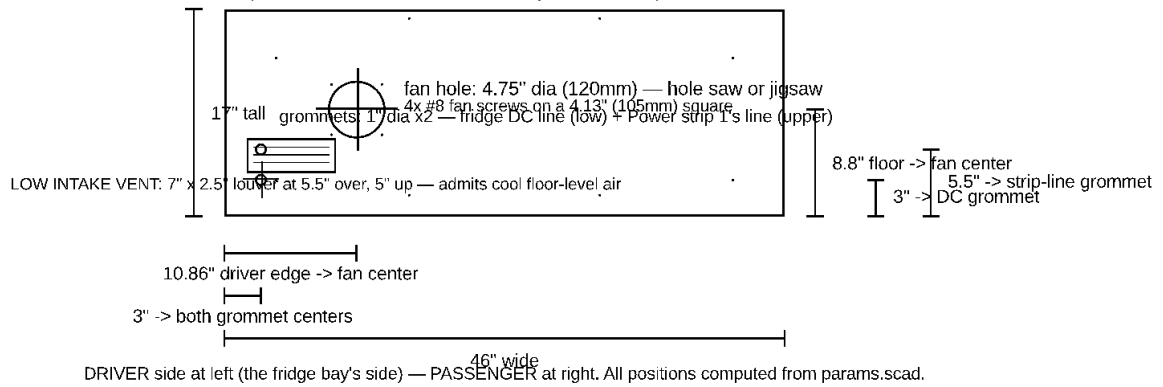
DELTA 3 and WAVE 3 stowage detail

Panel C Front Wall

The one wall any panel gets, as a flat pattern with **every hole dimensioned**: the 120mm intake-fan hole (with its 105mm screw square) centered on the fridge bay, ONE 1" grommet (fridge DC line low — Power strip 1 now plugs into the REAR outlet and never crosses this wall), and the 8 perimeter mounting screws (2 now landing in the cube-frame bottom rail). Panels A and B have no walls or skirts at all, Panel C's sides stay open, and its tailgate face needs no wall — the fridge, the open utility bay between the units, the kitchen unit, and the kitchen drawer face fill it completely:

PANEL C FRONT WALL — 3/8" ply, flat pattern (the ONLY wall on any panel)

Mounts on Panel C's front (B-facing) face, floor to rail underside. Intake fan bolts over the big hole (blows IN); the low louver is a passive cool-air scoop below it.
 Panel A/B: no walls or skirts anywhere. Panel C sides: open. Panel C tailgate face: no wall — fridge + open utility bay + kitchen + kitchen drawer fill it.
 8x #8 x 1-1/4" perimeter screws: 2 into each front leg + 2 into the top rail + 2 into the bottom rail



Panel C front wall detail

1. Overview

Vehicle constraints (hard limits — everything below is designed inside them)

The 2nd row seats are being **removed entirely** (not just pushed forward), with the 3rd row folded flat as before. **Every figure in this table was measured on the actual vehicle on Aug 1 2026** (owner), with the 2nd row out and the 3rd row folded as it sits in camper mode — see Appendix A for the survey sheet those numbers came off, including what each one changed.

Constraint	Value	Notes
Max interior length	93.75"	Closed rear hatch to the front seatbacks, 2nd row removed — MEASURED V1 (Aug 2026), 2¼" under the old estimate
Max floor width	49"	Between the wheel wells (54" forward of them) — MEASURED V2
Max height	42"	Load floor to headliner, mid-van; 37" back at the gate — MEASURED V3
Vent intrusion	3.5" per side	Rear lower heat vents, at floor level only — legs must stay clear; the raised deck may overhang — MEASURED V4, 1" per side deeper than assumed
Hatch curvature reserve	0"	Nothing needs reserving: the build never rises high enough to meet the hatch glass/trim curvature — MEASURED V5
Usable width at the sleeping plane	50"	Wall-to-wall at ~18.5" up (49.5" at ~22.5") — MEASURED V7. The walls do NOT flare to the ≥53" this design assumed , which is what cut the bed platform to 49"

What the measurement changed (Aug 1 2026): the old 96" was an estimate extrapolated from a 72" figure taken with the 2nd row still installed. The real empty interior is **93¾"** — 2¼" shorter. That would have forced every panel to shrink, except that V5 measured the hatch-curvature reserve at **zero** and handed back the 2" that was being held at the tailgate end, so the panel train only lost ¼" **net**. **Panel C absorbed all of it (36" → 35¾")** — forced by two of the model's own guards: Panels A and B must stay equal (shared step diagrams / cut list), and $A + B \text{ is } \text{bed_frame_length}$, so trimming either would change the 58" bed rails for no reason. Panel C's ~7" of fore-aft slack around the fridge is where the quarter inch comes from. The one casualty of the zeroed reserve: the pantry drawers were allowed to sit 0.3" proud of the deck edge into it, so `pantry_len` grew 14" → 14.5" and the deck now carries them fully. All `assert()` guards in `params.scad` pass with the measured numbers in place.

That leaves a **93.75" usable length** (V5 reserves nothing at the hatch) and a **42" floor-level width** (full 49" available at deck height, 50" up at the sleeping plane). The math: Panel A 29" + Panel B 29" + Panel C 35.75" = **93.75" of panels** — **Panel A sits flush with the front seatbacks (no open floor left over)**, filling `usable_length` exactly. Unlike the earlier "three panels + a separate rear row" layout, there's no length lost to a dedicated fridge/kitchen zone — the fridge and kitchen unit both live in the void *underneath* Panel C's deck, sharing space with the same legs that hold up the mattress. The rear pantry claims the last 14.5" of Panel C's own 35.75" (the tailgate-most strip, right next to the fridge/kitchen void), leaving **79.25" of continuous sleeping run** (Panel A + Panel B + the first 21.25" of Panel C) for the mattress — the HEST Dually Long (78" long) rides here with ~1.25" of spare at the head end. The measured interior *did* come in shorter than the 96" estimate, and the `assert()` guards in `params.scad` are what forced the ¼" out of Panel C and the ½" into `pantry_len` rather than letting the train silently overflow. These limits live at the top of `params.scad`: bump any dimension past the envelope and every render fails loudly instead of silently overflowing the van.

Rear liftgate opening (MEASURED — owner, Aug 1 2026)

Every module has to physically pass through the liftgate opening to be lifted in and out — that's the entire point of the modular design, so this matters as much as fitting once installed. These were web-sourced estimates (48" × 36"); both are now tape measurements, and **both came back better than the estimate**.

Constraint	Value	Notes
Gate opening width	50"	Narrowest point — MEASURED V6a (was estimated 48")
Gate opening height	37"	Clear height at the centre of the opening — MEASURED V6b (was estimated 36")
Height out at the corner radius	20.5"	The corners round off hard — MEASURED V6b. Carry modules through the middle of the gate, not flat against one side

Fit check (each module carried upright, legs down, the way it sits once installed):

Module	Passes through as	Width margin	Height margin
Panel C	46"W × 18.5"H	2" per side	18.5" to spare
Panel A, Panel B	46"W × 17.75"H	2" per side	19.25" to spare
Bed platform (Component 2)	49"W × 58"L × 3/4"H	enters tilted diagonally (gate diagonal ~62")	—
Rear pantry drawer units (prefab, Component 1)	~12.1"W each	easily	a few lb each — lifts out one at a time

The fridge and kitchen unit never need to pass through the gate opening at all — they live inside Panel C's own void as bought products, installed once and left in place, not lifted in and out. The tight spot is no longer width — the measured 50" opening gives the 46" boxes **2" per side**. It is the **corner radius**: only 20.5" of height survives out at the rounded corner, so an 18.5"-tall box walked in flat against one side will foul long before it runs out of the 37" available in the centre. `params.scad` asserts against the measured numbers the same as every other hard limit. The rear pantry's prefab drawer units are separate pieces from Panel C by nature — loosen the cam strap and each ~12.1"-wide unit lifts out and through the gate trivially (Component 1).

Side door openings (FULLY MEASURED Aug 2026 — aperture 35" × 45", 29" ever clear, opening starts at the front seatback)

The side doors only matter for **Panel A**: its right drawer (DELTA 3) pulls out the passenger door and its left bay (WAVE 3) is reached through the driver door — the door openings sit over the old 2nd-row footprint, which is exactly where Panel A lives. **Panel B is beyond the door openings entirely** (owner-confirmed), which is why it has no drawers: nothing pulled sideways from it could clear a door, so its bay is top-loaded deep storage instead. Panel C's void houses the fridge and kitchen unit, both pulling out through the open tailgate. Measured Aug 2026 (V8) — and the news is worse than the estimate.

Constraint	Value	Notes
Usable clear width at the door's real stop	29"	MEASURED V8c — 11" tighter than the 40" assumed . This is the number that governs; <code>side_door_opening_width</code> = 29
Side door aperture	35" × 45"	MEASURED (Aug 2026) — the opening itself. The door parks 6" short of its forward edge, which is why only 29" is ever clear
Door panel (not the opening)	50" × 58"	Measured, but these are the door's own outside dimensions — recorded so they aren't mistaken for aperture figures
Door opening's fore-aft position	Y = 0	MEASURED (Aug 2026) — the front seatback is essentially even with the opening at build height; if anything the seatback intrudes ~¼" into it. <code>side_door_y0</code> = 0 is now a measurement, not the old guess

Reach caveat, now sharper: a 29" clear opening won't expose the full 58" drawer run (Panel A + Panel B) from one position — the reachability check in `params.scad` reports Panel A reachable and **Panel B effectively unreachable from the side**. That is survivable only because Panel B was already designed as a top-loaded bay with no drawers; nothing has to change. Put your most-used gear in Panel A's drawers, and re-run the check once `side_door_y0` is actually measured.

Layout

- Rear pantry — prefab drawer cluster + pot bay** (owner, July 2026): the custom plywood pantry is **replaced by bought prefab drawers** — nothing built, and ~27 lb lighter. A **2×2 array of IRIS USA 12"W stackable drawers** (Home Depot #500163, two 3-packs) sits on the **tailgate end of Panel C's deck**, in the same tailgate-end footprint the plywood pantry used, now 14.5" deep (the units' 14.3" depth is carried entirely by the deck — V5 measured the hatch reserve at zero, so there is nothing left to hang 0.3" proud into; the sleeping run is 79.25"). The cluster is **24.2" W × 14.3" D × 16.8" H**, leaving **~21.8" of open deck** on the passenger side for a **pot/pan crate** (~13" milk crate — the pots' own 11×11 box drops inside) and the relocated power/wiring. It's held by a **cleat pocket + one cam strap across the drawer fronts** (which also keeps the drawers shut) — no fasteners into the units, so each drawer unit **lifts straight out** and clears the gate. **Gone with the plywood pantry:** the cam-lever clamp set, the 2 steel L-angle sway braces, the fiddle-lip/lash-strap retention kit, and the **enclosed bed cubby** — so **Power strip 1 and the ROLL bubble level move to the deck edge** in the 19" bay (still reached from the bed). See the Rear Pantry render and Section 6, Component 1.
- One continuous sleeping deck**, 46" wide × 93.75" long (Panels A + B + C, flush with the front seatbacks — no gap), deck surface sitting **18.5"** above the van floor: **the horizontal ply is recessed INTO the rail plane** (deck recess, owner July 2026 — Panel C's deck drops between its rails onto 3/4"×3/4" bearer cleats, flush with the rail tops; Panels A/B instead run **16.25" legs**, so the 3/4" bed platform resting ON their rails tops out at the same 18.5" plane). That buys 3/4" of headroom: **19.5" sitting headroom** over the 4" mattress (V3 measured the cabin at 42", not 44" — so the deck recess is now carrying more of the load than it was designed to). Panel C's `leg_height` stays 17", driven by the fridge's mounted stack (see below), not by the folded 3rd-row well depth — confirm 17" clears your actual well before cutting.
- Panel A and Panel B share the same frame, and neither has a top of its own**. Panel A has a center divider splitting its bay in two, reached through the side doors: the right (passenger) side is the one real drawer (DELTA 3), the left (driver) side is WAVE 3 open storage (the unit is too wide for a boxed drawer — Section 1). **Panel B has NO drawers and no divider** — the sliding-door openings sit over the old 2nd-row footprint (Panel A), not over Panel B, so nothing pulled sideways from Panel B could clear a door. Its whole bay is deep storage instead, loaded from above by lifting out **either half of its two-piece bed top** (long-term/bulky items you don't touch at camp). The slatted bed platform (Component 2, 15 slats between 1x4 side rails, 58" × 49" overall — cantilevered 1.5" past the boxes each side, 3/4" thick; **cut back from 52" by V7**) spans Panel A + B ONLY and ENDS at the B/C seam: Panel C's own fixed deck is at exactly the same surface height, so the two meet flush and the mattress's last ~20" rides that deck (an 80" platform would have to sit ON that deck — 3/4" too high). It rests DIRECTLY on Panel A/B's top rails as one flush 3/4" plane (slats pocket-screwed between the side rails). Leveling happens at the leg feet down at the floor — each leg is cut 1" short and carries a hand-adjustable leveling foot.

SPLIT INTO THREE LIFT-OUT PIECES (owner, Aug 2026). It used to be one 58" piece that lifted off whole — which meant that reaching Panel B, the only bay with no side access at all, involved wrestling a 49"×58" frame out and finding somewhere in a packed van to put it. It is now three pieces:

- Panel A's section** — 29" × 49", one piece, **screwed down and permanent** (owner, Aug 2026). It never needed to come out: Panel A's two bays are both reached through the side doors — the DELTA 3 drawer pulls out the passenger side, the WAVE 3 bay is reached by hand from the driver side. Fixing it is a real simplification, not just one less loose part. It becomes a **screwed-down diaphragm** across Panel A's rails, putting back some of the torsional stiffness the design gave up when Panels A and B lost their plywood tops (the diagonal corner braces now have help rather than carrying that alone); it is the **fore-aft datum** the two loose halves locate against; and it **cannot rattle**, so it needs no anti-rattle pads. The trade: Panel A's bay loses its from-above deep-cleaning route and is side-door-only. Acceptable — the 29" clear side gap exactly spans Panel A's 29" length, and the DELTA drawer (25" fore-aft) comes out through it.
- Panel B's TWO HALVES** — 29" × 24.5" each, **split on the centreline**, each lifting out on its own at ~6 lb. Lift the mattress clear and take out only the half on the side you're standing at.
- A centre bearer in Panel B's frame** — 26" of 3"-wide 2×2 (two 2×2s side by side, or one 2×4 laid flat) running fore-aft at the centreline, its top flush with Panel B's long rails. The halves' inner rails land on it, 1½" of bearing each — the same as their outer rails get from the long rails.

Why halves rather than one hinged leaf (which is what this replaces): **24.5" fits back out through the 35" side door, and 49" never could** — a full-width piece could only ever leave via the tailgate, which defeats the point of working from a side door. Halves also mean the mattress only has to be lifted clear of half the width, there's no hinge, lid stay or opening-angle limit (a hinged 29" leaf could only reach ~67° before hitting the headliner — there's just 23.5" above the deck), and **the**

centre bearer halves the deck's unsupported span over Panel B, 46" → ~22". Panel B never had a centre divider where Panel A always did, so the sleeping surface is stiffer than it has ever been. The costs: two loose parts to set down instead of one captive lid, and ~1.7 lb over the hinged version.

⚠ **Neither scheme gets you around the mattress.** It's one 78" piece and Panel B is the *middle* third of the bed — fold the head end back and it lands on Panel B; fold the tail end forward and it lands there too. The mattress comes off, or gets folded clear, first. Halving the width is what makes that manageable rather than a two-person job.

Sourcing: five 42" slats (Panel A) and ten 17.5" slats (the halves, five per 8ft board) plus six 29" side rails — **seven 1x4 x 8ft boards, crosscuts only** (a 1x4 is already ¾"x3½", the slat spec, so nothing gets ripped), plus 52" of 2x2 for the centre bearer. No bought slat kit: they come stapled to webbing or riveted to metal side frames, neither of which survives being cut down (Section 4 + Component 2). - **Trade-off: no top means no enclosure — answered with CUBE FRAMES.** Removing the plywood tops loses torsional rigidity and a dust barrier. Two fixes are now specified: diagonal corner braces up top, and **bottom rails closing each frame into a box** (underside 1" up — dropped to the leg bottoms, just clear of the leveling feet, for the tallest box section and the lowest floor-edge curb) on every face that can take one — Panel A's two END faces (its sides must stay open for the drawer/WAVE 3), ALL 4 faces of Panel B (the full cube), and Panel C's FRONT face (its tailgate face stays open for the appliances; its fixed top + new front wall already stiffen it). A closed frame racks far less than rails + brackets alone. The bed frame's slats still have gaps — small items can fall through into the bays below. - **Panel C has no drawers.** Its under-deck void instead holds two bought products side by side across the 46" width: a **BougeRV Rocky 40 (41-quart, dual-zone)** — 17.72" side left-right, reversible lid (manual \$4.4), optional detachable B240 battery at the tailgate-facing end — on a heavy-duty slide, against the driver-side REAR CORNER leg (1.5" in from the edge); and a **JAGAHABA slide-out camp kitchen** (26"L x 20"W x 11.8"H closed, with its own 2-burner stove space and built-in slide), against the passenger-side rear corner leg — its shelves swing out on that side. (Panel C's rear leg pair sits at the TRUE corners — inset legs would stand exactly in the appliances' slide paths, a collision the Rocky 40's extra width exposed. The rear-corner floor vents were checked Aug 2026 and do NOT reach the leg area, so those legs land on solid floor.) **Both pull straight out the open tailgate** (not a side door). Neither is something this plan builds from plywood — both are bought, standalone units. Unlike everything else in this build, both are **strapped to the van's crash-rated 3rd-row striker loops** via a shared no-drill anchor board (Section 8 — the owner ruled out drilling the vehicle) — too heavy to rely on the same "rests unbolted" approach as the sleeping panels. The gap between them is an **open utility bay — no door** (an earlier revision hung a hinged door with a louver there; the owner cut it as pointless: the louver only existed because the door trapped the exhaust air, and the door itself just slowed down reaching the switches). The exhaust fan blows the fridge bay's warm air INTO the bay and straight out toward the tailgate, and the control panel (switches + surge protector) mounts at the back of the bay on a backer board hung from the deck underside — everything electrical is reachable by just reaching in. The 2 utility bins that live in the bay's spare volume get hook-and-loop tape under them, since there's no longer a door holding loose items in. - **A HEST Dually Long mattress (78" x 50" x 4", solid foam — no air chambers)** rides on the **49"** cantilevered platform: **25" of width per person** (vs 23" at the old 46") and ~1.25' of spare length parked at the head end beside the rear-pantry cluster. The platform overhangs the 46" boxes by 1.5" per side — the boxes stay at 46" (floor vents + liftgate pass-through), and the mattress lives ~18.5–22.5" up. ⚠ **V7 (Aug 2026) measured wall-to-wall up there at 50" / 49.5", not the ≥53" this assumed — the platform came down from 52" to 49", and the 50"-wide mattress is now a ZERO-CLEARANCE fit between the walls that also overhangs its own frame by ~½" per side.** Solid foam tolerates both, but **dry-fit the foam in the van before cutting the platform to final width** — Appendix A, V7. Budget fallback: the DIY 2-layer foam build (Section 4/Component 9), same 50x78 footprint. - Panel legs sit inset **3.5"** from the deck's side edges so they land clear of the floor-level vent intrusion (**MEASURED V4, Aug 2026 — was 2.5"**; the vents eat 1" per side more than assumed). The deck itself still overhangs the vents harmlessly at height. - **Power:** the fridge is 12V-native, but now runs off the EcoFlow DELTA 3 stack (Panel A) via its own dedicated DC cord rather than the van's rear accessory outlet — see Section 1 for why. **The cooktop, Power strip 1, and the DELTA 3's own AC charging cord each get their own dedicated run to the front console now** (previously the cooktop and Power strip 1 shared one line) — see Section 5 for all three routes. - **Fire safety + CO safety (owner-placed):** the fire extinguisher and the low-level CO monitor are **deliberately not located in this plan** — both are owner-supplied and will be positioned manually once the build is in the van. Two reminders that survive from the earlier analysis: the design has no propane at all (electric induction cooktop), so the only combustion-gas risk is the Sienna's own engine exhaust while idling parked; and a generic 70ppm home-style CO detector is too slow for that job — use a low-level unit (alarms at 9/25ppm within ~60s). - **Camp power/cooling (EcoFlow):** an EcoFlow DELTA 3 Plus + Smart Extra Battery (~48 lb combined) stows unstacked, side by side, in Panel A's **right (passenger-side) drawer** — a normal drawer, not strapped down like Panel C's appliances, since 48 lb is well within what the drawer slides already handle and this gear is never used while driving. Within the drawer, the **DELTA 3 Plus sits outboard** (nearest the pull wall — it's what the WAVE 3 actually plugs into, used whether or not the extra battery is along) and the **Smart Extra Battery sits inboard** (grabbed less often, for extra runtime only). The **WAVE 3 portable AC/heater stores in Panel A's LEFT (driver-side) bay as open storage, not a drawer** — it's 20.4" wide, too wide for a boxed drawer's 19" clear interior, but the raw bay (20.75", no box walls to eat into it) fits it with ~0.35" to spare. It rests directly on the bay floor on 2 UHMW glide strips, reached by hand through the driver's side door — Panel A ends up with only one actual drawer (right/DELTA 3), since the left side gave up its drawer to make room for the WAVE 3. For camp use, the unit gets carried to wherever it actually runs (Panel C's tailgate deck or the front seat — see the sleeping-configurations note below); it's storage-only in Panel A, not used in place there. - **Side-door access got tighter, not looser (V8, Aug 2026).** The usable clear opening at the door's real stopping point measures **29"** — not the ~40" estimated — against Panel A + Panel B's combined 58" drawer run, so barely half that run can land within reach from any one door position. The door's fore-aft position is **now measured too:** the opening starts level with the front seatback, i.e. at **Y = 0** (owner, Aug 2026). So the 29" clear gap runs from 0" to 29" — **exactly spanning Panel A's 29" length.** Panel A is therefore reachable end to end, and Panel B (29–58") gets **zero overlap — blocked outright** behind the van's own body structure. **Both design decisions now rest on measurements rather than a placeholder:** the DELTA 3 stack belongs in **Panel A**, and Panel B correctly has no drawers, which is why its bed top is two lift-out halves instead (Component 2). The coincidence is worth noticing — the door runs out precisely at the A/B seam, so there is no margin either way, and the DELTA drawer (25" fore-aft, centred in the 29") clears it with ~2" a side. Both units are charged at camp via shore power or solar — this design does not route van power to either (see Section 5). - **Why the fridge (and its fan system) now run off the DELTA 3 stack, not the van's rear outlet.** The earlier design kept the fridge on the Sienna's rear 12V accessory outlet specifically to avoid two costs: a long cord run across all 3 module seams, and competing with the WAVE 3 for the DELTA 3's battery budget. This design accepts both costs deliberately, for a concrete reliability win: **the fridge keeps running when people are away from the vehicle, regardless of the van's ignition state** — the original design's whole fridge-on-the-van's-outlet approach had exactly one open risk (many factory 12V outlets cut off with the key out, which would mean the fridge stops cooling every time the van is parked and locked), and putting the fridge on the DELTA 3 instead makes that risk moot rather than just hoping the outlet turns out to be always-hot. **The real numbers make this hold up:** the DELTA 3 Plus + Smart Extra Battery combined is 2048Wh; the Rocky 40 is rated 60W max / 45W ECO (dual-zone, compressor duty-cycled) — realistically ~450–1050Wh/day depending on mode, partition use, and ambient temp, so a full charge runs the fridge alone for roughly **2 to 4.5 days** unattended, worst case to best case. And recharging is fast: the DELTA 3 Plus's AC input maxes out at 1500W, an almost exact match for the Sienna's front console outlet (**confirmed 1500W**) — a full 0-100% charge takes about 56 minutes, so even a short errand run tops it back off. EcoFlow's DELTA 3 line supports charging and discharging at the same time (running the fridge while AC-charging is normal, it just slows the charge rate somewhat), so there's no need to choose one or the other. **The fan system moves to the DELTA 3 too, not just the fridge** — the fans exist to vent the compressor's heat, so they need to stay available under the exact same conditions the fridge does (parked, people away, ignition off); leaving them on the van's outlet while moving only the fridge would defeat the point. The old rear 12V accessory outlet is simply unused by this design now — free for anything else you'd like to plug in there later. **One real trade-off to plan around:** the front console circuit now has 3 things wanting it (the cooktop, Power strip 1, and the DELTA 3's AC charging, which alone can draw up to the circuit's full 1500W) — see the shared-circuit note in Section 5. - **Modular lift-out**

design: Panel A, Panel B, and Panel C are each built as an independent, self-supporting module — own frame, own 4 legs (the fridge and kitchen unit are bought products with no frame of their own, and aren't lifted out as part of this; the rear pantry isn't its own module either — see below). All three now lift straight out and drop straight back in without touching the others (for deep cleaning or reconfiguring — day-to-day storage access is through the drawers/shelves, not by removing a module). With no tops or skirts, each panel is gripped by its exposed 2x2 top rails (no routed hand-holds needed — those were dropped along with the router jig), and anti-rattle bumpers + alignment pins + hand-released seam draw-latches at the A/B and B/C seams keep things fast to install/remove, quiet in transit, and clamped into one rigid beam (Component 5) — see Section 8. The rear pantry's prefab drawer units just sit on Panel C's deck behind a cleat pocket and one cam-buckle strap (Component 1) — loosen the strap and any unit lifts straight out, contents and all.

WAVE 3 sleeping configurations: tent vs. no tent

The WAVE 3 is **stored** in Panel A's left bay either way (see the DELTA 3/WAVE 3 stowage detail, Renders) — these two configurations are about where it gets carried to and set up for actual camp use, and where its hoses vent. Both use the same unit and DELTA 3 stack.

With a tent (tailgate open): carry the WAVE 3 from its Panel A storage bay back toward the tailgate, setting it down on the mattress-covered part of Panel C's deck just forward of the rear-pantry cluster (the pantry's 14" occupies the tailgate-most strip of Panel C, so the WAVE 3's spot sits a few inches forward of that) — still blowing toward the open tailgate so the shared van+tent air volume gets conditioned together. A cheap non-slip mat under it keeps it from creeping while it runs. Its intake/exhaust hoses route the extra short distance past the pantry cluster and through the open tailgate gap to true outside air — the open tailgate itself is the path outside, no window seal needed. Power cord runs from the DELTA 3 stack in Panel A's right drawer, the length of Panels A/B/C.

No tent (sleeping in the Sienna alone, tailgate closed): with the tailgate shut for security/weather, there's no gap to vent hoses through and nothing to usefully blow into at the tailgate — so the WAVE 3 instead moves to the front of the van:

- Move the WAVE 3 forward to the front passenger seat (seatback reclined or folded flat, whichever the unit sits more level on) — the only open seat/floor space once Panels A/B/C fill the rear cargo area. A non-slip mat or a simple bungee strap around the seat frame keeps it from sliding off during setup; it isn't a permanent mount, just an occasional-use position.
- Vent the hoses out a front window using the [EcoFlow WAVE Series Car Vent Kit](#) — a real EcoFlow accessory, **\$39, officially compatible with both WAVE 2 and WAVE 3**: a 47.2" x 27.6" waterproof, UV-resistant nylon cover that velcros over the window opening (window cracked, not removed) with sealed duct pass-throughs for both hoses. 15-second install, no permanent modification, fully reversible.
- With the front window sealed around the hoses and every other window/door shut, the WAVE 3 conditions the **entire sealed cabin**, not just the back — likely more effective in this configuration than the tent setup, since the whole volume it's blowing into is the one you're sleeping in.
- Power cord routes back to the DELTA 3 stack in Panel A's drawer — roughly the same length run as the induction cooktop's own forward cord (Section 5), just the opposite direction (front seat instead of tailgate). Budget for an extension or a quick-disconnect if the stock WAVE 3 cord doesn't reach.
- **This configuration seals the cabin more tightly than the tent setup, with no open tailgate providing passive fresh-air exchange — the owner-placed low-level CO monitor (see the safety note above; Section 4) matters even more here.** Confirm it's active and audible from the sleeping position before using this configuration, same engine-exhaust-while-idling risk as always, just less passive ventilation to dilute it.

2. Dimensions & Layout

Section	Length	Width	Notes
Rear pantry — prefab drawer cluster + pot bay (on Panel C's last 14" + 0.3" into the hatch reserve)	14.3" (of Panel C's last 14" + 0.3" into the hatch reserve)	24.2" cluster + ~21.8" bay = 46"	Bought, not built — a 2×2 IRIS drawer cluster (24.2" × 14.3" × 16.8") on the driver side + a rigid ~13" pot crate in the ~21.8" open deck; held by a cleat pocket + a cam strap (each unit lifts out). Power strip 1 + the ROLL bubble level relocate to the deck edge here.
Panel A	29"	46"	No top of its own — capped by the bed frame's screwed-down section (Component 2); right (DELTA 3) drawer through the passenger door, left bay is WAVE 3 open storage through the driver door (no drawer box)
Panel B	29"	46"	No top of its own — capped by the same bed frame; no drawers, no divider, no skirts (the side doors don't reach it): a bare 2x2 frame whose bay is deep storage, loaded from above by lifting the platform + mattress
Panel C	35.75" (21.25" mattress-covered + 14.5" rear pantry)	46"	Keeps its fixed top (recessed flush between its rails on bearer cleats — deck plane 18.5") — the mattress-covered part only reaches ~21.25" into it (the rear pantry claims the last 14.5", Section 1), and the fridge/kitchen void underneath needs the enclosure regardless; no drawers
Fridge (BougeRV Rocky 40, in Panel C's void)	28.74" deep (incl. handles)	17.72" wide	15.79" tall — drives <code>leg_height</code> via the mounted stack: 0.5" tray hang + 3/8" tray + fridge = 16.67", leaving 0.33" running clearance under the tailgate end rail; on 24" VADANIA slides mounted on the SIDES (nothing under the tray), against Panel C's driver-side rear corner leg (1.5" in from the edge)
Kitchen unit (JAGAHABA, in Panel C's void)	26" deep (closed)	20" wide	11.8" tall; against Panel C's passenger-side rear corner leg (1.5" in from the edge) — shelves swing out that side; own built-in tailgate slide

Drawer dimensions (Panel A's single right-side DELTA 3 drawer — its left bay is WAVE 3 open storage, and Panel B and Panel C have no drawers):

Drawer dimension	Value	Notes
Travel (pull-out direction)	20"	Matches a standard 20" full-extension slide
Depth (fore-aft)	25"	Fits between the panel's front and back legs
Height	13.75"	Inside the 16.25" leg-height storage bay (A/B legs, deck recess), under the removable bed platform (Panel A has no fixed deck of its own)

`leg_height` (17") is driven by the fridge's mounted stack — 0.5" tray hang + 3/8" tray + 15.79" fridge = 16.67", leaving 0.33" of running clearance under the tailgate end rail (the slides mount on the tray's SIDES; nothing sits under it) — not by the folded 3rd-row well depth. **That 17" is Panel C only: Panels A/B run 16.25" legs since the deck recess, so the platform on their rails lands flush with Panel C's recessed deck at the one 18.5" plane.** Measure your actual well and confirm 17" clears it before cutting — if your well is shallower, the fridge (and therefore every leg on the platform) still needs the full 17" of standing height regardless, so the platform would simply sit a bit higher off the true floor than the well alone would require.

Walls & skirts, by panel

The panels are open 2x2 frames — plywood skins exist only where something needs one:

Panel	Walls/skirts	Why
Panel A	None	Both bays face the side doors — a skirt would block the only access. Passenger side: DELTA 3 drawer. Driver side: WAVE 3 open bay.
Panel B	None	Nothing to see or reach — it sits under the platform between A and C. Bare frame, deep storage from above.
Panel C — front (B-facing)	The ONE wall: 3/8" ply, 46" x 17"	Mounts the 120mm intake fan, holds the low intake louver, and passes the fridge DC line (one 1" grommet — Power strip 1 feeds off the rear outlet, not through this wall) — every opening dimensioned in the Panel C Front Wall render.
Panel C — sides	None	The van wall is ~1" away; the exhaust fan still pulls a net flow across the fridge, so side leakage doesn't matter.
Panel C — rear (tailgate)	None needed	Fully occupied already: fridge face + the open utility bay + kitchen unit face + kitchen drawer face.

With no tops and no skirts, every panel's top rails are exposed — **grip those to lift a panel out**. The old routed hand-hold holes (and the router jig for them) are gone from the plan.

Component coordinates

Two coordinate conventions, each picked to match how you'd actually stand at the access point in question — full reasoning and diagrams in `fridge_install_detail.scad` / `rear_view.scad`:

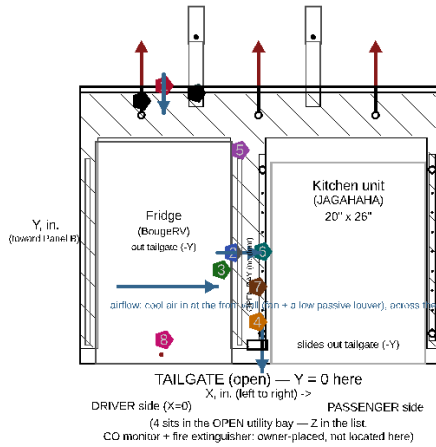
- **Panel C** (fridge + kitchen system, AND the rear pantry — it sits on Panel C's deck, sharing this same coordinate frame): origin (0,0) at Panel C's **tailgate-facing DRIVER-side corner**, floor level. X increases toward the passenger side (0-46"), Y increases forward toward the Panel B seam (0-35.75"). Z increases

up from the van floor. The rear pantry occupies Y 0-14.5 (the tailgate-most 14.5", above deck level) — the SAME Y range the fridge/kitchen's own tailgate-facing ends sit in, just at a different Z height (Z 18.5+ for the pantry vs. Z 0-16ish for the fridge/kitchen below deck).

- **Panel A, Panel B:** origin (0,0) at that panel's **own front-left corner** (front = toward the front seats). X increases right (0-46"), Y increases toward the tailgate (0 = that panel's own length). Z from the van floor.

PANEL C — 46" wide x 35.75" deep, looking down from above

2 steel rail tongues (2"x3/16") BOLT to the rear ends of the 2nd-row FLOOR RAILS (seat anchorage — no new holes; F8;
failback: butt the striker-row step, F7) — forward restraint is steel-to-steel to the van's own rails (Sec. 6)
3 ratchet straps -> 3rd-row STRIKER LOOPS (crash-rated, ~10-14" fwd of Panel C — position UNVERIFIED, F4)



Component
Position / fastener

- 1 Intake fan (120mm) — blows IN
X=10.86 Y=35.75 (Panel C's FRONT wall) Z=8.8
4x M4x20 machine screws over the wall's fan hole (Front Wall render)
- 2 Exhaust fan (120mm) — blows INTO the open utility bay
X=19.72 Y=14.4 Z=8.8
4x M4x20 machine screws, 10mm bolt circle, into a plywood fan ring
- 3 NTC temp sensor
X=18.22 Y=12.2 Z=8.8
adhesive thermal pad of 1x44 screw throughs bracket tab
- 4 Control panel enclosure
X=21.5-24.3 Y=2 Z=6.5-12.5 — in the OPEN utility bay (no door: reach in)
backer board (3/4" solid) hung from the deck; 4x #8x1" screws
- 5 Anchor-board rail strips (2 of the board's 3, under the slide rails)
mat + 3/4" ply band on each rail line BESIDE the tray (side-mount) — part of the ONE-piece board, not separate strips; the fixed rails' risers bolt to them
1/4-20 machine screws into T-nuts from below — NO holes in the van (Section 8)
- 6 Kitchen tie-down: L-track + 4 stud D-rings
on the board's kitchen-side strips (utility-bay gap + the 1.5" band at the panel edge)
4 ratchet straps (400lb WLL) criss-crossed over the top into the D-rings
- 7 WAVE 3 hose/cord hook
in the open bay — screwed up into the deck underside, kitchen side; bundle the hoses so nothing swings out
1x heavy-duty hook, #8x1.5" screw — stows hoses/cord when not in use
- 8 Fridge hold-down strap D-rings (x2)
tray side apron, near the tailgate end — hooks to the fridge's 2 end handles
cam strap, snug not tight — secures fridge TO its tray (the anchor board secures the tray to the van); side profile in fridge-side-detail
- 9 Board bridge (same piece) + 2 steel rail tongues
the board's full-width front edge at Y=29.25 — one continuous 45"x33" comb with the strips, NO joints; 2"x3/16" flat bars run fwd and BOLT T to the 2nd-row floor rails' rear ends (F8)
forward restraint = steel-to-steel to the van's own seat rails, no new holes (failback: butt the striker-row step, F7)
- 10 Striker straps (x3)
bridge D-rings -> the 3rd-row seat striker loops, fwd of Panel C (crash-rated; position UNVERIFIED, F4)
400lb WLL ratchet straps — rearward + tilt restraint; re-tension after the first drive

All coordinates measured from Panel C's tailgate-facing left corner, floor level (see file header).
NO-DRILL (Section 8): no new holes in the van. The board straps to the 3rd-row striker loops and its tongues
bolt to the 2nd-row floor rails' rear ends. Cut the board's outline only AFTER the Section 0 F1 F8 survey.

Fridge install detail

Component	Panel	X (in.)	Y (in.)	Z (in.)
Kitchen unit (JAGAHABA)	C	24.5–44.5	0–26	0–11.8
Fridge (BougeRV Rocky 40)	C	2–19.7 (slide rails flank at ~0.6–1.5 and ~20.2–21.2)	0–28.74	0.5–16.67 (hanging tray + fridge)
Intake fan (120mm)	C	10.86 (on the front wall)	~35 (Panel C's front wall)	8.8
Exhaust fan (120mm)	C	19.7 (fridge's kitchen-facing wall)	~14.4	8.8
NTC temp sensor	C	~18.2 (just inside the fridge's exhaust wall)	~12.9	8.8
Control panel (switches, surge protector)	C	21.1–25.1	~2 (at the tailgate end of the open utility bay)	6.5–12.5 — in the open utility bay, on a backer board hung from the deck underside
Power strip 2 (cooktop)	C	~34.5–37.5	~26 (kitchen unit's front face)	~1.5
Right drawer (the only one)	A	23.75–43.75	2–27	0–13.75
Center divider	A	22.25–23.75	1.5–27.5	0–16.25
Rear pantry: 2x2 IRIS drawer cluster (bought)	C	driver 0–24.2	0–14.3	18.5–35.3
Rear pantry: pot/pan crate + hold-down cleats/strap	C	passenger 24.2–46	0–14	18.5 (on the deck)
Power strip 1 + ROLL bubble level (relocated to the deck edge)	C	~33 (passenger bay)	on the deck, back edge	~19.25–20.25 (just above the deck)

Coordinates for the fixed/structural items (drawers, panels, fridge, kitchen) come directly from `params.scad` / `platform.scad`'s real geometry. Power strip 1/2 positions are schematic markers in the 2D diagrams (approximate, not separately modeled in the 3D assembly) — confirm exact placement against the real frame once built.

Fridge clearance vs. the manual (CR04001, saved user manual): BougeRV asks for **200 mm of free air at the compressor end and 100 mm at the sides** — free-convection numbers the enclosed bay can't offer (the van wall is ~1" away). The forced cross-flow above is the substitute, and the manual makes the geometry work in its favor: the compressor, louvers, control panel, and power port are **all at ONE end** (manual-confirmed — this answers measurement row #4), and Component 7 faces that end toward the tailgate. So intake air enters at the front wall, washes down the bay past the fridge, and the exhaust fan on the kitchen-facing wall pulls it out across the louvered end into the open utility bay, where it exits straight toward the tailgate (no door in the way — the old cabinet door and its louver were cut). Two things keep that substitution honest: the louvered end must stay unblocked (the plenum strip in front of the appliances stays empty — see the access notes below), and the NTC probe + controller exist precisely to prove the airflow is doing its job — if the fans can't hold the bay near ambient, the fix is more airflow, not less clearance. Confirm the louver faces on the physical unit when it arrives (measurement row #4).

How under-deck storage is accessed, by panel

- **Panel A:** a **center divider** splits its under-deck void in two. The right (passenger) side is the one real drawer on a 20" full-extension slide (the DELTA 3 stack); the left (driver) side is **WAVE 3 open storage** — no box, no slide, the unit rests on 2 glide strips, reached by hand through the driver's side door. Both van side doors are used, both for Panel A only.
- **Panel B: no drawers, no divider** — the side-door openings don't reach it. Its whole bay is deep storage: lift the platform + mattress off and load from above. **The SPARE TIRE stows here** (RJ-MODINI kit, flat on 3" cleats at the axle — the best weight placement in the van, Component 3) with **2× Sterilite 28-Qt lidded totes restacked on top** and the jack kit beside it — you lift out whole labeled totes instead of rummaging a loose pile (plain lidded totes on purpose: with no side access, a slide-drawer front would be a wasted feature). Use it for the long-term/bulky stuff you don't touch at camp.
- **Panel C:** no drawers at all. Its entire under-deck void is split between the fridge (left/driver side, on its own slide) and the kitchen unit (right/passenger side, on its own slide) — both bought products, both accessed by pulling them out through the **open tailgate**, not a side door. There's no "storage bay" here in the drawer sense; the fridge and kitchen units *are* the storage, permanently installed rather than removable containers. (There's a ~7–10" strip of under-deck space in *front* of the stowed appliances, but it's deliberately **left empty** — it's the intake fan's cool-air plenum, and it's boxed in with poor access, so filling it would choke the fridge cooling. The found-storage additions instead reclaim the dead headroom above the DELTA 3 and WAVE 3, and the open utility bay's spare volume — see Component 8.)
- None of the 3 panels needs to be lifted out for routine storage access — lifting a panel (gripping its exposed top rails) is only for deep cleaning or reconfiguring, per Section 8. (Panel B's bay is the exception in spirit: loading it means lifting the platform, though never the panel itself.) The rear pantry's drawer units are the one exception with routine lift-off access: loosen the cam strap and any unit lifts out in seconds — day-to-day restocking just opens the drawers, nothing needs removing.

Material options & upgrades (owner review, July 2026)

The base build is **Baltic birch plywood + pine framing**, chosen for cost and availability. Two optional upgrades trade a little weight and money for durability — worth it in targeted spots, overkill everywhere:

- **Plywood is already Baltic birch** for every panel that matters (Panel C's deck, the drawer/kitchen boxes, the fridge tray). No change needed — birch is the dent- and screw-holding win people usually ask for, and it's already specified.
- **Poplar (or ash/maple) for the top rails.** Pine dents, and the *one* place that matters is the exposed **2x2 top rails you grip to lift each module** — those get handled constantly. Swapping just those rails to poplar (a hardwood) resists denting and holds biscuits/screws noticeably better than pine, which suits the biscuit joinery. Cost: poplar is ~20% denser (a few lb of added lift weight across the rails) and pricier per board — so upgrade the **grip rails only**, and leave the hidden framing and the under-mattress 1x4 slats as pine (weight and cost win there, and they never show or get bumped). Budget ~\$25–40 extra.
- **Aluminum angle for the diagonal corner braces.** The Panel A/B corner braces can be aluminum L-angle instead of steel/wood — lighter (these modules are carried) and rust-proof. Budget ~\$10–15 for the aluminum swap.

Neither upgrade changes any dimension in the cut list below — same sizes, different stock. The weight deltas are folded into the note in **Appendix F (Weight Budget)**.

Weight-reduction swaps (APPLIED, July 2026). Separately, and pulling the *other* way, the owner asked to lighten the build wherever safe — so the plywood was thinned wherever it isn't carrying real load, and the bed dropped from 10 to 8 slats. This is already reflected in the cut list above and in Appendix F (~21 lb off the structure, build ~449 → ~424 lb):

- **Battery-drawer walls, fridge tray, Panel C front wall → 3/8"** (was 1/2") — non-structural; the drawer corners get glued + biscuited, the tray gets a glued edge frame.
- **Kitchen-drawer cheeks → 1/2"** (was 3/4"); **bed platform → 8 slats** (fine under the solid-foam mattress). *Partly given back in Aug 2026: splitting the platform into three lift-out pieces put it at 15 (shorter) slats over 6 rails, plus Panel B's new centre bearer — ~+5.5 lb all in. See Component 2.*
- **The rear pantry went further than thinning — it's not plywood at all anymore:** the prefab IRIS drawer cluster (~15 lb) replaced ~42 lb of built shelving outright (Component 1).
- **Kept 3/4":** Panel C's deck — it carries sitting load. **Note the tension with the poplar upgrade above:** poplar *adds* ~20% weight, so if you're chasing weight, keep poplar to the grip rails only or skip it.

Honest trade: this reshuffles the plywood so it now needs 3/8" material (~+\$50 for two 4x4 handy panels) even though it's lighter — a poor cost-per-pound, chosen for easier module lifts, not payload (which already has margin, Appendix F).

Plywood — 1 sheet 3/4" + 1 sheet 1/2" + two 3/8" 4x4 handy panels, Baltic birch (or shop-grade)

Weight-swap note (owner, July 2026): the plywood was thinned wherever it isn't carrying real load, and the rear pantry left plywood entirely (prefab drawers — Component 1). Only Panel C's deck stays **3/4"** (it carries sitting load); the kitchen boxes and the battery-drawer bottom are **1/2"**; the three purely non-structural pieces — the battery drawer's walls, the fridge tray, and Panel C's front wall — are **3/8"**. This adds 3/8" material to the buy (**two 4x4 handy panels, ~\$50** — corrected Aug 2026 from "a half-sheet, ~\$25", which was too little material) while the 3/4" and 1/2" sheets are now only lightly used (spare stock). **Panel A/B have no tops** (Component 2), and **Panel B has no drawers**.

3/4" sheet — Panel C's deck plus the Section 8 anchor board (the plywood pantry is gone — Component 1 is prefab drawers). Even with the anchor board on it this sheet is still under half used: rip the cleats from the offcut, and keep the rest for the control-panel backer board (Section 6) and repairs:

Piece	Qty	Dimensions
Panel C top (deck)	1	33" x 43" — drops BETWEEN the rails onto bearer cleats, flush with the rail tops (deck recess: 3/4" more headroom)
Anchor board (Section 8)	1	ONE comb-shaped piece, 46" x 33" overall — full-width bridge 46" x 6" with three strips 27" long running back from it (2.5" / 4.65" / 1.5" wide, at X = 0, 19.35 and 44.5). Face grain along the 33" direction; ~1/2" fillet at each inside corner. Its two gaps fall out as usable 16.85" x 27" and 20.5" x 27" offcuts. Final dimensions only after the F1–F8 floor survey
Deck bearer cleats	4	3/4" x 3/4" strips: 2 x 33" + 2 x 40" (ripped from the offcut; screwed to Panel C's rails' inner faces, tops 3/4" below the rail tops)
Rear-pantry hold-down cleats	~4	~1" x 1" x 12–14" strips (ripped from the offcut; cab-side + both sides of the drawer cluster)

1/2" sheet — the kitchen drawer box + its 2 hanging cheeks and the battery drawer's bottom (the plywood-pantry carcass is deleted — that frees most of this sheet):

Piece	Qty	Dimensions
Kitchen drawer box	5 pieces	bottom 18" x 26" ; 2 sides 26" x 4"; front/back 17" x 4" (4.5" exterior height). Widened Aug 2026 — was a 16" box; the assembly was sitting 2" shy of the kitchen's inboard edge for no reason
Kitchen drawer hanging cheeks	2	26" x 5.45" (1/2" — screwed up into Panel C's recessed deck, flanking the drawer)
Battery drawer bottom (Panel A right)	1	20" x 25" (1/2" — the base under the 48 lb stack)

3/8" — TWO 4x4 handy panels (weight swap) — the four non-structural pieces. They total **18.0 sq ft**, so a half 4x8 (16.0 sq ft) does **not** cover it — that was an arithmetic error, corrected Aug 2026. Two handy panels give 32 sq ft with room to spare; see the cutting layout above for which part goes on which panel:

Piece	Qty	Dimensions
Battery drawer side wall	2	25" x 14.5" (3/8" — glue + biscuit the corners)
Battery drawer front/back wall	2	20" x 14.5" (3/8")
Fridge tray	1	17.72" x 28.74" (3/8" + two 1x3 side aprons, 28¾" each, glued+screwed to its edges — the slides' moving members screw to them and their top edges are the fridge's anti-shift lip)
Panel C front wall	1	46" x 17" (3/8") — 120mm fan hole + two 1" grommet holes, positions in the Panel C Front Wall render
WAVE 3 glide strips (scrap, not from these sheets)	2	20" x 1", UHMW or laminate offcut

Frame lumber — 2x2 pine (or 1"x1" aluminum L-channel), sold in 8ft (96") lengths

Panel A, Panel B, and Panel C each get their own independent perimeter frame — 2 long rails + 2 end rails sized to that module's own footprint — plus its own 4 legs. No rail is shared between modules, which is what makes each one lift straight out on its own. **The rear pantry needs no frame lumber at all** — it's bought drawer units sitting on Panel C's already-built deck. The fridge and kitchen unit need no frame lumber either.

Piece	Qty	Cut length	Notes
Panel A long rails	2	29"	
Panel B long rails	2	29"	
Panel C long rails	2	35.75"	
End rails (all 3 modules)	6	46"	2 per module
Center divider	1	26"	Panel A only — splits its bay into drawer (right) + WAVE 3 (left) runs, see Step 2 diagram. Panel B and C have none.
Legs — Panel C	4	16"	cut 1" short; the leveling foot makes each an effective 17" (Leg Leveling Foot Detail). REAR pair at the true corners
Legs — Panels A/B	8	15.25"	cut 1" short; foot brings each to an effective 16.25" — 3/4" shorter than Panel C (deck recess: the platform on A/B's rails lands flush with C's recessed deck)
Bottom rails (cube frames)	5x 46" + 2x 26"	see notes	underside at 1" (dropped to the leg bottoms): Panel A both END faces (2x46), Panel B all 4 faces (2x46 + 2x26), Panel C FRONT face only (1x46) — 2x 2" screws + glue into each leg

Total linear footage needed: 958" (4@29 + 2@36 + 6@46 + 1@26 + 4@16 + 8@15.25, plus the cube-frame bottom rails: 5@46 + 2@26). Eleven 8ft (96") boards hold 1,056" — a ~9% margin for kerf and layout waste. **Buy 12 boards** to allow for a mis-cut.

Hardware sizing

Item	Size needed
Drawer slides	1 pair, 20" full-extension — Panel A's right (DELTA 3) drawer, mounted between the drawer box and the frame's long rail / center divider. Panel B has no drawers (side doors don't reach it)
Drawer catches	1, simple friction catch or small turn latch — Panel A's drawer
WAVE 3 glide strips	2, UHMW or laminate scrap, ~20" x 1", glued/screwed to Panel A's left bay floor — cuts friction sliding the unit in/out by hand (no slide hardware there)
Wood screws	1.25" for plywood-to-frame, 2" for frame-to-frame joints
Corner brackets	2" or 3" steel L-brackets, 12 total (4 per module x 3 modules — the rear pantry has no frame of its own, no brackets needed)
Cable grommets	1" diameter, 7 needed (1 cooktop line, 1 Power strip 1 line — both short rear-outlet hops, 3 fridge DC line, 2 DELTA 3 AC charging line — see Section 5)
Hand-hold holes	REMOVED — with no tops or skirts, every panel's 2x2 top rails are exposed; grip those to lift. (The router jig went with them.)
Anti-rattle bumper strips	2 strips (one at each lift-out seam: A/B, B/C — the pantry and fridge/kitchen aren't lift-out modules, so no seams of their own). Adhesive-backed felt or closed-cell foam weatherstrip
Alignment dowel pins	4 pins (2 per seam x 2 seams), 3/8" dowel rod, ~3/4" long, friction-fit into a matching hole on each module face
Seam draw-latches	4 over-center draw latches (2 per seam x 2 seams), stainless/zinc, ~3" body. Mounted LOW on the bottom-rail band (~1.75" up), over the inset leg line, one on each side of each seam. Base plate on one module, hooked bail + lever on its neighbour — flip to clamp the two tight against the bumper, flip to release. Alignment pins LOCATE the modules; these CLAMP them into one rigid beam (with the bed platform tying the A/B tops). Hand-released, so the modular lift-out is preserved — see Component 5
Rear pantry — prefab drawer cluster	4x IRIS USA 12"W stackable drawers (Home Depot #500163, two 3-packs), 12.1" x 14.3" x 8.4" each, stacked 2 wide x 2 high (24.2" x 14.3" x 16.8", ~13 lb). Contains its own contents; each unit lifts out for the gate
Rear pantry — hold-down	~1" birch cleats (cab side + both sides, ripped from the 3/4" offcut) + 1 cam-buckle strap across the drawer fronts + 2 flush deck D-rings (strap anchors) + anti-rattle felt tape + 2 Velcro ties linking top units to bottom
Pot/pan bay	1 rigid ~13" milk crate (the pots' 11x11 box drops inside) in the ~21.8" of open deck beside the cluster + a corner cleat + a bungee
Panel B deep-storage totes	2x Sterilite 28-Qt under-bed lidded totes (23.5" x 16.9" x 5.9"), restacked ON the spare in Panel B's bay — containerized top-load storage (lift the platform, lift out whole totes)
Spare tire stowage	RJ-MODINI kit (T155/85R18 on an 18x4 steel wheel, 28.5" dia x ~6.4" stored, ~40 lb w/ 2-ton jack) FLAT in Panel B's bay at the axle: a cradle skid (3x ~3" cleats screwed to 2 battens, from the offcut) over a non-slip liner offcut + 1" cam strap to 2 footman loops on the bottom rails + tool case nested in the wheel — install/access diagrams on the Spare Tire Stowage render
Panel A/B diagonal corner braces	4 used (2 per panel, Panels A + B) from the purchased Sumnacon 4" 8-pack (4 spares), steel flat-strap — recovers some of the racking rigidity lost by removing each panel's own top
Leg leveling feet	12 total (4 per panel x 3 panels), 3/8-16 screw-in threaded insert (7/16" OD coarse outer thread — the Anwenk kit's "T-nut") + leveling stud with a 1-3/8" pad, fixed hex collar, and ~2" star-knob hand grip. Legs are CUT to 16" and drilled in the bottom end grain (1/2" dia x 7/8" deep — the maker's spec; test-fit the insert in offcut first) — the foot brings each back to an effective 17" (Leg Leveling Foot Detail render). Tip a corner slightly and spin the knob to level
Bed platform side rails	6, all 29" x 3.5" x 3/4" (1x4 pine) — one pair per lift-out piece: Panel A's section (set 49" apart outside-to-outside, 42" slats between them) and Panel B's two halves (each set 24.5" apart, 17.5" slats between them). One flush 3/4" plane cantilevered 1.5" past the boxes per side (Bed Platform Detail render). Width set by measurement V7 (Aug 2026) — was 52"/45"; split into three lift-out pieces Aug 2026 — was one pair of 58" rails
Bed platform slats	15: 5 x 42" (Panel A section, ~2.9" gaps) + 10 x 17.5" (five per Panel B half, ~2.9" gaps), all 3.5" x 3/4" 1x4 pine — two 42" or five 17.5" per 8ft board
Panel B centre bearer	1, 26" x 3" x 1.5" — two 2x2s side by side (or one 2x4 laid flat), fore-aft at Panel B's centreline between its end rails, top flush with its long rails. Carries the inner edge of both bed halves (1.5" bearing each) and halves the deck's unsupported span, 46" → ~22" (NEW, Aug 2026)
Ryobi R-series biscuits	R1 (thin stock / 1/2" ply), R2, R3 (3/4" ply) — buy an R1/R2/R3 assortment for the DBJ50. Wood glue (already in the BOM) goes in every biscuit slot + mating face

Joinery & Fasteners (Ryobi DBJ50 biscuit guide)

Every wooden joint's method — biscuit vs. screw — with the exact biscuit spacing is on the **Joinery & Fastener Guide** render below. The short version, using the Ryobi DBJ50 detail biscuit joiner (R1/R2/R3 biscuits, 0–9/32" depth, 1-1/2" 6-tooth blade, 90°/45° fence):

- **Biscuits where clean plywood faces meet** (hidden strength + self-alignment, no screw heads on show faces): every drawer/tray box corner — the battery drawer and the kitchen drawer (**2x R1 per corner, centered on the ½"/⅜"**). (The plywood-pantry carcass that used to be biscuited here is gone — Component 1 is now prefab drawers.) **Rule for any biscuited joint: first biscuit 2" from each end, then ~6" on center, always centered on the stock thickness; glue the slot and the mating face, then clamp.**
- **Screws (+ glue) for everything into the 2x2 frame or hardware:** all frame corners get **2x 2" screws + a steel corner bracket** (a biscuit would blow out the 1.5" stock — the one place to skip biscuits); Panel C's fixed top and front wall, the base cleats, and the kitchen-drawer cheeks are screwed to the frame; slides and threaded inserts use their own hardware.

- **No fasteners:** the adjustable shelf rests on pins, and the DELTA 3 tray + utility-bay bins just drop in (the bay bins get hook-and-loop tape — the bay is open to the tailgate).

JOINERY & FASTENERS — Ryobi DBJ50 detail biscuit joiner (R1 / R2 / R3 biscuits)

Legend: Peru ovals = biscuit slots (glue + biscuit, centered on thickness) Crimson = screws (+ glue) Grey = 2x2 frame

1 — SPACING RULE (any biscuited joint, edge-on): 2" from each end, then ~6" o.c., centered on thickness

— REFERENCE: any shelf into a side panel (3/4" ply): 3x R3 biscuits/end at 2" / 7" / 12" from the front, centered on 3/4", glue + clamp

4" 1.5" — BOX CORNERS (1/2" ply -> R1): 2x R1 biscuits/corner + glue, all 4 corners — Panel A drawer, kitchen drawer, fridge-tray apron
bottom sits in a glued rabbet (or glue + brads) — clean box, no visible fasteners

— FRAMES (2x2 pine, rail/leg/bottom-rail/divider): SCREWS ONLY — a biscuit would blow out the 1.5" stock
2x 2" screws + glue per corner + a steel corner bracket (diagonal corner braces screw on top of the frame)

5 — PLY TO FRAME (Panel C top / front wall / cleats / cheeks): SCREWS + glue
1-1/4" screws ~8" o.c. into the rail + glue (front wall: 8 screws per the Front Wall render; cheeks/cleats: 2" screws ~6")

COMPONENT	JOINT	METHOD (biscuit sizes are Ryobi R1/R2/R3)
Panel A drawer + kitchen drawer boxes (1/2)	box corners	2x R1 biscuits/corner + glue; bottom in a glued rabbet
Fridge tray + slide apron (1/2 + pine)	edge / face	R1 biscuits OR 1-1/4" screws + glue
Kitchen-drawer hanging cheeks <-> deck (3/4)	face-to-face	2" screws every ~6" up into the deck
WAVE 3 overhead shelf <-> 1x1 cleats	shelf-on-cleat	cleats screwed to the frame; shelf screws down to the cleats
Panel C fixed top <-> 2x2 rails	ply-to-frame	1-1/4" screws every ~8" + glue (R2 optional, corners, to align)
Panel C front wall <-> frame	ply-to-frame	8x #8 x 1-1/4" screws (exact positions in the Front Wall render)
ALL 2x2 frames (rails/legs/bottom rails/dividers)	2x2 corners	2x 2" screws + glue + a steel corner bracket (NO biscuit)
Bed platform slats <-> 1x4 side rails	end-to-edge	2x 1-1/4" pocket screws/end (OR 1x R2 biscuit + glue, cleaner)
DELTA 3 tray / utility-bay bins / adjustable shelf	none	drop in / rest on pins — no fasteners

Joinery and fastener guide

4. Bill of Materials (BOM)

Item	Qty	Est. Unit Cost	Est. Total
3/4" Baltic birch plywood, 4x8 sheet (Panel C deck + the rear-pantry hold-down cleats ripped from the offcut — now largely spare stock: backer board + repairs)	1	\$65	\$65
Rear pantry — IRIS USA 12"W stackable drawers, Home Depot #500163 3-pack (buy 2 packs; use 4, 2 spare)	2 packs	\$30.29	\$61
Rear pantry hold-down — 1 cam-buckle strap + 2 flush deck D-rings + anti-rattle felt tape + Velcro ties (cleats ripped from the 3/4" offcut)	1 set	\$26	\$26
Pot/pan bay — rigid ~13" milk crate + bungee (e.g. 1 of an ORGANIZE-IT 4-pack; in the ~21.8" bay)	1	\$15	\$15
Panel B totes — Sterilite 28-Qt under-bed lidded (23.5" x 16.9" x 5.9"), restacked on the spare	2	\$12	\$24
RJ-MODINI spare kit — 2011-2026 Sienna incl. Hybrid (T155/85R18 steel, 28.5" dia + 2-ton jack + wrenches/sockets + cases) — \$349.95 on Amazon	1	\$350	\$350
Spare hold-down hardware: 2x 1" footman loops (from a 4-pack, screws included) + 1" x 6 ft cam-buckle strap (Masterwise multi-pack, purchased — a ~5.5 ft over-stack run, so 6 ft reaches; spares cover the pantry + DELTA 3 drawer straps); skid + liner from offcuts	1 set	\$24	\$24
1/2" Baltic birch or shop-grade plywood, 4x8 sheet (kitchen drawer box & cheeks + battery-drawer bottom — the plywood pantry is deleted, so this sheet is now mostly spare)	1	\$50	\$50
3/8" Baltic birch plywood, half-sheet — weight swap (battery-drawer walls + fridge tray + Panel C front wall)	1	\$25	\$25
2x2 pine (8ft lengths) or aluminum L-channel	12	\$8–15	\$96–180
Panel drawer slides, 20" full-extension (1 pair — Panel A's right/DELTA 3 drawer; Panel B has no drawers, Panel A's left bay has none either)	1 pair	\$16	\$16
Drawer catches (friction catch or small turn latch)	1	\$3	\$3
WAVE 3 glide strips (Panel A's left bay floor, UHMW or laminate scrap)	2	\$3	\$6
Corner brackets (frame joints, 4 per module x 3 modules — the rear pantry has no frame, no brackets needed)	12	\$1	\$12
Wood screws (1.25" and 2", assorted box)	2 boxes	\$10	\$20
Anti-rattle bumper strip (adhesive felt/foam weatherstrip roll)	1	\$10	\$10
Alignment dowel pins (3/8" dowel rod, cut to 4 pins)	1	\$3	\$3
Seam draw-latches (over-center, 4 total — both sides of the A/B and B/C seams)	4	\$4	\$16
Utensil bins + non-slip liner (loose, inside the drawers/bins as desired)	1 set	\$12	\$12
Wood glue	1	\$8	\$8
Ryobi R-series biscuit assortment (R1/R2/R3) for the DBJ50 — carcass, shelf, drawer & box joints (see the Joinery & Fastener Guide)	1	\$12	\$12
Panel A/B diagonal corner braces (2 per panel, A + B = 4 used — Sumnacon 4" 8-pack, 4 spares, PURCHASED)	1 pack	\$9	\$9
Leg leveling feet — Heavy Duty 3/8-16 furniture levelers w/ T-nut kit, 4-pack (330 lb per foot; buy 3 packs = 12)	3 packs	\$12	\$36
Foot hand-grip knobs — Peachtree 3/8-16 thru-hole star knobs, ~2" dia, 4-pack (1 per foot, jam-locked on the bolt; buy 3 packs = 12) + 12x 3/8-16 jam nuts	3 packs	\$12	\$36
Bed platform lumber — 1x4 x 8ft pine boards (five 42" slats + ten 17.5" slats + six 29" side rails; crosscuts only)	7	\$5	\$35
Kitchen drawer slide pair — 24" full-extension side-mount, 100lb class	1	\$16	\$16
Kitchen drawer magnetic catch (box + cheeks come from the plywood sheets' spare area)	1	\$3	\$3
RV bubble levels — Larbeti stick-on 2+2 pack, bar + T-shape (PITCH on the platform's driver-side rail edge, ROLL on the rear-pantry deck edge) — PURCHASED July 2026	1	\$10	\$10
RV leveling blocks, Lynx-style 10-pack + 1 wheel chock (per-site leveling at the WHEELS — see the Block Calculator note, Appendix E)	1	\$40	\$40
HEST Dually Long mattress (hest.com/products/dually — 78" x 50" x 4" solid foam, memory-foam top, washable waterproof cover INCLUDED; no air chambers)	1	\$530–590	\$530–590
— Budget fallback: DIY 2-layer foam (queen 4" HD base + 1.5" memory topper, both cut to 50" x 78" with a serrated/electric knife + spray adhesive + a separate waterproof cover) — swaps in for the HEST at ~\$185–255 total	(alt)	(\$185–255)	(\$0 in total below)
Cable grommets (1 cooktop line, 1 Power strip 1's line — both short hops to the rear outlet now, 3 fridge DC line, 2 DELTA 3 AC charging line) — EASYEAH 20-pack covers all 7, PURCHASED	1 pack	\$10	\$10
16 AWG extension cord (household AC-rated — cooktop to the REAR outlet, short + slack loop for the slide)	1	\$8	\$8
16 AWG extension cord (household AC-rated — Power strip 1 to the REAR outlet beside it, short)	1	\$8	\$8

Item	Qty	Est. Unit Cost	Est. Total
Fridge DC line — OYMSAE 12-ft 16AWG SAE-to-SAE 2-pack: the cord AND its 2 seam quick-disconnects in one part, PURCHASED July 2026 (don't buy cord and connectors separately)	1 pack	\$22	\$22
16 AWG cord (household AC-rated — DELTA 3 AC charging line, front console to Panel A, ~3-4ft)	1	\$8	\$8
Per-module surface wire raceway — a CordMate II kit for the 3 AC lines (~\$28) + a 123" open-slot raceway kit for the DC line (~\$18, screws included), AC and DC in separate channels + screw-mount cord clips	1 set	\$46	\$46
SAE quick-disconnect connectors (2 for the fridge DC line's 2 seams — DC-rated, matches the fridge's actual 12V circuit)	2	\$8	\$16
Fused (15A) male car-plug → SAE adapter — DELTA 3 end of the fridge line (P1 ; its fuse is the line's upstream protection). Any 16AWG fused SAE adapter kit works	1	\$9	\$9
SAE → female 12V socket pigtail — fridge end (P7 ; the fridge's own cord plugs into it). Often sold in the same kit as the P1 adapter	1	\$9	\$9
18AWG red/black hookup wire, ~10 ft (in-bay branch runs, P6–P12)	1	\$8	\$8
Crimp terminal assortment — #10 heat-shrink rings (fuse-block studs) + 0.25" female spades (switch tabs)	1 kit	\$12	\$12
WAGO 221-413 lever nuts (fan feed + fan return splits — P12)	1 pack	\$10	\$10
2-way AC outlet tap (REAR outlet — Power strip 1 + the cooktop share it)	1	\$5	\$5
BougeRV Rocky 40 (CR04001), 41QT dual-zone (bougeRV.com Rocky V3.0 — specs verified against the saved user manual: 28.74" x 17.72" x 15.79", 40.6 lb (18.4 kg), 60W max / 45W ECO, reversible lid, optional B240 battery) — PURCHASED July 2026 direct from bougeRV.com (New Rocky V3.0 41QT, JULY7 coupon; the 30-Qt Amazon unit ordered by mistake is cancelled)	1	\$400–500 (verify invoice)	\$400–500
Heavy-duty 24" full-extension drawer slide pair — VADANIA #D76 locking, 379 lb, PURCHASED July 2026 (locks closed AND extended; supersedes the original 200 lb non-locking spec; loaded fridge can realistically hit 60–90 lb). Rails mount VERTICALLY on the tray's sides — never under it (see the fridge slide detail)	1 pair	\$79	\$79
Steel riser angles for the fridge slides, ~2"x2"x3/16" x 24" (1 per fixed rail, drilled to match — bolted atop the anchor board's rail-line strips with 1/4-20 screws into T-nuts; the board raises the rails ~0.85", and the moving member screws lower on the tray apron to compensate, Section 8)	2	\$12	\$24
1x3 pine, 6 ft (rip the fridge tray's two 28¾" side aprons from it)	1	\$6	\$6
JAGAHABA wooden slide-out camp kitchen , right-side variant (kitchen faces out the passenger side), built for a 2-burner stove (confirmed product link — no listed price found in research; treat as an UNVERIFIED estimate until you check the live listing)	1	\$300–350	\$300–350
COOKTRON Portable Induction Cooktop 2 Burner w/ Removable Iron Cast Griddle Pan — confirmed exact product, 18.1"x9.1"x4.3" (fits the JAGAHABA's 23"x15.7"x5.7" stove tray with margin on every axis); see Section 7 for why the Duxtop 9620LS was dropped	1	\$189.99	\$189.99
120mm 12V fan + PWM temperature controller w/ NTC probe (W1209-style, fridge-bay cooling)	1	\$15–20	\$15–20
Snap-in louvered RV vent — 1 low intake (front wall, cool-air scoop; the old cabinet-door exhaust louver went away with the door — the utility bay is open now)	1	\$6	\$6
12V fused distribution block (control panel) — PURCHASED July 2026: Nilight 6-way blade fuse block w/ negative bus, waterproof cover, label stickers, fuse assortment included (no separate ATO fuse pack needed)	1	\$17.98	\$17.98
Illuminated toggle switches (SW1 fridge, SW2 fans, SW3 spare — the old "cooktop" label predates the verified-outlet round; the cooktop is AC, switched at Power strip 2)	3	\$5	\$15
Small electrical project enclosure (control panel housing)	1	\$10–15	\$10–15
Windmill Smart Fan 2-in-1 air circulator (personal comfort, Power Strip 1) — PURCHASED July 2026 , swapped in for the planned Claymore V600+	1	\$69	\$69
US Cargo Control aluminum L-track, 48" 4-pack , ≥2,000 lb/ft — cut down and screwed to the anchor board's kitchen-side strips + bridge (Section 8, no-drill; the Sienna's factory cargo hooks are rated for cargo nets only, not this). 2 of the 4 lengths are spares	1 pack	\$95	\$95
L-track single-stud fittings with D-ring, WLL 1,333 lb, 10-pack — 4 kitchen strap corners + 3 bridge striker-strap points, 3 spares	1 pack	\$45	\$45
Anchor board — 3/4" ply comb, ONE piece 46" x 33" (bridge + 3 strips continuous, no wood joints — cut from the 3/4" sheet's spare, so no plywood to buy; Section 8, assembly sheet V1/V5) + 1/4-20 machine screws & T-nuts ~30 ea	1	\$20	\$20
Steel flat bar 2" x 3/16" — the 2 forward rail tongues, cut 16" each (F8a: rail ends at 42" from the hatch)	1x 36" bar	\$15–25	\$15–25
¼" pin per tongue — a ¼" bolt through the tongue, nutted on its own accessible face, plain end pointing DOWN into the steel cap's ¼" hole. No lug, no bend	2	\$1 ea	\$2
Hold-down per tongue — a clip hooked over the cap's side skirts, or a saddle clamp on the rail just forward of it. Essential: a vertical pin lifts straight out	2	~\$5 ea	\$10
Washers/shim stack — the 0.15" step under each tongue's lap on the bridge	2 sets	\$1	\$2
¼"-20 bolts + washers + nylocks — 2 per tongue through the bridge	4	\$1 ea	\$4

Item	Qty	Est. Unit Cost	Est. Total
Non-slip rubber utility mat, under the anchor board's strips	1	\$25	\$25
Blue medium-strength threadlocker (Loctite 242 or equivalent — on the anchor board's machine screws)	1 bottle	\$8	\$8
Erickson ratchet straps (01415), 400 lb WLL / 1,200 lb break, 1" x 10' (4: kitchen unit, criss-crossed over the top to the board's L-track; 3: bridge D-rings to the 3rd-row striker loops; 1 spare)	2 packs of 4	\$32	\$64
Fridge hold-down — 1 low-profile cam strap + 2 screw-eye D-rings (hooks to the fridge's 2 end handles, down to the tray's side apron — keeps the fridge from lifting off the tray in a hard stop)	1 set	\$10	\$10
Cable clips for the fridge DC line's slide-travel slack (3, screwed to the slide's fixed outer rail — same VIPMOON-type clip as the cooktop run, buy links below)	3	\$1	\$3
Tension rod + blackout curtain fabric	1	\$40	\$40
Wood sealant/varnish	1	\$20	\$20
Sandpaper, misc finishing supplies	—	\$15	\$15
6-outlet power strip, individual switches (Power strip 1 + Power strip 2) — CRST metal, PURCHASED July 2026	2	\$25	\$50
UMETRE 9-Piece Detachable-Handle Induction Cookware Set — nests flat, its 11x11 box drops into the pot crate (Component 1); induction-compatible, dishwasher-safe — PURCHASED July 2026 (bought in place of the Cook N Home stainless set)	1 set	\$45	\$45
PeaceOut Toyota Sienna Sliding Doors Bug Nets — Sienna-specific, sold as a pair (driver + passenger sliding doors), no-see-um mesh, magnetic + hook-and-loop attachment (temporary, no screws), carry bag included — lets both side doors stay open for ventilation while sleeping without bugs getting in. Real listed price is in CAD; \$ figure here is an approximate USD conversion — confirm current exchange rate before ordering.	1 pair	~\$197 (CAD \$269.99)	~\$197
DELTA 3 drawer hardware: 1x1 pine locating cleats, 2 screw-eye D-rings + cam strap, 1" cable grommet (Panel A right drawer — see Renders)	1 set	\$15	\$15
WAVE 3 hose/cord storage hook, in the open utility bay (screwed up into the deck underside at the bay's kitchen side; bundle the hoses with their strap so nothing swings — see Renders)	1	\$4	\$4
Non-slip mat, WAVE 3's point-of-use surface (Panel C's deck or the front seat — not the Panel A storage bay, see Renders)	1	\$12	\$12
FOUND STORAGE — DELTA 3 drawer top tray (reclaims ~2.2" of dead headroom above the DELTA 3 stack — was ~3.3" before the deck recess: cables, the DELTA 3's own cords, dongles) — shallow lift-out bin	1	\$8	\$8
FOUND STORAGE — WAVE 3 bay overhead shelf (reclaims ~2.2" above the WAVE 3 — was ~2.9" before the deck recess: hoses, remote, flat soft goods; the unit still slides out beneath it) — 1/2" ply on 1x1 cleats, from offcut/scrap	1	\$5	\$5
FOUND STORAGE — utility-bay bins (the open bay's spare volume beside the control panel: dish soap, sponges, utility odds) + hook-and-loop tape under each (no door holds them in anymore)	2	\$4	\$8
EcoFlow WAVE Series Car Vent Kit — official accessory, WAVE 2/3 compatible, velcro window seal + duct pass-throughs, no permanent modification (no-tent sleeping configuration, Section 1) — PURCHASED July 2026 (order EFUS-426448, \$40)	1	\$40	\$40
SRS airbag emulators, Sienna 2nd row — PURCHASED July 2026 : 0651 Round x2 + 5522 "Square" x2 (shipped). Click into the van-side yellow floor plugs when the seats are out so the airbag system reads normally (Section 9). VERIFY before seat-removal day that the 5522 mates with the Sienna's rectangular floor plug — the original spec called for a 6134 Rectangle	1 set	~\$60	~\$60

EcoFlow DELTA 3 Plus + Smart Extra Battery and WAVE 3 (already owned, not priced here) — see the DELTA 3/WAVE 3 stowage detail in Renders and Section 1.

Recommended purchase links — slides + electrical (checked July 2026)

- **Rear pantry (prefab):** IRIS USA 12"W stackable drawer 3-pack, Home Depot #500163 (buy 2 packs; 2x2 cluster + 2 spares) + a 1"×6' cam-buckle strap 4-pack + flush deck D-rings + felt tape + VELCRO ONE-WRAP 8"×1/2" ties.
- **Panel B deep-storage totes:** Sterilite 28-Qt under-bed lidded tote x2 — plain lidded totes on purpose (Panel B has no side access, so slide-drawer versions would waste money).
- **Spare tire:** RJ-MODINI kit, 2011-2026 Sienna incl. Hybrid (\$349.95) — T155/85R18 on an 18x4 steel wheel with the factory 60.1mm hub-centric bore; at **28.5" rolling diameter it's the closest match to the ~29.1" OE tires** (~2%, vs ~4.8% for the T145 kits — gentlest on the AWD systems). Includes a model-specific 2-ton jack, lug + ratchet wrench, 4 sockets, gloves, and cases (the tool case nests inside the wheel barrel). Skid + liner come from offcuts already in the build. (Alternatives compared: ValueSpare ~\$270-385 steel/27.7", Modern Spare ~\$350-460 alloy/27.7".)
- **Spare hold-down hardware:** NVAHV 1" stainless footman loops, 4-pack w/ screws (~\$10, need 2) + Cartman 1" x 12 ft cam-buckle lashing straps, 6-pack — **PURCHASED July 2026**: footman loops (2 packs) + a Masterwise 6-ft x 1" cam-strap multi-pack bought in place of the Cartman 12-footers (the spare's over-stack run is ~5.5 ft, so 6 ft reaches; spares cover the pantry + DELTA 3 drawer straps).
- **Leveling blocks:** Lynx Levelers 10-pack — the "legos"; pair with the [Sienna Block Calculator](#).
- **Bubble levels:** **PURCHASED July 2026** — Larbeti stick-on 2+2 pack (bar + T-shape, self-adhesive): PITCH bar on the platform rail edge, ROLL on the rear-pantry deck edge, 2 spares. (Replaces the planned RV Designer E409 screw-ons — the adhesive mount skips two screw holes in the poplar rail; degrease the surfaces first.)

Specific products verified against this plan's specs, full URLs shown for copy-paste. Quantities refer to what THIS build needs; several listings are multi-packs. The AITRIP W1209 3-pack originally recommended went **Currently unavailable** — the HiLetgo 2-pack below is the same W1209 board with case, in stock. The low-level CO monitor is owner-supplied (battery-powered, see the BOM note above) and deliberately absent here.

Item (plan spec)	Product	Link
Panel drawer slides — only 1 pair needed now (Panel A's right/DELTA 3 drawer; Panel B has no drawers — the side doors don't reach it — and Panel A's left bay is WAVE 3 open storage, Section 1), 20" full-extension, 100 lb class — PURCHASED (July 2026)	GlideRite 20" 100 lb ball-bearing slides, 5-pair pack (\$46.99; 4 spares now — one could serve as the kitchen drawer's pair if you go 20" instead of 24" there)	https://www.amazon.com/20-inch-Extension-Ball-Bearing-Drawer-Over-Travel/dp/B07KFLS2C9
Fridge slide — 1 pair, 24" — PURCHASED (July 2026)	VADANIA #D76 (VD2576) 24" industrial heavy-duty WITH LOCK — 379 lb, 3" (76mm) rail, 3-fold full extension, locks both closed (transit) and extended (loading), 1 pair (\$79.20). The lock is an upgrade over the original non-locking 200 lb spec. INSTALL NOTE: the 76mm rail wants ~3" of flat vertical face — the rails stand VERTICALLY, one flanking each side of the tray: the moving member screws to a 1x3 apron glued+screwed along each tray edge, and the fixed member screws to a steel riser angle bolted to the no-drill anchor board (Section 8; the board raises the rails ~0.85" — the moving member screws lower on the apron so the tray keeps its height). NEVER mount them flat under the tray — undermount adds ~1.2" of height the 17" void doesn't have, and the fridge would hit Panel C's end rail (the mounted stack clears it by 0.33").	https://www.amazon.com/VADANIA-Industrial-Extension-Bearing-Widening/dp/B08C56Z6GF
Fridge slide, premium alternate	Hettich 24" full-extension, 500 lb (~\$70+)	https://www.amazon.com/Drawer-Slide-Extension-Capacity-Hettich/dp/B0016LEBBG
120mm 12V fans — need 2 (intake + exhaust)	GDSTIME 120mm x 25mm 12V dual-ball-bearing fan, ~\$9 — sold singly, ORDER QTY 2. (The UMLIFE 2-pack link originally here started redirecting to a 60mm 5V USB variant — verify 12V / 120x120x25mm on the spec table before buying any fan listing.)	https://www.amazon.com/GDSTIME-Bearings-Brushless-Cooling-Exhaust/dp/B00N1Y4BMA
W1209 temp controller + NTC probe (need 1; 2nd is a spare)	HiLetgo 2pcs W1209 12V with case + waterproof probe (~\$10)	https://www.amazon.com/HiLetgo-Temperature-Controller-Thermostat-One-Channel/dp/B07VDWRZKB
Fused 12V distribution block — PURCHASED (July 2026)	Nilight 6-way blade fuse block w/ negative bus, waterproof cover, label stickers + fuse assortment included (\$17.98 — replaces the Blue Sea 5025 pick and the separate ATO fuse pack)	https://www.amazon.com/Nilight-Negative-Standard-Waterproof-Automotive/dp/B09NPQBXC6
Illuminated toggle switches — need 3 — PURCHASED (July 2026)	Amppper 12V 20A illuminated round rockers, 10-pack (\$8.99)	https://www.amazon.com/Illuminated-Rocker-Switches-Amppper-Terminals/dp/B08ZPY5D9L
Control panel enclosure	LMioEtool IP65 hinged ABS junction box, 5.9" x 3.9" x 2.8" (~\$12)	https://www.amazon.com/LMioEtool-Waterproof-Dustproof-Universal-Electrical/dp/B0CJJCKXK3
16 AWG AC cords — cooktop → REAR outlet + Power strip 1 → REAR outlet (both short, no seams) + DELTA 3 charging → front console	DEWENWILS 3ft 16AWG SJTW indoor/outdoor extension cord, ETL listed, 2-pack (~\$13) — 1 pack covers the two short rear-outlet hops; add a 2nd cord for the DELTA 3 charging line (or size one to your console distance)	https://www.amazon.com/DEWENWILS-Extension-Weatherproof-Outdoor-Grounded/dp/B0891VDLV5
16 AWG DC cord run + 2 seam disconnects (fridge line — this one really is 12V DC, DELTA 3 to fridge, so the SAE connector IS the right part here)	OYMSAE 12 ft 16AWG SAE-to-SAE quick-disconnect cables, 2-pack — the SAE plugs ARE the seam disconnects (~\$14)	https://www.amazon.com/OYMSAE-Extension-Disconnect-Connector-Automotive/dp/B082JLGJCR
16 AWG cord, ~3-4ft (DELTA 3 AC charging line, front console to Panel A — household AC-rated, no seam crossing)	DEWENWILS 3ft 16AWG SJTW indoor/outdoor extension cord, ETL listed, 2-pack (~\$13) — same product as the cooktop line above, 1 pack covers this with a spare	https://www.amazon.com/DEWENWILS-Extension-Weatherproof-Outdoor-Grounded/dp/B0891VDLV5

		oor-Grounded/dp/B0891VDLV5
2-way AC outlet tap (REAR outlet — Power strip 1 + the cooktop share it; the front console outlet is the DELTA 3's alone)	GE Outlet Extender Wall Tap, UL listed, 125VAC (~\$5-13)	https://www.amazon.com/GE-Adapter-Grounded-Outlets-52203/dp/B00006IBFC
Leg leveling feet — 12 needed (buy 3 four-packs); 3/8-16 leveling feet WITH T-nut kit, 330 lb per foot	Heavy Duty Furniture Levelers w/ T-Nut Kit, 4-pack (~\$12/package)	https://www.amazon.com/Furniture-Levelers-Adjustable-Leveling-Cabinets-Sofa/dp/B081ZT4Q4G
Foot hand-grip knobs — 12 needed (buy 3 four-packs), jam-nutted onto each foot bolt as the easy hand grip	Peachtree PW6103 Star Thru-Hole Knob 3/8-16, 4-pack (~\$12/package) + 12x 3/8-16 jam nuts	https://www.amazon.com/Star-Thru-Hole-Knob-Woodworking/dp/B000UH7Y84
1" cable grommets — need 7 (the 20-pack covers all pass-throughs) — PURCHASED (July 2026)	EASYEAH 1" rubber grommets, 20-pack (\$9.99)	https://www.amazon.com/Diameter-Hole%E2%BC%8C-Rubber-Synthetic-Protection-Double-Sided/dp/B085S6S6KH
6-outlet power strips, individual switches — need 2	CRST 6-outlet metal strip, individual switches, 1200J (~\$25 each)	https://www.amazon.com/CRST-Individual-Protectors-ECO-Friendly-Protector/dp/B097JH3W3W
Cord clips — ~10-12 along the cooktop cord run	VIPMOON 100x 1/4" R-type nylon screw-mount clamps w/ screws (~\$9) — screw-mount, NOT adhesive: stick-on clips let go under vehicle vibration and temperature swings. Size up to 3/8" if the SAE cable measures over ~0.28"	https://www.amazon.com/VIPMOON-Mounting-Fastener-Electrical-Management/dp/B07HGZ83JB

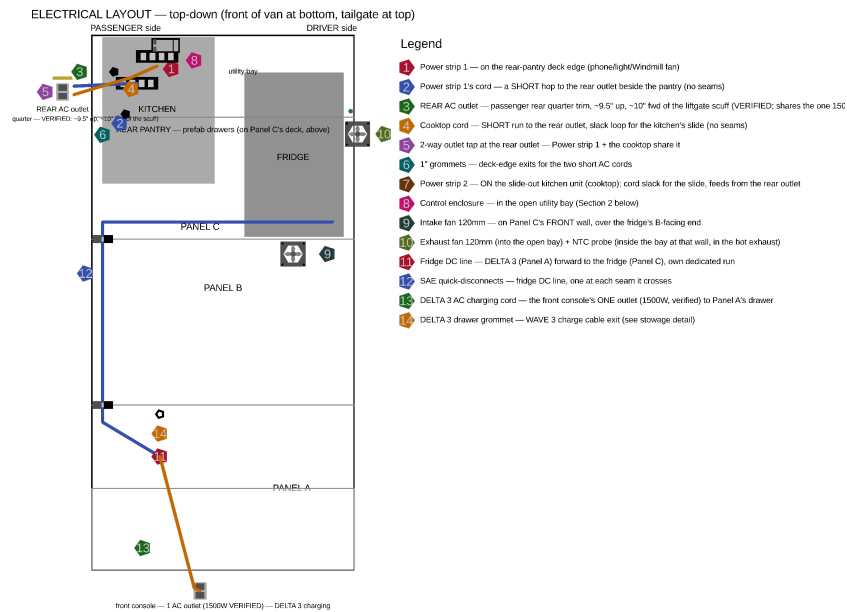
Slides + electrical together run **~\$290-305**. These links satisfy the corresponding BOM rows above — don't double-count.

Estimated total (summing the BOM table above): ~\$3,520–3,890 with the fridge + kitchen included, or ~\$2,820–3,040 if you already own equivalents — the owner-supplied CO monitor and fire extinguisher aren't priced here. The itemized BOM table is the authoritative figure; the breakdown below traces where the money goes and how the design's cost evolved. It includes ~\$111 for the bed platform and its leveling — six 1x4 boards (~\$30, weight swap dropped it from seven), 12 leg leveling feet w/ T-nuts (3 four-packs, ~\$36), 12 star-knob grips + jam nuts (~\$36), and the Larbeti stick-on bubble-level 2+2 pack (~\$10, purchased: pitch on the rail edge, roll on the rear-pantry deck edge); ~\$19 for the kitchen drawer hung above the kitchen unit (slide pair + catch — its box and cheeks come out of the plywood sheets' spare area); ~\$24 for Panel A/B's diagonal corner braces, added to recover some of the racking rigidity lost when their tops came off, ~\$60 (purchased) for the SRS airbag emulators (Section 9), ~\$43 that Power strip 1's old full-length cord run used to add (8ft cord + 2 grommets + 2 inline connector pairs — since deleted by the verified-outlet round below), and ~\$52 that the old plywood pantry's retention system used to add (fiddle lips, lash straps, bins, liner, pins, net) — since removed with the prefab-drawer swap below. The July-18 round adds ~\$191: a Lynx-style leveling-block 10-pack + chock (+\$40 — per-site leveling moves to the wheels, driven by the Block Calculator), Kipp cam levers replacing the old plywood pantry's hex bolts and wing nuts (+\$60 — later removed entirely by the prefab-drawer swap), three extra 2x2 boards for the cube-frame bottom rails (+\$24 mid-range), cable grommets (a \$10 20-pack covers all 7 pass-throughs), a passive louvered cool-air intake scoop in the front wall (+\$6 — a matching exhaust louver in the old cabinet door was cut along with the door itself; the utility bay is open now), a pantry upgrade — an adjustable bottom-bay shelf plus the fiddle-lip/lash-strap/bins/liner retention system replacing the old nets-everywhere approach (a single soft-goods net stays) (+\$33 net over the old \$12 net), and 4 over-center seam draw-latches that clamp the three modules into one rigid beam (+\$16); Panel B losing its drawers entirely (owner-confirmed: the side doors don't reach it — its bay is top-loaded deep storage now) saves ~\$86: two slide pairs, two catches, and a whole 1/2" plywood sheet, less ~\$8 for Panel C's new front wall hardware coming out of that freed sheet. Panel A's left bay going to WAVE 3 open storage instead of a 4th drawer nets out close to a wash: -\$19 (one less drawer slide pair, one less catch, no more plywood tray) offset by the glide strips and a non-slip mat. Moving the fridge and fan system onto the DELTA 3 stack (instead of the van's rear outlet) adds ~\$79: a 2nd cord (fridge DC line, Panel A to Panel C) and its own charging cord (front console to Panel A), the grommets and disconnects both need, and a 3-way splitter when 3 lines wanted the console's power (later downsized to a 2-way rear-outlet tap). Relocating the old plywood pantry onto Panel C's deck instead of its own module saved ~\$20-34: one fewer module's worth of frame lumber (2 fewer 2x2 boards) and corner brackets, since it no longer needs a frame of its own. This is above the old built-from-plywood design's ~\$1,245–1,452 mainly because of three swaps that trade DIY labor (or an inadequate factory feature) for bought reliability: a real BougeRV Rocky 40 + JAGAHABA kitchen (~\$700–850 combined, Rocky price UNVERIFIED) instead of building fridge/kitchen boxes from the same plywood as the panels, the fridge cooling/control electronics (2 fans, controller, sensor, surge protector — new this redesign, ~\$85–120 combined), and the no-drill securing hardware — L-track + stud fittings + rated straps + the anchor board's steel and mat (~\$270, up ~\$150 from the drilled E-track plan it replaced when the owner ruled out holes in the vehicle, Section 8) — standing in for the factory cargo hooks once those turned out to be rated for cargo nets only, not securing a 45-90lb item. The mattress is now a bought HEST Dually Long (~\$530–590, cover included) — the DIY 2-layer fallback (~\$185–255) stays in the BOM as the budget swap. The plywood was ~\$115 across 2 sheets (one 3/4", one 1/2") after Panel A/B lost their tops, Panel B its drawers, and the pantry moved onto Panel C's deck — down from 4 sheets earlier. The July-2026 **weight swap** then thinned the non-critical panels to 1/2"/3/8" (~21 lb off the structure, Appendix F) and the bed to 8 slats: it shaves ~\$5 on the 1x4s but adds a 3/8" half-sheet (~\$25), so plywood is

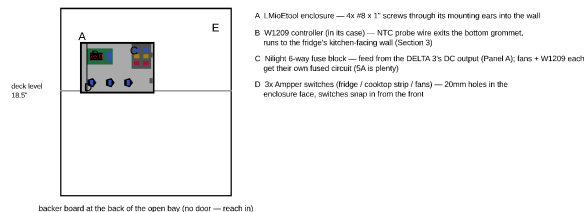
now ~\$140 across a 3/4" sheet (lightly used), a 1/2" sheet (full), and a 3/8" half-sheet — a poor cost-per-pound, but the owner prioritized easier module lifts. A later review round also adds the per-module surface wire raceway — a CordMate II kit for the AC lines + an open-slot kit for the DC line, AC/DC separated (~\$46, Section 5) — replacing the loose cord clips for cleaner, serviceable runs. Two optional material upgrades are documented but NOT in the total above — poplar top-rails (~\$25–40) and aluminum corner braces (~\$10–15), see Section 3's "Material options & upgrades." A verified-outlet round (the van has exactly **2 AC outlets: 1 front console + 1 back passenger area**) moves the cooktop and Power strip 1 onto the REAR outlet — deleting all 4 inline AC seam connectors (~\$32), shortening both AC cords (~\$14), downsizing the splitter to a 2-way tap (~\$3), and dropping 3 grommets (~\$18): **~\$65 and two fewer failure points per lift-out**. A later round then **swaps the plywood pantry for a prefab drawer cluster** (Component 1): out go ~\$137 of pantry mounting + retention hardware (cam levers, 2 steel braces, fiddle-lips/straps/net, half-round edging) and ~42 lb of plywood/hardware; in come 4 IRIS 12"W stackable drawers + a cleat/strap hold-down + a pot crate (~\$115, ~15 lb) — roughly cost-neutral and ~27 lb lighter, with nothing to build. A companion change containerizes Panel B with Sterilite 28-Qt lidded under-bed totes so its top-loaded bay unloads as whole labeled boxes. Finally, an **RJ-MODINI spare kit** (+\$350, +~40 lb — 28.5" diameter, the closest OE match of the kits compared) stows flat in Panel B at the axle — chosen over a hitch basket (which would cost ~\$180-400, add its own 30-60 lb, and lever ~130+ lb onto the rear axle) — displacing 2 of the 4 totes (~\$24, ~4 lb) (Appendix F).

5. Power Cord Routing

Every purchased electrical component at its real position — icon-based top-down layout, the control-cluster mounting elevation, and the fan installation detail (screw pattern + airflow direction):



CONTROL CLUSTER — in the open utility bay on its backer board, elevation (looking forward from tailgate)



FAN MOUNTING — face-on, both fans identical hardware



Cut a 4.5" round hole (or 4.5" square) in the wall; fan mounts over it with 4x #8 x 1-1/4" screws through the corner holes. Grill goes on the AIR-ENTRY side. Arrow on the fan's side frame shows its blow direction — point it per the labels above. Wiring: both fans in parallel off the W1209 relay (Section 5 wiring schematic).

Electrical layout

The fridge (and its fan system) now runs off the DELTA 3 stack, not the van's rear outlet — see the "Why the fridge and fan system now run off the DELTA 3 stack" note in Section 1 for the full reasoning (reliability while people are away from the vehicle, backed by real runtime/recharge numbers). This does mean the fridge needs a cord after all, unlike the previous design's direct-plug simplification — a DC line from the DELTA 3 (Panel A's right drawer) forward to the fridge (Panel C), crossing the same 2 module seams the cooktop's cord does. The Sienna's rear 12V accessory outlet is simply unused now — nothing in this design plugs into it, so it's free for anything else later.

Cord management — run each module's cords in a surface raceway, not loose clips (owner, July 2026). Instead of loose zip-tied/clip-covered runs, route the cords through a **segmented plastic surface raceway (wire duct / Wiremold-style)** — a short length screwed to the inside of each module's frame, one channel per panel, meeting at the existing seam disconnects. Two rules keep it correct: (1) the raceway must stay **segmented per module** (a separate length in A, B, and C) — NOT one continuous conduit spanning the seams, or a panel can't be lifted out; the inline AC connectors / SAE DC disconnects still live at each seam. (2) Keep the **120V AC and 12V DC lines in separate channels** (a divided duct, or two small ducts side by side) — don't bundle them together. This protects the cords from abrasion, makes every run traceable, and lets you pull or add a wire later without fishing it through the frame — a real serviceability win over clipped cords for a small added cost (see the raceway line in Section 4's BOM). Where a run has to cross into a module's interior bay, a snap-cover raceway lifts open without unscrewing.

The van has exactly 2 AC outlets (VERIFIED): 1 in the front center console (1500W) + 1 in the back passenger area. That rear outlet transforms the routing: the two tailgate-end AC loads (the cooktop and Power strip 1) plug into it with **short cords and ZERO seam crossings** — the old full-length runs with their inline seam connectors are gone. The front console outlet is dedicated to the DELTA 3's charging cord. **Four cords need routing in total, each on its own independent line:**

Cooktop line (Power strip 2 → REAR outlet) — short

Power strip 2 rides the slide-out kitchen unit so it travels to wherever you're actually cooking — the JAGAHABA extends to ~70" out the tailgate in use. Its cord now just hops to the **rear AC outlet in the back passenger area**, a few feet away. This is a full 120V AC household circuit, so use a household-rated extension cord.

1. Drill one 1" grommet hole in the frame rail at the kitchen unit's corner post nearest the rear outlet.
2. Run the short cord from the strip, through the grommet, to the rear outlet's 2-way tap (shared with Power strip 1 — below).
3. Leave a **slack loop** sized to the kitchen's full ~70" slide extension, coiled near the stowed position, so the cooktop keeps power at the cook position without strain.

Power strip 1 line (rear pantry deck edge → REAR outlet) — short

Power strip 1 (phone/device charging, reading light, the Windmill fan) mounts at the **rear pantry's deck edge** in the open bay (Component 1) — which sits directly beside the rear outlet. One short 120V AC cord:

1. Drill a 1" grommet at the deck edge where the cord drops off the pantry bay.
2. Run it to the rear outlet's 2-way tap. Leave ~1 ft of slack so lifting the pantry's drawer units never requires unplugging anything.

Fridge DC line (DELTA 3 → Panel C fridge bay)

The fridge's DC line runs from the DELTA 3 Plus's DC output (Panel A's right drawer) back toward the tailgate, through Panel B, into Panel C's fridge bay. Unlike the cooktop and Power strip 1's lines, this one really is 12V DC, so an automotive/solar SAE quick-disconnect is the right connector type here. Like the cooktop's line, it crosses the same 2 remaining module seams (Panel A → Panel B, Panel B → Panel C), so it needs its own quick-disconnects too. This is the trade-off Section 1 already flagged: reintroducing a long cord run in exchange for the fridge no longer depending on the van's ignition state.

1. Drill a 1" grommet hole in Panel A's frame rail near the DELTA 3 drawer, and one more at each of the 2 seams it crosses (Panel A → B, Panel B → C) — 3 total, reusing the same rail-channel approach as the cooktop line (a dado groove or a zip-tied cord under a clip-cover strip both work).
2. Install a quick-disconnect connector at each of the 2 seams so Panel A and Panel B still lift out independently.
3. Terminate the cord at the fridge's DC input inside Panel C's void, near the control enclosure (Component 7).
4. Leave a few inches of slack at the DELTA 3 end so opening/closing that drawer doesn't strain the connection.
5. **At the fridge end, clip the slide-travel slack loop to the slide's FIXED outer rail — not the tray, and not left loose** — 2-3 screw-mount cable clips (same VIPMOON-type clips as the cooktop line, Section 4 BOM) along the rail's inside face. The outer rail never moves, so the clipped loop stays out of the pinch point between the tray and the rail as the tray slides home; a loose coil left to dangle can get caught under the tray or crushed against the rail on the way in.

Every connection point, P1–P12 (matches the numbered markers on the wiring diagram, Renders)

The ONE 16AWG SAE line runs DELTA 3 → fuse block; **both** circuits branch at the Nilight block, and every black (–) return lands on its negative bus:

#	Where	What lands there — terminal, connector, crimp
P1	DELTA 3 Plus 12V car-power outlet (Panel A right drawer)	Fused (15A) male car-plug → SAE adapter — hand-plug, no tools. That fuse is the whole line's upstream protection
P2	Panel A/B seam	SAE quick-disconnect #1 — the OYMSAE cord ends mate; zip-tie a strain-relief loop each side of the seam
P3	Panel B/C seam	SAE quick-disconnect #2 — same part; both disconnects hang slack, never taut
P4	Panel C front wall, low driver side	1" grommet — the cord just passes through into the fridge bay, no termination
P5	Nilight 6-way fuse block, main studs (in the utility bay)	Red → +IN stud, black → –IN stud: strip 3/8", crimp #10 heat-shrink ring terminals, M5 nuts snug. F1 = 10AATO (fridge), F2 = 3AATO (fans), F3–F6 spare
P6	SW1 "FRIDGE" — Ampper rocker, 3 spade tabs	F1's 16AWG red → PWR tab; LOAD tab → pigtail red; GND tab → neg bus (powers the glow ring). 0.25" female spade crimps on all 3
P7	SAE → female 12V socket pigtail	SAE end mates toward SW1's LOAD side; the socket zip-ties in the utility bay near the fridge's kitchen-facing wall
P8	Fridge DC input jack	The fridge's OWN 12V cord: plug → socket, plug → fridge. Slack loop for the 24" slide, clipped to the FIXED rail (step 5 above)
P9	SW2 "FANS" — Ampper rocker	F2's 18AWG red → PWR tab; LOAD tab → W1209 "+12V"; GND tab → neg bus
P10	W1209 controller — 4 screw terminals	+12V ← SW2 LOAD; GND ← neg bus (18AWG black); K0 ← short jumper from +12V; K1 → fan feed wire. Strip 1/4", screws snug. Settings: P0 = H mode, fans ON ~95°F / OFF ~85°F
P11	W1209 2-pin probe header	NTC probe plug clicks on; the probe TIP zip-ties inside the bay at the fridge's kitchen-facing wall, next to the exhaust fan
P12	WAGO 221-413 lever nuts (×2)	Nut A: K1 feed → both fan REDS. Nut B: both fan BLACKS → one 18AWG black → neg bus. Levers fully closed, tug-test every wire

Wire + termination spec: main run = the OYMSAE cord's own 16AWG; in-bay branches = 18AWG red/black hookup wire. Crimps: heat-shrink #10 rings (block studs) and 0.25" female spades (switch tabs) — crimp, tug-test, then shrink. SW3 is a spare on the panel (its old "cooktop" BOM label predates the verified-outlet round — the cooktop is 120V AC, switched at Power strip 2's own outlet switch).

The fan system's own wiring (fuse/switch/PWM controller/fans, Section 1) taps off this same DELTA 3 source rather than a separate line — see the fridge wiring schematic in Renders. The rail-channel routing is illustrated in the [Step 8 diagram](#) — it now carries only this DC line, since both AC loads plug into the rear outlet

instead.

DELTA 3 AC charging line (front console → Panel A drawer)

A second, independent cord charges the DELTA 3 stack itself from the front console's AC outlet while driving — this is what makes the fridge's long runtime (Section 1) actually sustainable trip after trip, instead of draining down with no way to recharge until you're back home. Like the cooktop and Power strip 1's lines, this is a 120V AC household cord (charging the DELTA 3's AC input), not a DC connector. Panel A is now the module closest to the front seats (the pantry sits at the opposite end, on Panel C — Section 1), so this cord **crosses zero panel seams**, just a short run through the open floor gap in front of Panel A.

1. Drill a 1" grommet hole in Panel A's front end rail, near the DELTA 3 drawer.
2. Run the cord forward, through the open floor gap (nothing built there — Section 1), to a grommet hole near the front of the platform.
3. Continue along the van floor track to the front console AC outlet — same routing as the other two forward-running lines.
4. Leave enough slack at the Panel A end that pulling the DELTA 3 drawer open doesn't strain the connection.

Outlet facts (VERIFIED): 2 AC outlets total, both fed by the van's single 1500W inverter. The rear outlet sits on the **passenger-side rear quarter trim: 22.5" above the cargo floor, ~10" forward of the liftgate scuff plate**, socket center **10" laterally in from the right sidewall** (MEASURED V9b, owner Aug 1 2026 — this **corrects the ~9.5" / ~16" figures** carried in earlier drafts), directly above the 12V battery access panel and below the rear cup holders. **That is 4" ABOVE deck level (18.5"), not below it — which makes the cord runs easier than planned:** both cords reach the socket from deck level instead of having to drop down the gap between the kitchen unit's outer face and the van wall. Plug access no longer depends on sliding the kitchen out. **Serviceability note: the 12V battery access panel sits under/behind the kitchen's passenger rear corner — keep it clear: the anchor board's kitchen-side strip stops short of it (Section 8), and battery service = unstrap + slide the kitchen out (Component 7).**

Shared-inverter wattage: every AC load shares one 1500W pool (confirmed — both outlets ride the same inverter). Plugging the cooktop + Power strip 1 into the rear outlet and the DELTA 3 charger into the front one cleans up the *cords*, but not the *budget*: the DELTA 3's AC input alone (up to 1500W at full X-Stream speed) can consume the entire rated capacity. Running the induction cooktop while fast-charging the DELTA 3 could oversubscribe the inverter. In practice this rarely collides — cooktop use happens at camp with the engine off, DELTA 3 charging while driving — but if you ever want both at once (e.g., cooking at a rest stop with the engine idling), cap the DELTA 3's AC charge rate in its app (EcoFlow lets you limit input wattage) to leave headroom.

The WAVE 3 is the one thing still fully self-contained. It draws from the DELTA 3 Plus (in Panel A's drawer) via its own charge cable, and isn't wired to any van circuit at all.

6. Illustrated Build Manual — By Component

Everything from Sections 3-4 reorganized around **10 self-contained components**, in the order you'd actually build them. Each component lists its parts, then walks through numbered steps in instruction-manual format: **the new parts for that step on the left, an exploded assembly view showing how they go in on the right** — drawn as woodworking-plan line art (white parts, black edges, letter labels keyed to the parts list, arrows showing insertion direction). Every piece dimension traces back to the cut list (Section 3) and every hardware item to the BOM (Section 4). Panels A and B share identical frame/divider/drawer construction, so those steps intentionally reuse the same diagrams in both components.

Cut every piece for every component up front, per Section 3's cut list, before starting Component 1 — sand all edges once cut.

Component 1: Rear Pantry — Prefab Drawer Cluster + Pot Bay

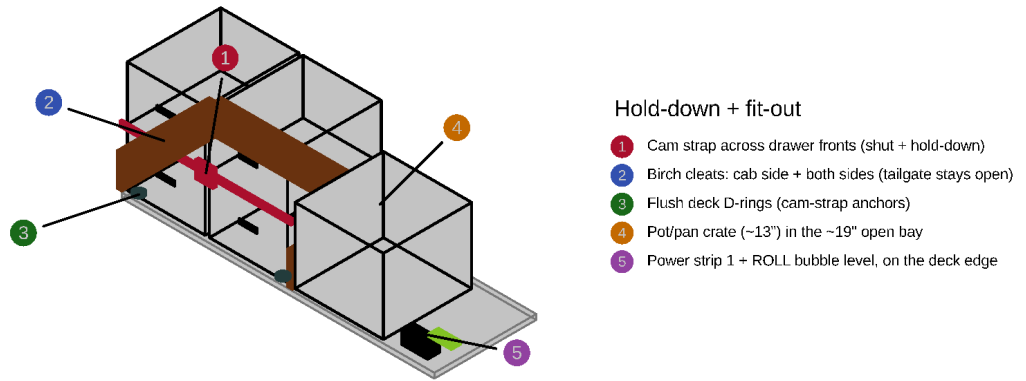
COMPONENT 1 — REAR PANTRY (PREFAB DRAWER CLUSTER + POT BAY)																	
FINISHED COMPONENT	ACCESSORY LIST <table border="1"> <tr> <td>A IRIS IRIS drawer x4</td> <td>B crate pot crate x1</td> <td>C pine cleat x3</td> <td>D cam strap x1</td> </tr> <tr> <td>E flush D-ring x2</td> <td>F felt tape x1</td> <td>G velcro tie x8</td> <td>H bubble level x1</td> </tr> <tr> <td>I strip power strip x1</td> <td colspan="3"></td> </tr> </table>	A IRIS IRIS drawer x4	B crate pot crate x1	C pine cleat x3	D cam strap x1	E flush D-ring x2	F felt tape x1	G velcro tie x8	H bubble level x1	I strip power strip x1				PART LIST <table border="1"> <tr> <td>01 IRIS unit 4x</td> <td>02 pot crate 1x</td> <td>03 pine cleat 3x</td> </tr> </table>	01 IRIS unit 4x	02 pot crate 1x	03 pine cleat 3x
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SWAPPED (owner, July 2026): the custom plywood pantry is replaced by bought prefab drawer units — no shelving to build, and ~27 lb lighter (out goes ~42 lb of plywood + mounting hardware; in comes ~15 lb of drawers + hold-down). A 2x2 array (2 wide x 2 high) of **IRIS USA 12"W stackable storage drawers** (Home Depot #500163, two 3-packs — use 4, keep 2 spares) sits on the **tailgate end of Panel C's deck** — the same tailgate-end footprint the plywood pantry used, now **14.5" deep** so the deck carries the 14.3" units outright (V5 measured the hatch reserve at zero — there is nothing left to sit 0.3" proud into), leaving a **79.25" sleeping run**. The cluster is **24.2" W x 14.3" D x 16.8" H**, which leaves **~21.8" of open deck on the passenger side** for a **pot/pan crate** plus the relocated power/wiring. Nothing is screwed into the units — a **cleat pocket + one cam strap** hold them, so each drawer unit **lifts straight out** (removable, clears the measured 50" liftgate easily). **Deleted with the plywood pantry:** the 4 cam-lever clamps, the 2 steel L-angle sway braces + their bolts, the base cleats, the adjustable shelf, and the entire fiddle-lip/lash-strap/bin/liner retention kit — the drawers now contain their own contents. **The enclosed bed cubby is gone too**, so **Power strip 1 and the ROLL bubble level relocate to the deck edge** in the 19" bay (still reached and read from the bed). Build Panel C (Component 4) first — the cluster just sits on its finished deck.

Parts needed:

- **IRIS USA 12"W stackable storage drawers** (Home Depot model #500163, gray, sold as 3-packs at ~\$30.29 — buy 2 packs) — **use 4 units**, 12.1" W x 14.3" D x 8.4" H each, stacked 2 wide x 2 high (2 spares for swap-in). The 14.3" depth sits 0.3" proud of the deck edge — inside the 2" hatch reserve
- Hold-down cleats — ~1" x 1" birch strips (from the 3/4" sheet offcut), ~3–4 ft, screwed to the deck around the **cab side + both sides** of the cluster (tailgate side left open)
- Cam-buckle tie-down strap — 1, ~1" nylon, across the drawer fronts to the deck D-rings (holds the drawers shut + snugs the stack)
- Flush-mount deck D-rings — 2 (or a short L-track offcut — the anchor-board 4-pack leaves spares, Section 8), screwed to the deck flanking the cluster
- Anti-rattle felt/cloth tape — for the interlocking rails between the stacked units; + 2 small Velcro ties linking the top units to the bottom
- Pot/pan bay — 1 rigid ~11" x 11" bin or crate (the pots' own cardboard box works short-term, but a rigid bin takes a strap and won't crush) + a corner cleat + a bungee
- Power strip 1 + the ROLL bubble level — relocated to the deck edge in the 21.8" bay (Power strip 1's cord run is Component 6; the PITCH bubble level still goes on the bed platform, Component 2)

Build it (mostly placement + hold-down — there's no shelving to make):



REAR PANTRY — prefab 2x2 IRIS drawer cluster (24.2" x 16.8") + pot crate, on Panel C's deck (Component 1)

Replaces the plywood pantry — ~27 lb lighter, nothing built. Each drawer unit lifts straight out once the cam strap is loosened.
Canned goods LOW, boxed dry goods UP. Cab-side cleat stops the forward slide under braking; the strap doubles as drawer-shut + hold-down.

Rear pantry layout

- Set the cluster.** Stack the 4 drawer units 2 wide x 2 high on the tailgate end of Panel C's deck, drawers facing the tailgate. Felt-tape the interlocking rails between the stacked units, and add a Velcro tie linking each top unit to the one below so the top row can't creep off under braking.
- Cleat pocket.** Screw ~1" birch cleats to the deck hugging the cluster's **cab-facing side and both sides** — leave the **tailgate side open** for drawer access. The cab-side cleat is the critical one: it stops the cluster sliding forward under braking.
- Strap the fronts.** Screw 2 flush D-rings to the deck flanking the cluster and run one cam-buckle strap across the drawer faces. Double duty — it keeps the drawers from sliding open under acceleration AND snugs the whole stack down. Unclick one end to open the drawers.
- Pot/pan bay + power.** In the ~21.8" of open deck beside the cluster, set a rigid ~13" crate for pots/pans (their 11x11 box drops inside), held by a corner cleat + a bungee. Route Power strip 1's cord and the 12V/wiring along the deck's back edge behind the bin, and mount Power strip 1 + the ROLL bubble level on the deck edge where you can reach and read them from the bed.
- Load it.** Dense/heavy items (canned goods) in the bottom drawers, boxed dry goods up top, pots in the bin — then cinch the front strap before driving.

Component 2: Panel A & Bed Platform

COMPONENT 2 — PANEL A & BED PLATFORM																														
FINISHED COMPONENT	ACCESSORY LIST <table border="1"> <tr> <td>A corner blk 1" x4</td> <td>B slat brace x2</td> <td>C wood screw 2" x16</td> <td>D leveling foot x4</td> </tr> <tr> <td>E 3/8-16 insert x4</td> <td>F slide 25" pr x1</td> <td>G flange patch x1</td> <td>H D-ring x2</td> </tr> <tr> <td>I cam strap 1" x1</td> <td>J grommet 1" x1</td> <td>K pocket scr 1 1/2" x16</td> <td>L discut RT x4</td> </tr> <tr> <td>M pine cleat x1</td> <td>N LVL bubble level x2</td> <td></td> <td></td> </tr> </table>	A corner blk 1" x4	B slat brace x2	C wood screw 2" x16	D leveling foot x4	E 3/8-16 insert x4	F slide 25" pr x1	G flange patch x1	H D-ring x2	I cam strap 1" x1	J grommet 1" x1	K pocket scr 1 1/2" x16	L discut RT x4	M pine cleat x1	N LVL bubble level x2			PART LIST <table border="1"> <tr> <td>01 end rail 40" 2x</td> <td>02 side rail 25" 2x</td> <td>03 leg ~8" 4x</td> <td>04 brn rail 45" 2x</td> </tr> <tr> <td>05 diverter 25" 1x</td> <td>06 diverter box 1x</td> <td>07 gl dz strip 2x</td> <td>08 bed rail 55" 2x</td> </tr> <tr> <td>09 bed slat 45" 10x</td> <td></td> <td></td> <td></td> </tr> </table>	01 end rail 40" 2x	02 side rail 25" 2x	03 leg ~8" 4x	04 brn rail 45" 2x	05 diverter 25" 1x	06 diverter box 1x	07 gl dz strip 2x	08 bed rail 55" 2x	09 bed slat 45" 10x			
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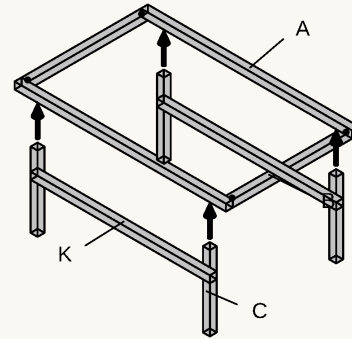
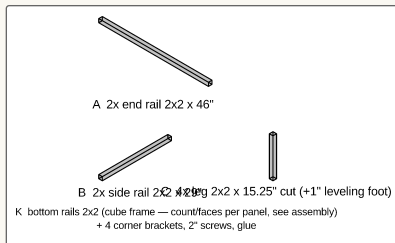
Panel A and Panel B share the same frame construction — neither has a top of its own anymore; the three-piece slatted bed frame caps both (see Steps 6-7 below, and the Bed Frame Detail render): Panel A's section is **screwed down** (it is also acting as a diaphragm — see Section 1), and Panel B's top is two centreline halves that lift out individually. **Panel B also gains a centre bearer** (Aug 2026) for those halves' inner edges — the one frame difference from Panel A's divider, which splits its bay instead. They diverge at the drawers: **Panel A's left (driver-side) bay is WAVE 3 open storage, not a drawer** — the WAVE 3 (20.4" wide) is too wide for a boxed drawer's 19" clear interior, so it rests directly on the bay floor instead, reached by hand through the driver's side door. Panel A ends up with only ONE actual drawer (right side, DELTA 3).

Parts needed:

- Long rails — 2, 29" cut, 2x2 pine
- End rails — 2, 46" cut, 2x2 pine
- Legs — 4, cut to 16", 2x2 pine (1.5" x 1.5" actual — plenty for this load); the leveling foot brings each back to an effective 17"
- Leg leveling feet — 4 (3/8-16 screw-in the leg's bottom end grain + leveling stud + 2" star knob jam-locked on the shaft above the stud's fixed hex collar), one per leg (Leg Leveling Foot Detail render)

- Bottom rails — 2, 46" cut, 2x2 pine (END faces only — the side faces stay open for the drawer/WAVE 3)
- Center divider — 1, 26" cut, 2x2 pine
- Corner brackets — 4
- Diagonal corner braces — 2 (recovers racking rigidity lost without a top)
- Drawer box — 1 (right/DELTA 3 side only), 5 pieces: bottom 20"x25" (1/2" ply), 2 side walls 25"x14.5" + 2 front/back walls 20"x14.5" (3/8" ply) — weight swap: walls 1/2" → 3/8", glue + biscuit the corners since it carries the 48 lb DELTA stack
- Drawer slide — 1 pair, 20" full-extension
- Drawer catch — 1, friction catch or small turn latch
- DELTA 3 drawer hardware (right drawer) — 1x1 pine locating cleats, 2 screw-eye D-rings + cam strap, 1" cable grommet
- WAVE 3 glide strips (left bay floor) — 2, UHMW or laminate scrap, cuts friction sliding the unit in/out by hand
- Bed platform lumber — seven 1x4 x 8ft pine boards: five 42" slats + ten 17.5" slats + six 29" side rails — crosscuts only, no hardware (Bed Platform Detail render)
- Panel B centre bearer — 52" of 2x2 (two 26" pieces side by side), out of the 2x2 frame stock
- RV bubble levels — Larbeti stick-on 2+2 pack (PURCHASED): PITCH bar mounts here, ROLL one goes to the rear-pantry deck edge (Component 1)

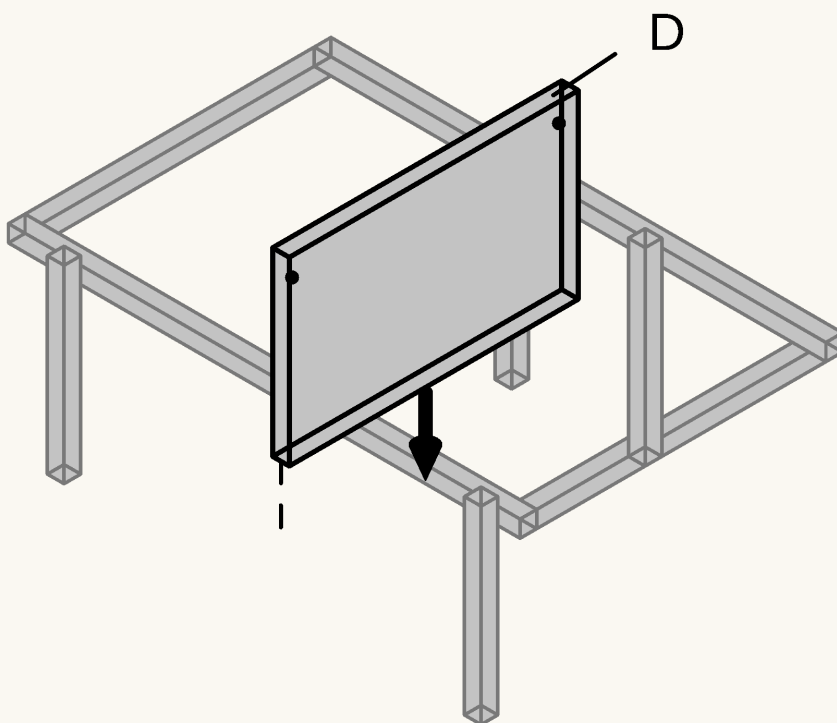
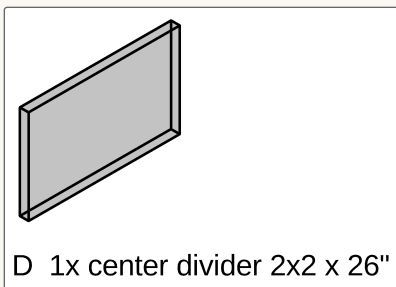
1



K: bottom rails (2x2, underside at 1") close the frame into a box — both end faces only (drawer + WAVE 3 exit the sides)

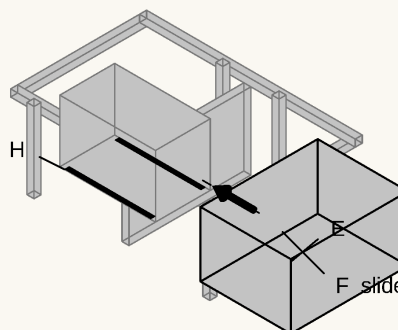
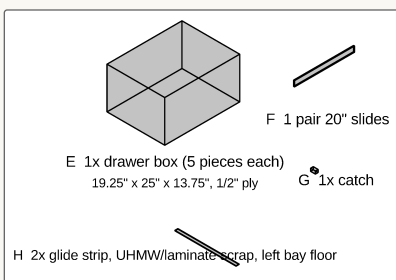
Build the frame: 2 side rails (B) + 2 end rails (A) joined with corner brackets and 2" screws, then the 4 legs (C) — CUT to 15.25" (Panels A/B run 3/4" shorter than Panel C: deck recess), drilled in the bottom end grain (1/2" dia x 7/8" deep, centered — maker's spec; test-fit in offcut) for the 3/8-16 screw-in threaded insert, so the leveling foot brings each leg back to an effective 16.25" (Leg Leveling Foot Detail render). Much easier to drill before assembly. Legs inset **3.5*** from the deck's side edges to clear the floor-level vent intrusion (V4, Aug 2026 — was 2.5"). Then close the bottom: 2 END-face bottom rails (K, 46", underside 1" up at the leg bottoms — Panel A's SIDE faces stay open so the drawer and WAVE 3 can exit) plus 2 diagonal corner braces up top — the part-cube racks far less than rails + brackets alone. (Panel B's frame differs only at the bottom rails: it closes all 4 faces.)

2



Drop the center divider (D) into the bay, centered on the panel's width — it splits the void into a left and right bay and carries one side of the right bay's drawer slide.

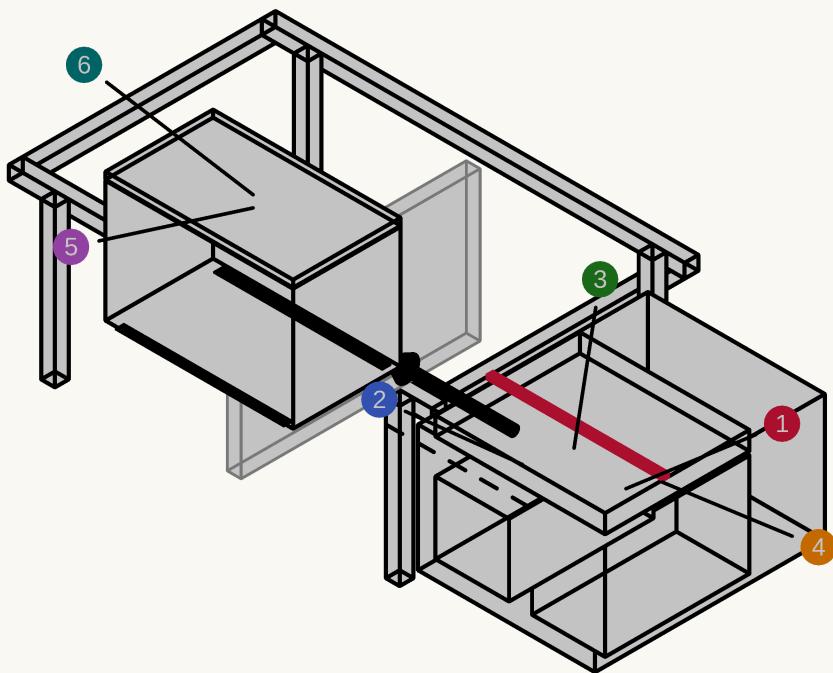
3



F slide: box-to-rail + box-to-divider

Left bay: WAVE 3 open storage (shown in place for reference) — no box, no slide

Build the right (DELTA 3) drawer box (E, 5 pieces, open top) — biscuit the corners with 2x R1 biscuits each + glue, bottom in a glued rabbet (Joinery & Fastener Guide) — and slide it onto its pair of 20" full-extension slides (F) — one slide side on the frame's outer rail, the other on the center divider. Add a catch (G) so it doesn't slide open in transit. On the LEFT bay, skip the drawer entirely: glue/screw down the 2 glide strips (H) along the floor instead — that's the WAVE 3's storage spot (Component 8).



4

DELTA 3 drawer hardware (parts list above); pine cleats, 2 D-rings + cam strap, 1" grommet

DELTA 3 STACK (right drawer) + WAVE 3 (left bay)

Panel A, exploded — Components 2 & 8

DELTA 3 Plus outboard (the WAVE 3 plugs into it), Extra Battery inboard; ~48 lb in a normal drawer — no E-track. The WAVE 3 rests in the raw bay on 2 glide strips.

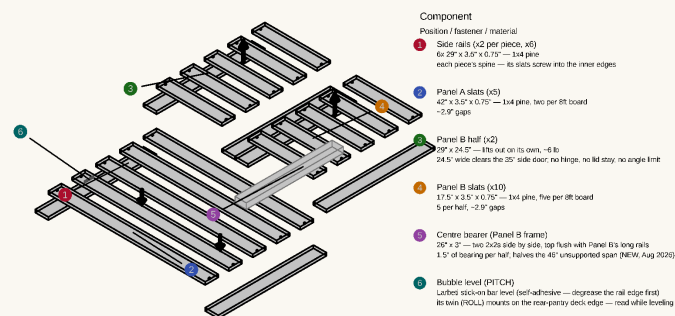
Found storage: a ~2.6" lift-out tray tops the DELTA stack, and a shelf @ 13.7" tops the WAVE 3 — it still slides out beneath.

Component / position

- 1 DELTA 3 Plus — 7.95"x15.7"x11.16", ~28 lb outboard (pull wall); cam-strapped over locating cleats
- 2 Smart Extra Battery — 8"x15.7", ~20 lb inboard — extra runtime only, grabbed less often
- 3 Lift-out tray — reclaims ~2.6" dead air cables, the DELTA's own cords, dongles
- 4 Cam strap + D-rings + 1" grommet strap over both units; WAVE 3 charge cable exits the grommet
- 5 WAVE 3 — 20.4"x11.7"x13.2", 33.7 lb raw bay, 0.35" clear — a boxed drawer would NOT fit
- 6 Overhead shelf @ 13.7" on 1x1 cleats flat soft goods + the WAVE 3's own hoses ride up here

In the right drawer, add the DELTA 3 fixtures: locating cleats glued/screwed to the drawer floor, D-rings + cam strap to hold the stack down in transit, and a cable grommet for the WAVE 3's charge cable. DELTA 3 Plus outboard (pull wall), Smart Extra Battery inboard — see Component 8.

seven 1x4 boards: 6 rails + 15 slats + pocket screws,
plus 52" of 2x2 for Panel B's centre bearer (parts list
above)



BED PLATFORM — 3 lift-out pieces, exploded (Component 2, 58" x 49" overall, 15 slats, 0.75" thick)

SPLIT Aug 2026: Panel A keeps one 29 x 49" section. Panel B's top is TWO 29 x 24.5" halves, split on the centreline, each lifting out on its own.

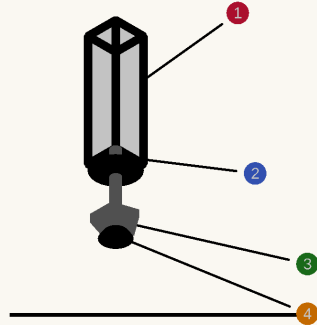
Panel B is top-load-ONLY — the 35 x 45" side door sits over Panel A and only 29" of it is ever clear. At 24.5" wide a half fits back out through that door; a 49" piece never would. Lift the mattress clear, take out only the half on the side you are standing at. The centre bearer also halves the deck's unsupported span over Panel B (46" -> ~22").

Crosscut everything from seven 1x4 x 8ft pine boards — nothing gets ripped, a 1x4 is already 3/4" x 3-1/2": six 29" side rails, five 42" slats, and ten 17.5" slats (two 42" or five 17.5" per board). **Build it as THREE lift-out pieces** (Aug 2026): Panel A's section is one pair of rails on edge-guides 49" apart outside-to-outside (was 52" — cut back by measurement V7) with its five 42" slats between them; each of Panel B's two halves is a pair of rails 24.5" apart with five 17.5" slats. Every gap lands at ~2.9". Pocket-screw each slat end into the rails' inner edges (2x 1-1/4" pocket screws per end; 2" corner braces work if you don't have a pocket-hole jig). Then add Panel B's centre bearer — two 26" 2x2s side by side, fore-aft at its centreline between the end rails, top flush with its long rails: it carries the inner rail of both halves (1-1/2" of bearing each) and halves the deck's unsupported span over Panel B. Last, **screw Panel A's section down** to its rails (2" screws every ~8", countersunk so nothing sits proud under the foam): it is permanent, it stiffens Panel A against racking, and it is the datum the two loose halves register against. Panel B's halves get no fasteners at all — they just lift, and at 24.5" wide either one will pass back out through the 35" side door. Everything sits in one flush 3/4" plane that ENDS at the B/C seam — Panel C's own deck is at the same height, so the sleeping surface stays flush and the mattress's last ~20" rides that deck. The platform adds only 3/4" to the stack. Last, stick one Larbeti bar level (self-adhesive — degrease the surface first) on the driver-side rail's outer edge at mid-span (the 3/4" face fits it exactly) — it reads fore-aft PITCH from the slider door while you turn the leg-foot knobs; its twin reads ROLL from the rear-pantry deck edge (Component 1).

Component

Position / fastener / material

- 1 Leg (bottom end shown)
2x2 pine (1.5" x 1.5" actual), cut to 16"
drill centered, 1/2" dia x 7/8" deep for the insert
- 2 3/8-16 threaded insert
driven UP into the leg's end grain
screw-in style, 7/16" OD coarse outer thread
- 3 Star knob, ~2" dia (hand grip)
3/8-16 thru-hole knob + jam nut on the bolt shaft
the adjustment — big enough to turn with the box tipped
- 4 Leveling glide bolt + pad
1.375" floor pad, +/-0.5" travel around 1" nominal
threads into the insert — leveling only, not height changes



LEG LEVELING FOOT — shown 3x actual size (12 total: 4 per panel x 3 panels)

Legs are CUT to 16" (Panel C) / 15.25" (A/B — deck recess) — the foot makes up the last 1".

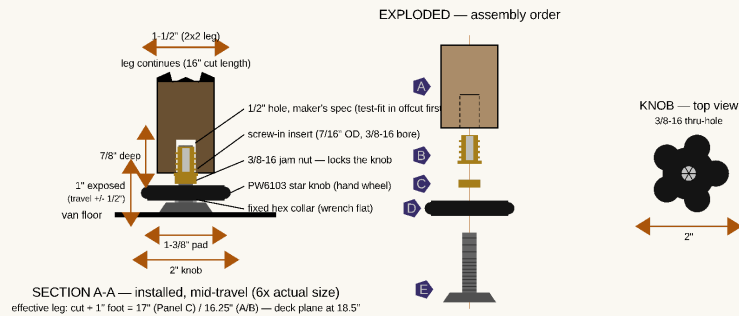
To level: kneel at the side door, tip that corner of the box slightly, spin the star knob. Every leg is exposed at floor level — nothing boxes it in. The platform above rests DIRECTLY on the box rails (no adjusters up top anymore).

LEG LEVELING FOOT — engineering drawing (parts as purchased, July 2026)

12 feet total (4 per panel x 3 panels): 3x Anwenk leveler 4-packs + 3x Peachtree PW6103 knob 4-packs + 12x 3/8-16 jam nuts

6

no new parts — the assembled platform from Step 5
(leveling feet already on the legs, Step 1)



ASSEMBLY

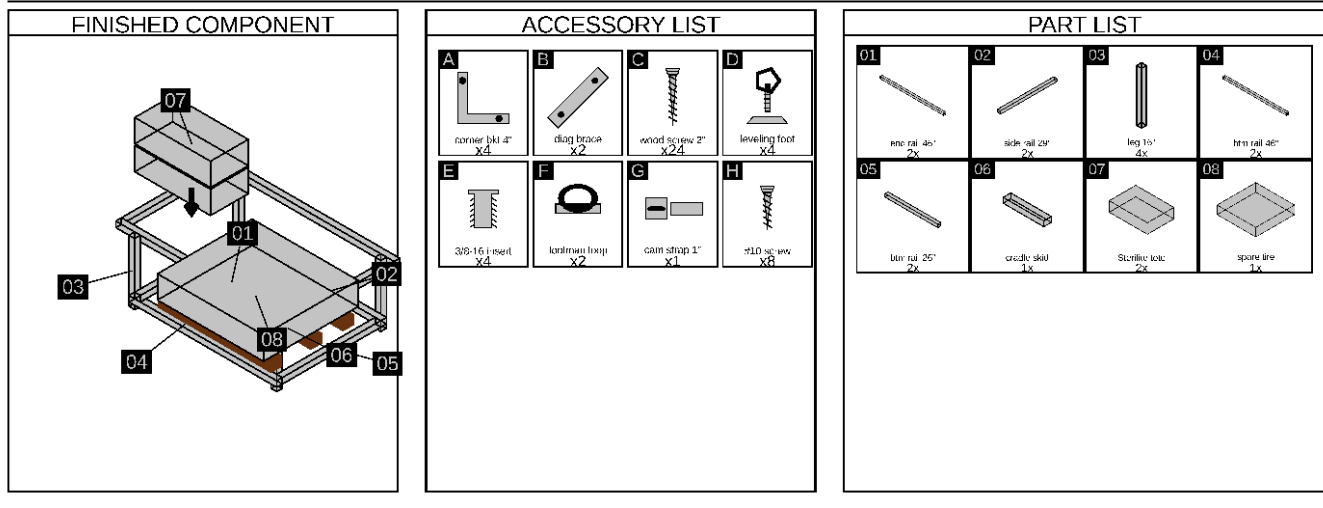
- A drill the leg's end grain: 1/2" dia x 7/8" deep, centered
 - B screw the insert in flush (drive it on a spare bolt + jam nut)
 - C spin a 3/8-16 jam nut ~1" up the stud
 - D thread the knob up to it: wrench the nut DOWN to lock them
 - E screw the stud into the insert to 1" exposure
- Locked together, knob + nut drive the stud: tip the corner, spin, done.
WHY the nut: jammed on the fixed collar instead, the knob locks one way only and unscrews when reversed. The collar = wrench flat for a seized foot.

Rating: 1,320 lb per foot (Anwenk) — 4 feet under the heaviest panel = ~5,280 lb capacity vs ~450 lb actual load.
NOTE: skip the kit's stick-on felt pads — the bare nylon pad grips the van floor better and doesn't shed.

Set the finished platform down directly on Panel A + B's top rails and Panel C's mattress zone, centered so it overhangs the boxes 3" per side — nothing fastens it, it rests there and lifts straight off for drawer-bay access (it enters/exits the van tilted diagonally through the liftgate). Then LEVEL the boxes at the floor: with a level on the platform, tip each box corner slightly and spin that leg's star-knob foot until the bubble centers along both axes. Re-check after the van is parked on level ground. **Foot-assembly rule (engineering drawing above): lock the knob with the JAM NUT** **wrenched down on TOP of it — never by jamming the knob against the stud's fixed hex collar.** Jammed on the collar, the knob only locks in the tightening direction — the first time you spin it the other way to lower a corner, it unscrews itself off the stud instead of driving it. Knob + jam nut locked together grip in BOTH directions, so one knob raises and lowers the leg. The hex collar's actual job: a wrench flat to break the foot loose if it ever seizes.

Component 3: Panel B

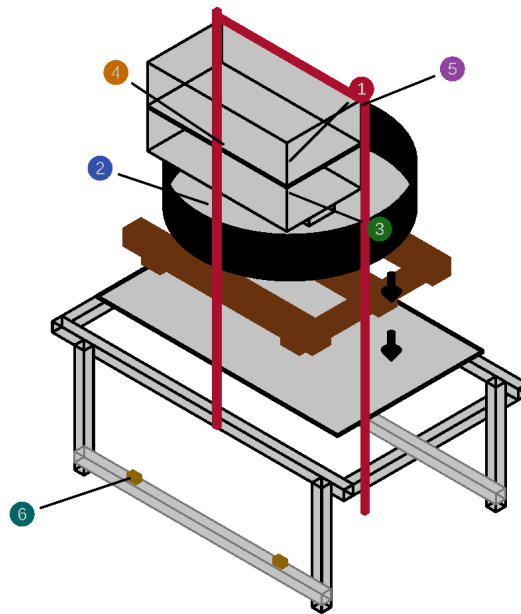
COMPONENT 3 — PANEL B



Same frame construction as Panel A — and that's the whole build. **Panel B has no drawers, no divider, and no skirts:** the sliding-door openings sit over Panel A's footprint, not Panel B's (owner-confirmed), so nothing pulled sideways from this panel could ever clear a door. Its bay is deep storage instead — lift the platform + mattress and load it from above. **The SPARE TIRE lives here** (RJ-MODINI kit, flat on 3" cleats at the axle — see the Spare Tire Stowage render), with **2x Sterilite 28-Qt lidded totes restacked on top** and the jack kit strapped beside it, so whole labeled boxes lift out instead of loose items. The Panel B detail render dimensions **every hole in the panel:** the 4 leg-bottom leveling-insert holes (1/2" dia x 7/8" deep, dead center of each leg's end grain) and the alignment-pin holes on both seam faces (2x 3/8" dia x 3/8" deep, 3" in from each side edge).

Parts needed:

- Long rails — 2, 29" cut, 2x2 pine
- End rails — 2, 46" cut, 2x2 pine
- Legs — 4, 16" cut, 2x2 pine (leveling feet make up the inch)
- Bottom rails — 4: 2x 46" + 2x 26" (all four faces — the full cube)
- Corner brackets — 4
- Diagonal corner braces — 2
- Deep-storage totes — 2x Sterilite 28-Qt under-bed lidded (23.5" x 16.9" x 5.9"), restacked ON the spare (plain lidded, NOT slide-drawer — no side access to use a drawer front)
- SPARE TIRE — RJ-MODINI kit for the 2011-2026 Sienna incl. Hybrid (T155/85R18 steel wheel, 28.5" dia x ~6.4" stored, ~40 lb incl. 2-ton jack/wrenches/cases — VERIFY weight + dims on arrival). Stows FLAT in this bay at the AXLE, on a **cradle skid**: 3x ~3" cleats screwed (glue + 1.25" screws) to 2x 1x3 battens, all from the 3/4" offcut — one piece, so the cleats can't wander, and tall enough to clear the 2.5" bottom-rail curb. NOTHING screws into the van floor: the skid just sits on a non-slip liner offcut (same roll as the drawer liner)
- SPARE INSTALL (once — see the exploded view on the Spare Tire Stowage render): (A) lay the liner offcut on the van floor mid-bay, (B) set the skid on it, (C) spare FLAT on the skid, **valve stem UP** (pressure checks without unloading), tool case nested in the wheel barrel, (D) 2 Sterilite totes restacked on top, (E) 1" cam-buckle strap OVER the whole stack, hooked to **2x 1" footman loops screwed (#10 x 3/4") to the inside faces of Panel B's bottom rails**
- SPARE ACCESS (emergency-only, no tools — 4 steps on the render): 1. lift the mattress + platform slats off; 2. lift out the 2 totes, still packed; 3. pop the cam buckle, lift out the tool case; 4. tilt the spare upright and roll it over Panel C's deck, out the gate
- NOTE: the spare does NOT fit above the kitchen (28.5" disc vs ~24" before the fridge blocks it) — Panel B is the only interior spot, and it's the best one: ~40 lb at the axle beats any hitch basket (which would lever ~130+ lb onto the rear axle)



Exploded stack (top -> bottom)

- 1 Spare, FLAT, valve stem UP (pressure checks in place)
- 2 Cradle skid: 3x 3" cleats on 2 battens (offcut), one piece
- 3 Tool case (2-ton jack + wrenches) nested in the wheel barrel
- 4 2x Sterilite totes restacked on top
- 5 1" cam strap over the whole stack
- 6 2x footman loops, #10x3/4" into the bottom rails (strap anchors)

Access (emergency-only, no tools)

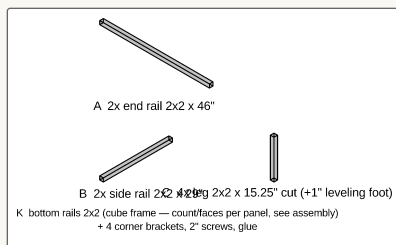
- 1 Lift the mattress + platform slats off
- 2 Lift out the 2 totes (they stay packed)
- 3 Pop the cam buckle, lift out the tool case
- 4 Tilt the spare upright, roll it over Panel C, out the gate

SPARE TIRE — flat in Panel B's bay at the AXLE (Component 3), exploded stack

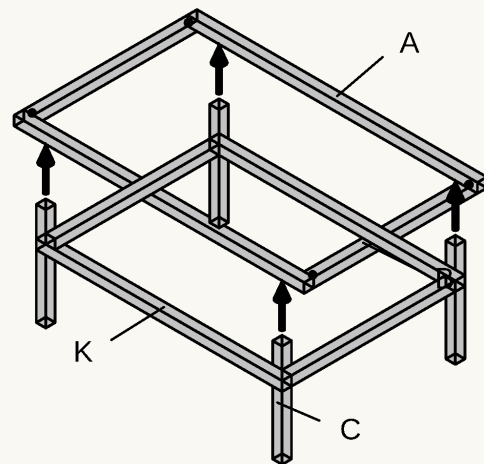
Best weight placement in the van — no hitch basket needed. NOTHING screws to the van floor: the skid sits on the liner, the strap ties to footman loops on the bottom rails.

Height stack: cleats 3" (clear the 2.5" curb) + spare 6.4" + tote 5.9" — fits under the 17.75" deck. Kitchen-drawer slot does NOT fit the 28.5" disc; the drawer stays.

Spare tire stowage



1

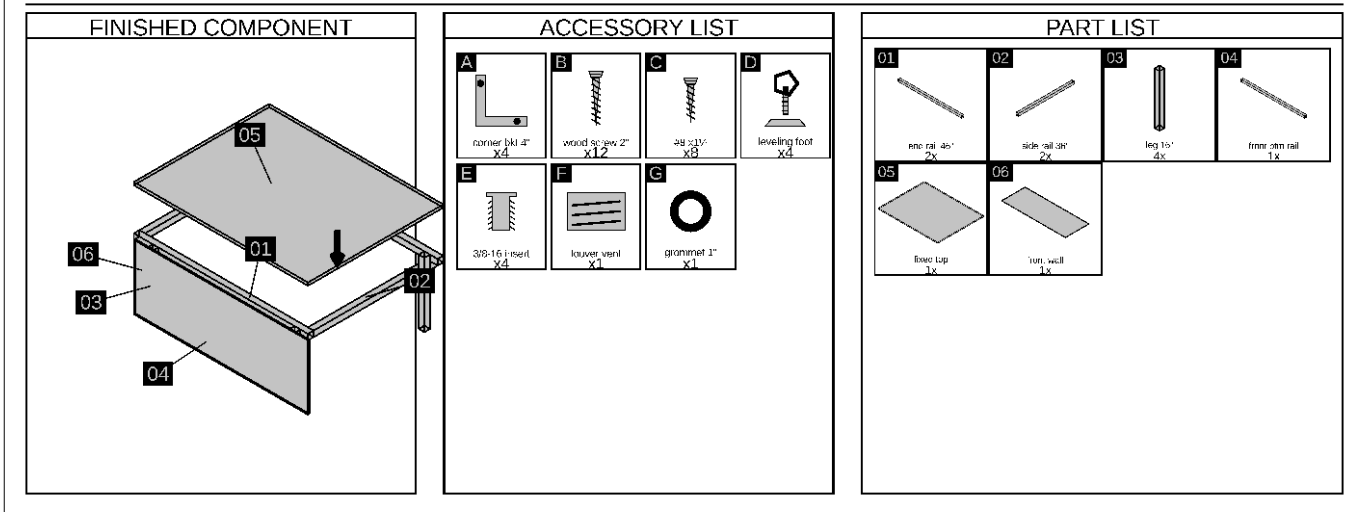


K: bottom rails (2x2, underside at 1") close the frame into a box — all 4 faces (full cube)

Build the frame — like Panel A's step 1: side rails (B), end rails (A), corner brackets, 4 legs (C) inset 3.5" from the deck edges (V4, Aug 2026 — was 2.5"; drilled for their leveling inserts BEFORE assembly), the 2 diagonal corner braces, and bottom rails on ALL FOUR faces (2x 46" + 2x 26", underside 1" up at the leg bottoms) — ****the full cube****: nothing exits Panel B sideways, so every face can close, and this is the stiffest frame of the three. That's the entire panel — no divider, no drawers, nothing else. Drill the alignment-pin holes when mating it to its neighbors (Component 5).

Component 4: Panel C

COMPONENT 4 — PANEL C

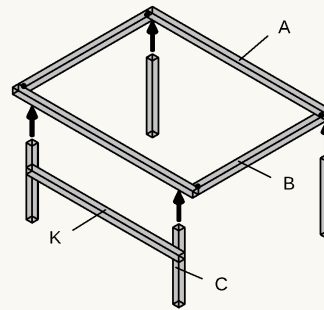
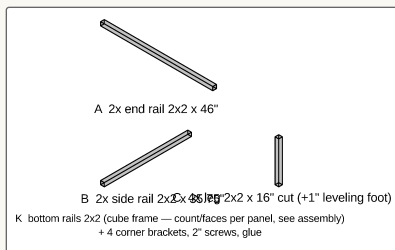


Build this one before Component 1 (Rear Pantry) — the prefab drawer cluster sits on this panel's finished deck at the tailgate end (cleat pocket + cam strap — no fasteners into the units), with no frame or top of its own.

Parts needed:

- Panel C top — 1, 33" x 43", 3/4" ply (drops BETWEEN the rails) + 4 bearer cleats (3/4" x 3/4": 2 x 33" + 2 x 40", from the outcut)
- Long rails — 2, 35.75" cut, 2x2 pine
- End rails — 2, 46" cut, 2x2 pine
- Legs — 4, 16" cut, 2x2 pine (leveling feet make up the inch; REAR pair at the true corners)
- Bottom rail — 1, 46" cut, 2x2 pine (FRONT face only)
- Corner brackets — 4
- Front wall — 1, 46" x 17", 3/8" ply, with the 120mm fan hole + two 1" grommet holes + the 7" x 2.5" low intake louver pre-cut (Panel C Front Wall render — every opening dimensioned)
- Louvered RV vent — 1 snap-in (low intake in the front wall)
- No divider, no drawers — the void stays fully open for Component 7 (Fridge & Kitchen Install)

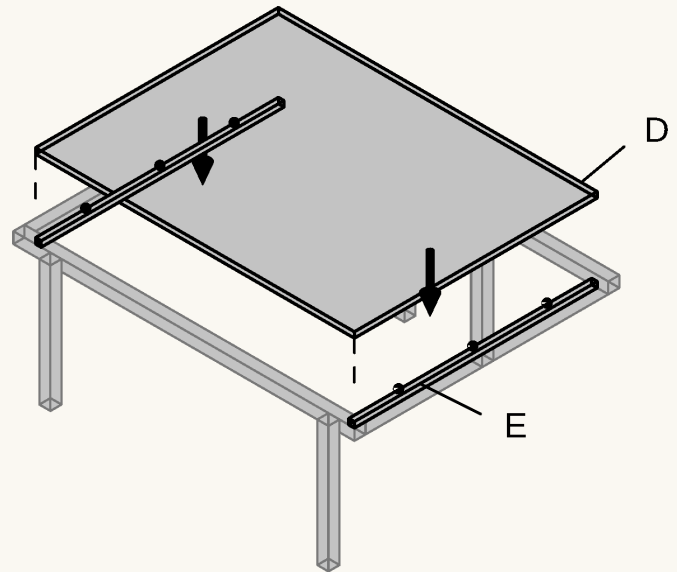
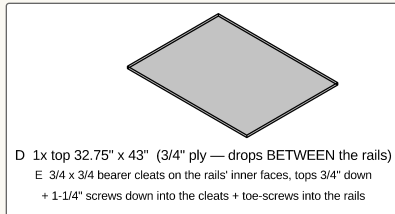
1



K: bottom rails (2x2, underside at 1") close the frame into a box — front face only (appliances exit the tailgate; rear legs at the TRUE corners)

Build the frame — same construction as Panels A/B, with 35.75" side rails (B): end rails (A), corner brackets, 4 legs (C) — FRONT pair inset 3.5", REAR pair at the TRUE corners (the fridge/kitchen slide paths pass exactly where inset rear legs would stand; the rear-corner floor vents were checked Aug 2026 and do NOT reach the leg area, so those legs land on solid floor). Add the FRONT-face bottom rail (46", underside 1" up at the leg bottoms) — the tailgate face stays open for the appliances.

2



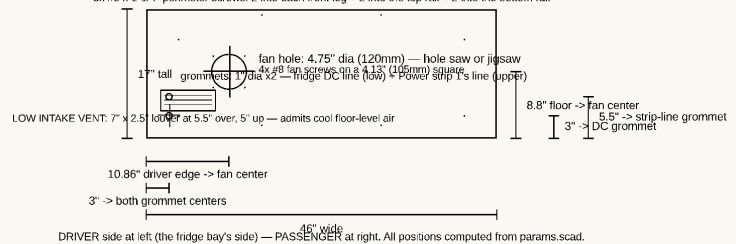
no divider — void stays open for the fridge + kitchen unit

Screw the bearer cleats (E) to the rails' inner faces — cleat tops exactly 3/4" below the rail tops — then drop the top (D) BETWEEN the rails onto them, flush with the rail tops (the deck recess: 3/4" more headroom), and screw it down into the cleats + toe-screw into the rails. NO divider, since the fridge and kitchen unit need the full width of the void underneath.

3

front wall: 46" x 17" 3/8" ply, holes pre-cut per the Panel C Front Wall render

PANEL C FRONT WALL — 3/8" ply, flat pattern (the ONLY wall on any panel)
 Mounts on Panel C's front (B-facing) face, floor to rail underside. Intake fan bolts over the big hole (blows IN); the low louver is a passive cool-air scoop below it.
 Panel A/B: no walls or skirts anywhere. Panel C sides: open. Panel C tailgate face: no wall — fridge + open utility bay + kitchen + kitchen drawer fill it.
 8x #8 x 1-1/4" perimeter screws: 2 into each front leg + 2 into the top rail + 2 into the bottom rail

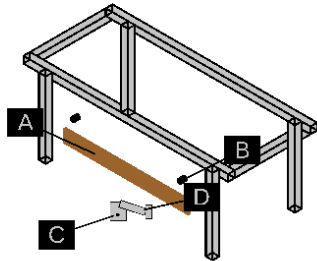


Cut the front wall and its openings to the render's dimensions — the 4.75" (120mm) fan hole centered on the fridge bay (10.86" from the driver edge, 8.4" up), one 1" grommet hole at 3" in, 3" up (fridge DC line — Power strip 1 feeds off the rear outlet, so this wall no longer passes its line), and the 7" x 2.5" LOW INTAKE LOUVER in the driver-side corner (5.5" over, 5" up) — a passive cool-air scoop that feeds the fridge the coolest floor-level cabin air, screened with a snap-in RV louver vent — then screw it to the front (B-facing) face: 2x #8 x 1-1/4" into each front leg + 2 into the top rail + 2 into the bottom rail (8 total). This is the ONLY wall on any panel; the intake fan bolts over the fan hole in Component 7. Panel C's sides stay open and its tailgate face is fully occupied by the fridge, the open utility bay, the kitchen unit, and the kitchen drawer.

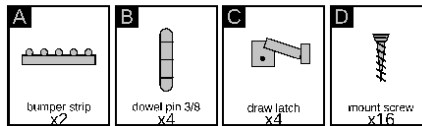
Component 5: Anti-Rattle Bumpers, Alignment Pins & Seam Draw-Latches

COMPONENT 5 — BUMPERS, ALIGNMENT PINS & SEAM DRAW-LATCHES

FINISHED COMPONENT



ACCESSORY LIST



PART LIST

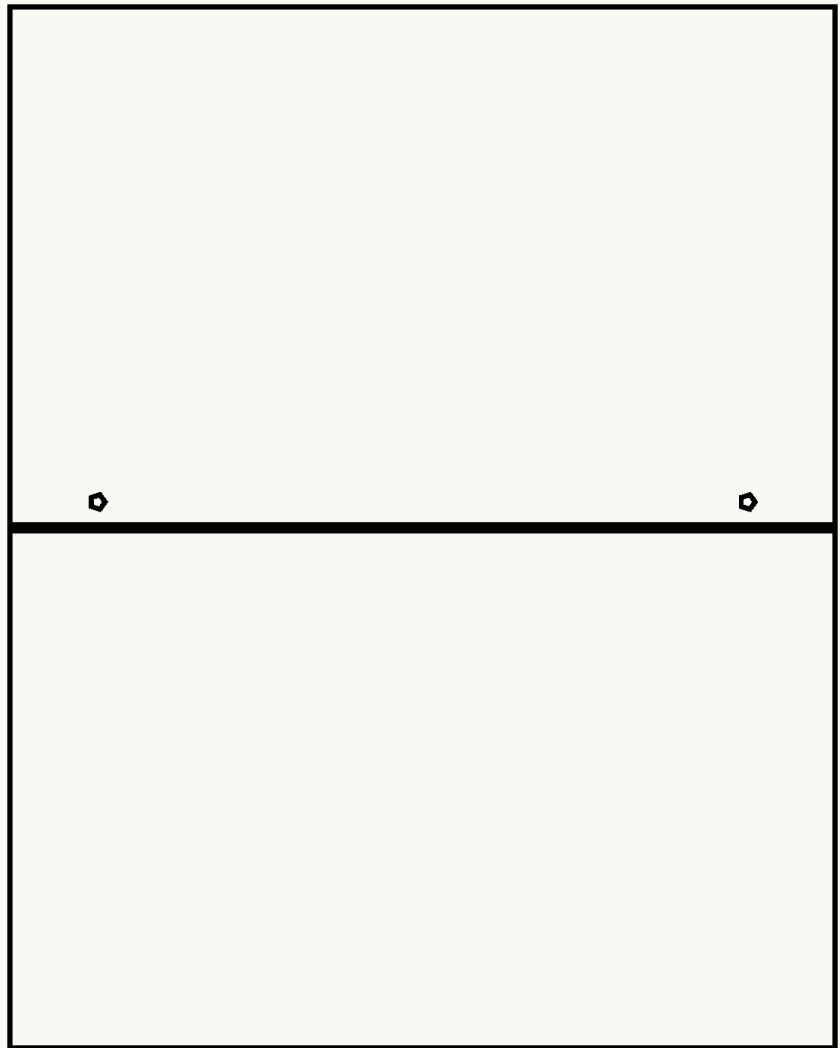
(no cut parts)
hardware mounts to existing panels

Parts needed:

- Anti-rattle bumper strips — 2 (adhesive-backed felt or closed-cell foam weatherstrip), one per seam
- Alignment dowel pins — 4 (2 per seam), 3/8" dowel rod, ~3/4" long
- Over-center draw latches — 4 (2 per seam × 2 seams), stainless/zinc, with their keepers and mounting screws

1

1 bumper strip + 2 dowel pins per seam (A/B seam and B/C seam)



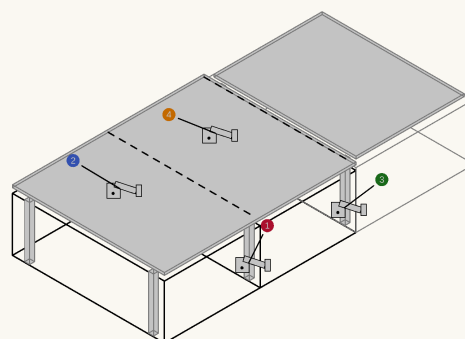
Bumper strip + 2 alignment pins

at every module seam (A/B shown)

At each seam, apply an adhesive bumper strip across the full seam face on one side of the joint, then drill matching 3/8" holes and friction-fit 2 alignment dowel pins — the panels register in the same spot every time they're lifted out and reinstalled, and don't rub or squeak in transit.

2

2 over-center draw latches per seam (one each side), mounted low on the bottom-rail band



DETAIL — one latch, two legs, bumper between base plate on one leg, hooked rail + base on the other. Flip to clamp. Flip to release (no tools)

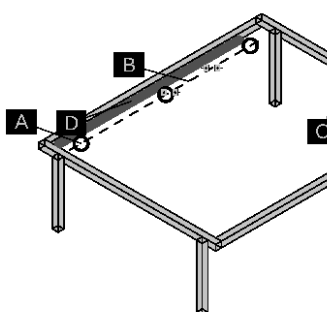
SEAM DRAW-LATCHES — clamp the 3 lift-out modules into one rigid beam (Component 5)
4 latches total: one each side of BOTH the A/B and B/C seams, low on the bottom-rail band (~1.75" up, hand-reach)
Alignment pins LOCATE the modules; the latches PULL them tight against the bumper strip. Hand-released — panels still lift out in seconds.
(front seats upper-left, tailgate lower-right; the bed platform already has the Panel A + B tops)

The alignment pins only *locate* the modules; the draw latches *clamp* them together. Mount one over-center draw latch on each **SIDE** of each seam (4 total: both sides of A/B and B/C), **low on the bottom-rail band (~1.75" up), over the inset leg line** — hand-reachable from the side door / tailgate. The base plate screws to one module's leg/rail, the hooked bail + lever to its neighbour's; flip the handle to pull the two modules tight against the bumper strip, flip to release. This ties the three boxes into one rigid beam: the bed platform already couples the Panel A + B tops, and a low latch completes a top-and-bottom couple that kills sway and rattle. It stays fully modular — the latches are hand-released, so any panel still lifts out in seconds. **Honest scope:** this buys rigidity and quiet, not a higher weight rating — the cube-framed boxes are already strong (5× margin at the feet). Don't rely on the latches to carry more load; their job is to stop relative motion between modules.

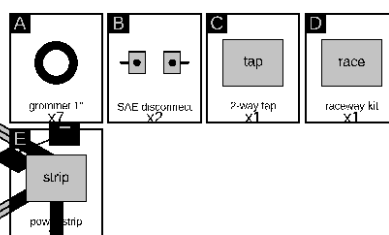
Component 6: Cord Runs — Cooktop, Power Strip 1, Fridge DC & DELTA 3 Charging

COMPONENT 6 — CORD RUNS (COOKTOP / STRIP 1 / FRIDGE DC / DELTA 3)

FINISHED COMPONENT



ACCESSORY LIST



PART LIST

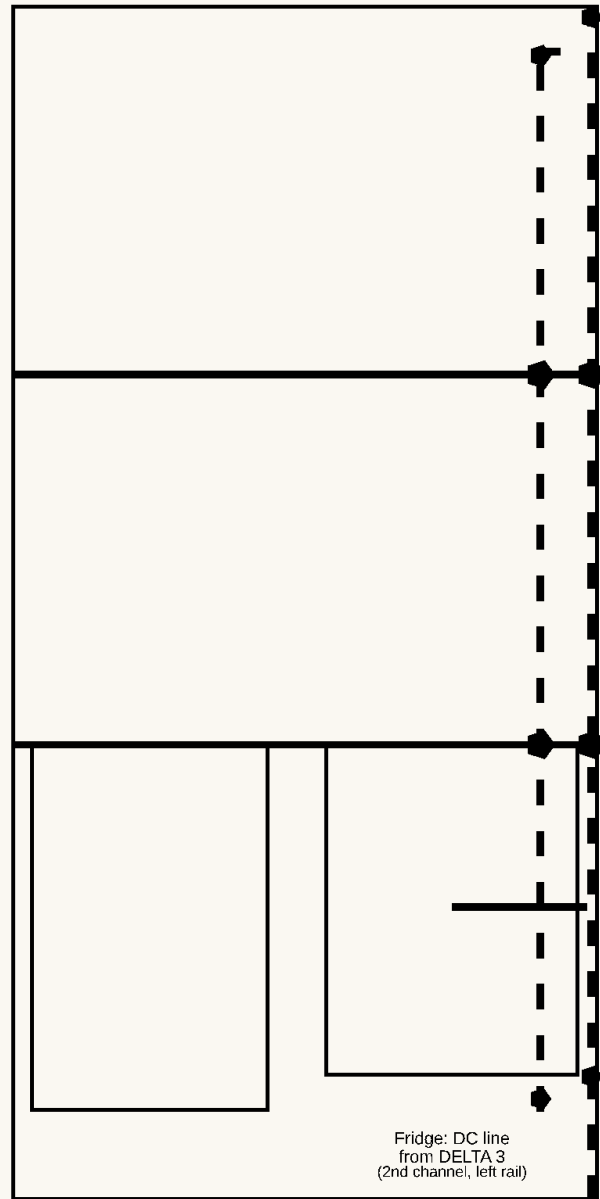
(no cut parts)
cords, grommets & connectors only

Parts needed:

- 16 AWG extension cord (household AC-rated), short (cooktop → REAR outlet, + slack loop for the slide)
- 16 AWG extension cord (household AC-rated), short (Power strip 1 → REAR outlet beside the pantry)
- 16 AWG DC cord (automotive/solar SAE-rated), ~8ft (fridge line — DELTA 3 in Panel A back to the fridge in Panel C)
- 16 AWG cord (household AC-rated), ~3-4ft (DELTA 3 AC charging line — front console to Panel A)
- Cable grommets — 7, 1" diameter (1 cooktop line, 1 Power strip 1 line, 3 fridge DC line, 2 DELTA 3 charging line)
- SAE quick-disconnect connectors — 2 (DC-rated, for the fridge DC line's 2 seams — the **ONLY** line that still crosses seams; both AC tailgate loads plug into the rear outlet)
- Fused (15A) car-plug → SAE adapter (P1, DELTA 3 end) + SAE → female 12V socket pigtail (P7, fridge end)
- 18AWG red/black hookup wire ~10 ft, crimp assortment (#10 heat-shrink rings + 0.25" female spades), 2× WAGO 221-413 lever nuts
- Per-module surface wire raceway (AC + DC separated) + screw-mount cord clips — 1 set
- 2-way AC outlet tap — 1 (REAR outlet: Power strip 1 + the cooktop share it — see Section 5)

1

cooktop's short cord + 1 grommet + slack loop (parts list above)



Power channels: cooktop (right rail) + fridge DC line (left rail)

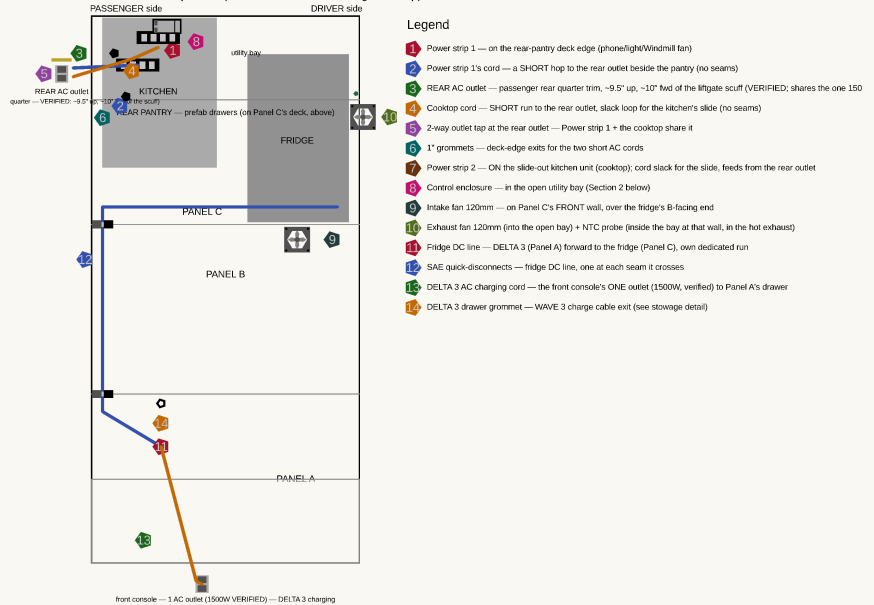
cooktop: kitchen unit -> Panel C -> B -> A -> console outlet

fridge: DELTA 3 (Panel A) -> B -> C -> fridge DC input

(large circles = quick-disconnect at each panel seam)

Drill the single grommet at the kitchen unit's corner post and run the cooktop's short cord to the REAR AC outlet in the back passenger area (a few feet — no seams, no inline connectors). Coil a slack loop sized to the kitchen's full ~70" slide extension near the stowed position so the cooktop keeps power at the cook position. Power strip 2 rides ON the slide-out unit. The outlet is at **22.5" up** (i.e. just above the deck plane), ~10" forward of the liftgate scuff on the passenger quarter trim, socket centre 10" in from the sidewall (MEASURED V9b, Aug 2026) — position the grommet to line up with it.

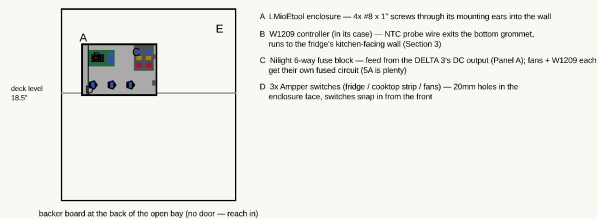
ELECTRICAL LAYOUT — top-down (front of van at bottom, tailgate at top)



2

Power strip 1's short cord + 1 grommet + the rear outlet's 2-way tap (parts list above)

CONTROL CLUSTER — in the open utility bay on its backer board, elevation (looking forward from tailgate)



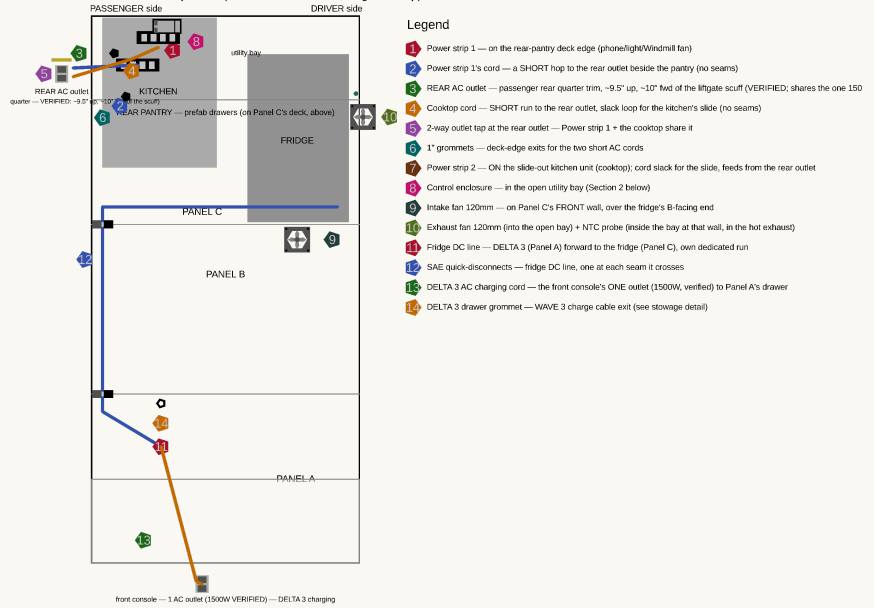
FAN MOUNTING — face-on, both fans identical hardware



Cut a 4.5" round hole (or 4.5" square) in the wall; fan mounts over it with 4x #8 x 1-3/4" screws through the corner holes. Grit goes on the AIR-ENTRY side. Arrow on the fan's side frame shows its blow direction — point it per the labels above. Wiring: both fans in parallel off the W1209 relay (Section 5's wiring schematic).

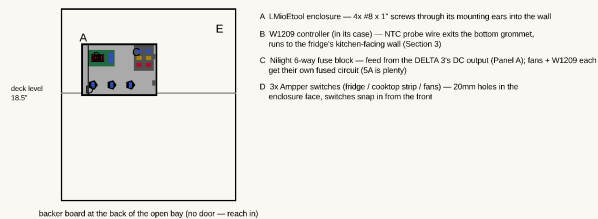
Power strip 1 mounts on the rear pantry's deck edge in the open bay (Component 1) — directly beside the REAR AC outlet. One grommet at the deck edge, one short cord to the outlet's 2-way tap (shared with the cooktop), ~1 ft of slack so lifting the pantry's drawer units never needs an unplug. The old full-length run to the front console — and its 2 inline seam connectors — are gone.

ELECTRICAL LAYOUT — top-down (front of van at bottom, tailgate at top)



3
fridge DC cord + fused car-plug → SAE adapter + SAE → 12V-socket pigtail + 3 grommets + 2 SAE disconnects (parts list above)

CONTROL CLUSTER — in the open utility bay on its backer board, elevation (looking forward from tailgate)



FAN MOUNTING — face-on, both fans identical hardware



Cut a 4.5\"/>

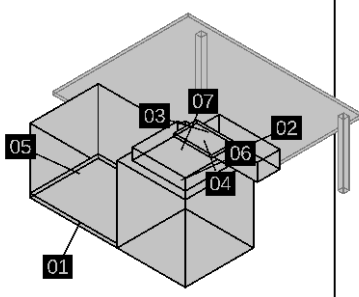
The fridge now runs off the DELTA 3, not the van's rear outlet (Section 1) — route its DC line from the DELTA 3's output (Panel A's right drawer) back through Panel B into Panel C, with an SAE quick-disconnect at each of the 2 seams it crosses, landing on the Nilight fuse block's main studs in the utility bay and branching from there to the fridge's DC input — every wire landing is numbered P1–P12 on the wiring diagram and in Section 5's connection-point table.

4
DELTA 3 charging cord + 2 grommets (no disconnect needed)

Run the DELTA 3's own AC charging cord from Panel A's drawer forward, through the open floor gap, to the front console's ONE outlet (1500W verified) — zero seams. It has that outlet to itself now; the cooktop and Power strip 1 live on the rear outlet. All AC loads still share the van's single 1500W inverter budget (Section 5).

Component 7: Fridge & Kitchen Install

COMPONENT 7 — FRIDGE & KITCHEN INSTALL

FINISHED COMPONENT	ACCESSORY LIST	PART LIST																				
	<table border="1"> <tr> <td>A  L-track + stud x7</td> <td>B  bolt + T-nut x30</td> <td>C  ratchet strap x7</td> <td>D  120mm fan x2</td> </tr> <tr> <td>E  PWM controller x1</td> <td>F  toggle switch x3</td> <td>G  fuse block x1</td> <td>H  magnetic catch x1</td> </tr> <tr> <td>I  lower vent x1</td> <td>J  slide 24\" x2</td> <td>K  wood screw x16</td> <td></td> </tr> </table>	A  L-track + stud x7	B  bolt + T-nut x30	C  ratchet strap x7	D  120mm fan x2	E  PWM controller x1	F  toggle switch x3	G  fuse block x1	H  magnetic catch x1	I  lower vent x1	J  slide 24\" x2	K  wood screw x16		<table border="1"> <tr> <td>01  fridge tray 1x</td> <td>02  kitchen drawer 1x</td> <td>03  drawer cheek 2x</td> <td>04  control packer 1x</td> </tr> <tr> <td>05  Rocky 40 fridge 1x</td> <td>06  kitchen unit 1x</td> <td>07  cooktop 1x</td> <td></td> </tr> </table>	01  fridge tray 1x	02  kitchen drawer 1x	03  drawer cheek 2x	04  control packer 1x	05  Rocky 40 fridge 1x	06  kitchen unit 1x	07  cooktop 1x	
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01  fridge tray 1x	02  kitchen drawer 1x	03  drawer cheek 2x	04  control packer 1x																			
05  Rocky 40 fridge 1x	06  kitchen unit 1x	07  cooktop 1x																				

Parts needed:

- BougeRV Rocky 40 (41QT dual-zone) + fridge tray (3/8" ply + glued 3/4" edge frame, 17.72" x 28.74")
- Heavy-duty 24" full-extension drawer slide pair — VADANIA #D76 locking, 379 lb (purchased)
- JAGAHABA slide-out camp kitchen + COOKTRON induction cooktop
- 120mm 12V fan + PWM temperature controller w/ NTC probe
- 12V surge protector/fused distribution block, 3 illuminated toggle switches, small electrical enclosure
- Anchor board (3/4" ply — ONE 46" x 33" comb: bridge + 3 strips continuous, no wood joints, Section 8) + 2 steel bearing tongues (2" x 3/16" flat bar) + non-slip mat + 1/4-20 machine screws & T-nuts
- Aluminum L-track offcuts + 7 stud fittings w/ D-rings (WLL 1,333 lb)
- 7 Erickson 400lb-WLL ratchet straps (4 kitchen criss-cross + 3 to the striker loops)
- 1 low-profile cam strap + 2 screw-eye D-rings (fridge hold-down, hooks to its 2 end handles)
- 3 screw-mount cable clips (fridge DC line slack, clipped to the slide's fixed outer rail)
- Utility bay fit-out — 2 low-profile bins + hook-and-loop tape (no door: the bay between the fridge and kitchen stays open)
- Kitchen drawer — box (1/2" ply, 18" x 26" x 4.5") + 2 hanging cheeks (1/2" ply, 26" x 5.45" — weight swap: was 3/4") + 24" full-extension slide pair (100lb) + magnetic catch

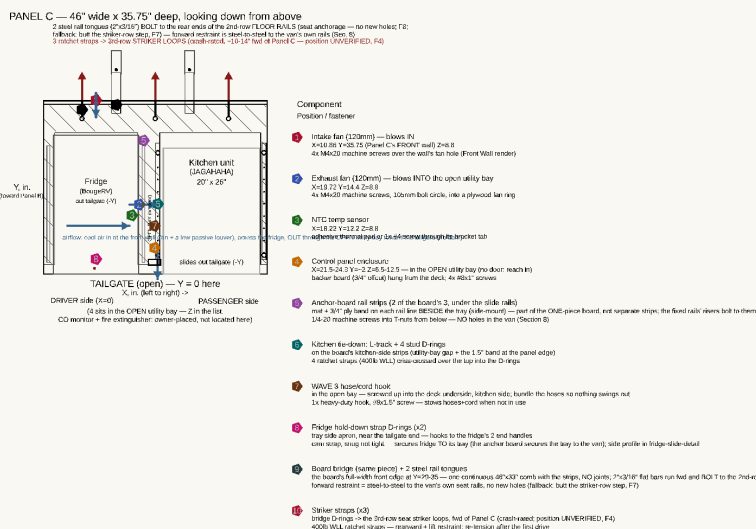
1

no parts — inspection only

The Appendix A floor survey (F1–F8) is DONE (Aug 1 2026) and it changed this component: all 3 striker loops ARE exposed and hookable with the 3rd row folded (~11" in from each sidewall, ~13" apart, 1.5" of clear height), the 2nd-row rail rear ends sit **42" from the hatch** so cut the tongues to that, and the load surface is folded seatback across the whole footprint — solid enough to bear on, so **no filler blocking** is needed. Two fallbacks are gone: there is no hard step face at the striker row (F7 — carpet over a soft ~1" step) and the 3rd row's seat-mount bolts are occupied (F6), so the rail ends plus the 3 straps carry the load. ALSO: the hybrid's 12V battery access panel sits at the passenger rear corner (under the rear AC outlet) — the kitchen-side strip stops short of it so the panel stays serviceable. Nothing gets drilled into the van at any point in this component.

2

anchor board + 2 steel tongues + L-track + 7 stud fittings + mat + 3 ratchet straps

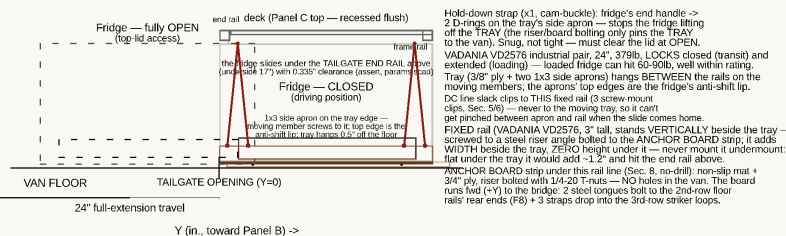


Bench-build the anchor board (Section 8): cut the comb as ONE piece — bridge and all three strips continuous, so there is no glue and no lap screws anywhere in it (grain along the strips, ~1/2" fillet at each inside corner) — then counterbore and seat the T-nuts flush from below, bolt on the riser angles, L-track lengths, bridge D-ring fittings, and the 2 steel tongues (blue threadlocker on every screw; cure ~24h). Set it in the van on its non-slip mat, bolt/clamp the 2 tongues to the rear ends of the 2nd-row floor rails (fallback: butt them against the striker-row step, F7/F8), notch Panel B's rear bottom rail where the tongues/straps cross under it, then hook and tension the 3 ratchet straps from the bridge D-rings into the 3rd-row striker loops. No holes in the van — the render above draws the board at the ASSUMED striker geometry; cut the final outline only after the Appendix A survey.

3

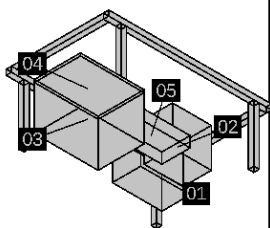
fridge + 3/8" ply tray (edge frame) + 24" 200lb slide pair

Fridge Slide Mechanism — side view (one of 2 parallel slides shown)
CLOSED (solid) vs fully OPEN (dashed) — pulled out through the tailgate for top-lid access



Mount the Rocky 40 on its plywood tray hanging BETWEEN the slide pair — rails vertical against the tray's 1x3 side aprons (side-mount, nothing under the tray); each fixed rail's steel riser angle is already bolted to the anchor board (step 2 — the board raises the rails ~0.85"; screw the moving members to the aprons lower in their overlap so the tray keeps its 0.5" hang), against Panel C's driver-side rear corner leg (1.5" in from the edge), pulling straight out the open tailgate between the corner legs. Face the compressor/battery end toward the tailgate (B240 swaps without a full slide-out), re-hinge the reversible lid to open toward the van wall, and let the unit stand 12 hours before first power-up (manual). Screw 2 D-rings into the tray's side apron level with the fridge's 2 end handles, and hook a low-profile cam strap over the top from handle to handle — snug enough to stop the fridge lifting off the tray, loose enough that it doesn't bind the lid at full slide extension (test this before loading the fridge). Connect its DC input to the fridge line routed in from the DELTA 3 (Component 6, step 3) — leave enough slack for the slide's full travel, and clip that slack loop to the FIXED outer rail (Component 6, step 5) so it can't get pinched when the tray slides home.

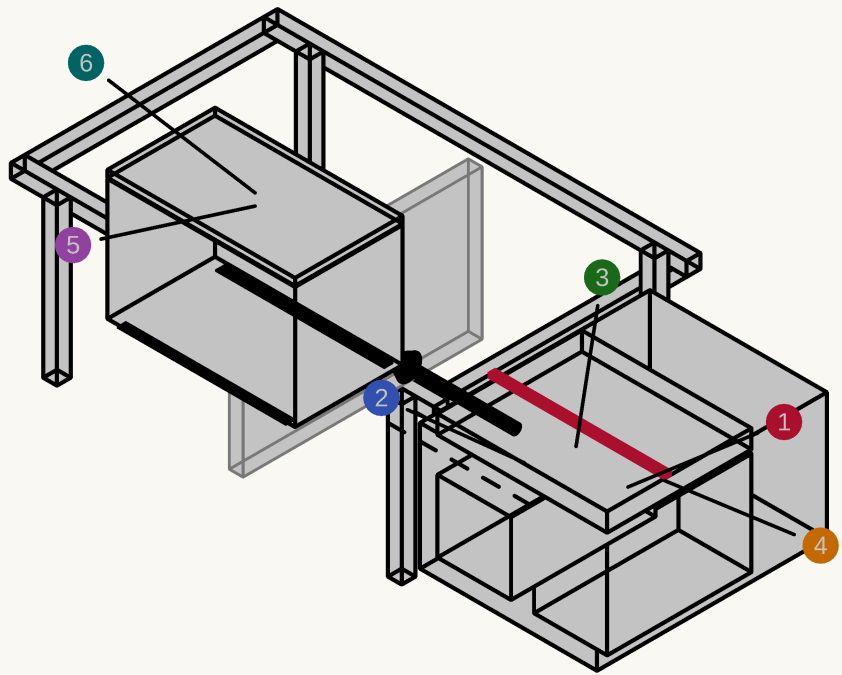
COMPONENT 8 — ECOFLOW STOWAGE (DELTA 3 + WAVE 3)

FINISHED COMPONENT	ACCESSORY LIST	PART LIST																
	<table border="1"> <tr> <td> A  hose/cord hook x1 </td> <td> B  non-slip mat x1 </td> <td> C  Car Vent. Kit x1 </td> <td> D  1x1 cleat x2 </td> </tr> <tr> <td colspan="4"> E  utility bin x2 </td> </tr> </table>	A  hose/cord hook x1	B  non-slip mat x1	C  Car Vent. Kit x1	D  1x1 cleat x2	E  utility bin x2				<table border="1"> <tr> <td> 01  DELTA 3 Plus 1x </td> <td> 02  extra battery 1x </td> <td> 03  WAVE 3 1x </td> <td> 04  overhead shelf 1x </td> </tr> <tr> <td colspan="4"> 05  drawer tray 1x </td> </tr> </table>	01  DELTA 3 Plus 1x	02  extra battery 1x	03  WAVE 3 1x	04  overhead shelf 1x	05  drawer tray 1x			
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01  DELTA 3 Plus 1x	02  extra battery 1x	03  WAVE 3 1x	04  overhead shelf 1x															
05  drawer tray 1x																		

Parts needed:

- EcoFlow DELTA 3 Plus + Smart Extra Battery, EcoFlow WAVE 3 (already owned, not priced)
- WAVE 3 hose/cord storage hook
- Non-slip mat (camp-use point of use — Panel C's deck or the front seat, not the storage bay)
- EcoFlow WAVE Series Car Vent Kit (no-tent sleeping configuration only)
- FOUND STORAGE: DELTA 3 drawer top tray (shallow lift-out bin) + WAVE 3 overhead shelf (1/2" ply + 1x1 cleats, from offcut) + 2 utility-bay bins (velcroed — Component 7 step 5)

Battery placement — low and counterbalanced (owner review, July 2026). Heavy batteries want to sit **low and centered**, and this design already gets both most of the way there: the DELTA 3 stack (~48 lb) rides in Panel A's floor-level under-deck drawer — about as **low** as it can go. On the **lateral** axis it looks off to one side (passenger), but the ~34 lb WAVE 3 sits in the **opposite** (driver) bay of the same Panel A, so the two heaviest movable items counterbalance each other — the whole build's lateral center of mass ends up only **~1.1" off centerline** (Appendix F), which is trivial on a ~4,500 lb van. So the "get it centered" concern is effectively already answered. A longer drawer sliding the DELTA to the true centerline was considered and **not adopted**: a 48 lb load cantilevered at the end of a long slide is tippy when open and hard on the hardware, it fouls the bed-platform cantilever near Panel B, and it wouldn't free clean storage (the drawer sweeps its own travel path). If you still want to nudge it inboard for free, shift Panel A's center divider a couple inches toward the driver side to re-center the DELTA bay. See **Appendix F** for the full weight-distribution analysis.



1

uses Panel A's right-drawer fixtures (Component 2, step 4)

DELTA 3 STACK (right drawer) + WAVE 3 (left bay)

Panel A, exploded — Components 2 & 8

DELTA 3 Plus outboard (the WAVE 3 plugs into it), Extra Battery inboard; ~48 lb in a normal drawer — no E-track. The WAVE 3 rests in the raw bay on 2 glide strips.

Found storage: a ~2.6" lift-out tray tops the DELTA stack, and a shelf @ 13.7" tops the WAVE 3 — it still slides out beneath.

Component / position

- 1 DELTA 3 Plus — 7.95"x15.7"x11.16", ~28 lb outboard (pull wall); cam-strapped over locating cleats
- 2 Smart Extra Battery — 8"x15.7", ~20 lb inboard — extra runtime only, grabbed less often
- 3 Lift-out tray — reclaims ~2.6" dead air cables, the DELTA's own cords, dongles
- 4 Cam strap + D-rings + 1" grommet strap over both units; WAVE 3 charge cable exits the grommet
- 5 WAVE 3 — 20.4"x11.7"x13.2", 33.7 lb raw bay, 0.35" clear — a boxed drawer would NOT fit
- 6 Overhead shelf @ 13.7" on 1x1 cleats flat soft goods + the WAVE 3's own hoses ride up here

Stow the DELTA 3 Plus outboard (pull wall — it's what the WAVE 3 plugs into) and the Smart Extra Battery inboard, side by side over the locating cleats, cam strap over both. Connect the fridge DC line (via its fused car-plug → SAE adapter — connection point P1) and the AC charging cord (both routed in Component 6) to the DELTA 3 Plus's 12V car-power outlet and AC input port — it now powers the fridge/fans continuously and gets charged from the front console while driving (Section 1).

2

uses Panel A's left-bay glide strips (Component 2, step 3) + storage hook

Slide the WAVE 3 into Panel A's left bay, resting on the 2 glide strips — no box, no slide hardware, just reach in through the driver's side door. Mount the hose/cord hook in the open utility bay (screwed up into the deck underside at the bay's kitchen side; bundle the hoses with their own strap so nothing swings out of the open bay) for when the unit is carried off to run at camp. **Found storage** (DELTA 3/WAVE 3 detail render, side sections): the DELTA 3 stack is only 11.16" tall in a 13.75" drawer, so drop a shallow lift-out tray in the ~2.2" of dead air on top for cables/cords; and the WAVE 3 is 13.2" in the 16.25" bay, so screw a thin shelf onto 1x1 cleats at ~13.7" — flat soft goods and the WAVE 3's own hoses ride up there while the unit still slides out beneath it. Neither touches the frame or the units' function.

3

tent config: non-slip mat only · no-tent config: non-slip mat + WAVE Car Vent Kit

Tent (tailgate open): carry the WAVE 3 from Panel A's bay to Panel C's deck, just forward of the rear-pantry cluster (which occupies the tailgate-most 14"), on the non-slip mat, and route its hoses past the pantry and through the open tailgate gap — no window seal needed. **No tent** (tailgate closed): carry it instead to the front passenger seat, on the same mat, and install the Car Vent Kit over a cracked front window for the hoses.

Component 9: Mattress

COMPONENT 9 — MATTRESS

FINISHED COMPONENT	ACCESSORY LIST	PART LIST
	A glue spray adhesive x1 B cover waterproof cover x1	01 HEST (DIY) 1x 02 firm base (DIY) 1x 03 memory top (DIY) 1x

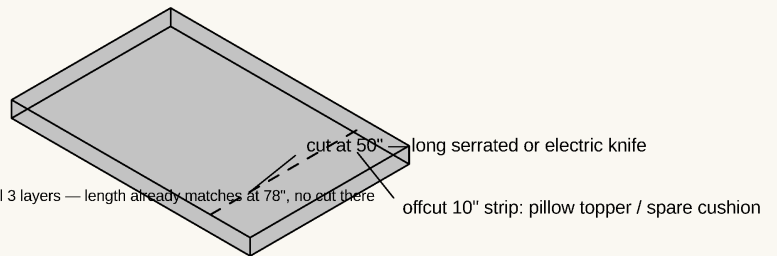
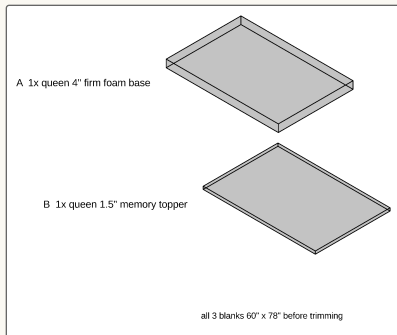
Primary path — BUY, no build: the HEST Dually Long (78" x 50" x 4") drops onto the platform as-is: solid foam, no inflation, washable waterproof cover included. Heads go at the rear-pantry end. Done.

Budget fallback — DIY 2-layer build (the steps below): same 50" x 78" footprint, ~\$185–255 all-in, 5.5" thick (costs ~1.5" of the headroom the HEST keeps).

Fallback parts needed:

- Queen 4" firm high-density foam base + queen 1"-2" memory foam topper
- Spray adhesive (or large fabric strips)
- Waterproof mattress cover

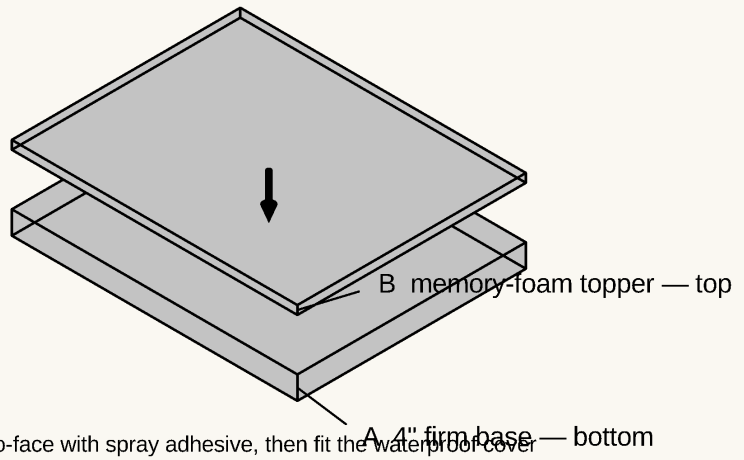
1



FALLBACK ONLY (skip if you bought the HEST). Buy both foam layers at queen size (60" x 80") and trim to 50" wide x 78" long with a long serrated or electric knife; the offcut strips work as a pillow topper or spare cushion.

2

the 2 trimmed layers from step 1
+ spray adhesive (or wide fabric strips)
+ waterproof cover, fitted after gluing



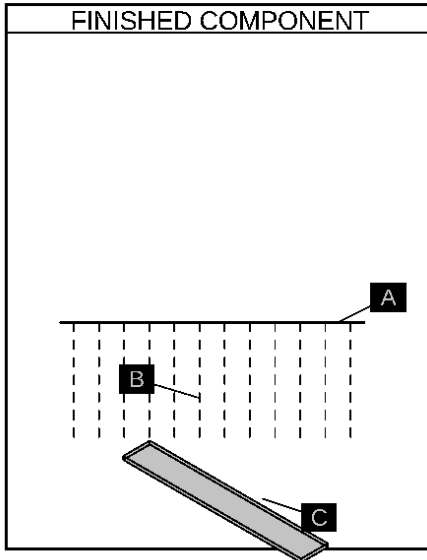
glue face-to-face with spray adhesive, then fit the waterproof cover

FALLBACK ONLY. Stack the firm base (A) with the memory-foam topper (B) on top; bond face-to-face with spray adhesive, fit the waterproof cover, and lay it on the platform.

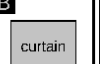
Component 10: Final Assembly, Curtain & Test Fit

COMPONENT 10 — FINAL ASSEMBLY, CURTAIN & TEST FIT

FINISHED COMPONENT



ACCESSORY LIST

A	B	C	D
			
tension rod x1	blackout curtain x1	wood sealant x1	sandpaper x1

PART LIST

(no cut parts)
finishing + test-fit only

Parts needed:

- Tension rod + blackout curtain fabric
- Wood sealant/varnish, sandpaper/misc finishing supplies

Steps:

1. Sand and apply 2 coats of wood sealant/varnish to all plywood surfaces (especially edges) to prevent moisture damage.
2. Install the curtain rod (tension rod + blackout fabric) between the front seats and the platform area for privacy.
3. Test fit everything: confirm the cooktop's and Power strip 1's cords reach the REAR outlet, the DELTA 3's charging cord reaches the front console outlet, confirm the fridge's DC line reaches the DELTA 3 in Panel A with its disconnects seated at both seams, and confirm the tailgate closes with everything loaded.
4. Do a full lift-out/reinstall dry run on Panel A, Panel B, and Panel C — gripping each by its exposed top rails, each should come free and drop back in without binding, rubbing, or forcing a bumper/pin out of place. For **Panel A and Panel B specifically**, also test lifting the one-piece bed platform off the box rails first (they no longer have a top or lid of their own — the platform is the only thing capping them), then confirm each panel's drawers pull smoothly once it's clear. For **the rear pantry specifically**, loosen the cam strap and confirm each drawer unit lifts off Panel C cleanly and drops back into its cleat pocket.

No diagram — finishing and test steps, not new construction.

7. Induction Cooktop & Power Strip Design

Recommended cooktop: COOKTRON Portable Induction Cooktop 2 Burner w/ Removable Iron Cast Griddle Pan

- **Real measurements confirmed from the JAGAHABA's own listing photos:** the stove tray is **23"L x 15.7"W, with 5.7" of clearance** when pulled out.
- The **JAGAHABA kitchen unit is built for a 2-burner stove** (per its own listing) — a single-burner cooktop would leave half its designed stove space unused.
- An earlier draft of this plan recommended the **Duxtop 9620LS dual burner** — its manufacturer manual gives its actual footprint as **23.9"L x 14.2"W x 2.4"H**, which is **0.9" too long for the 23" tray**. Dropped in favor of the option below.
- **Confirmed exact product** (real listing photos with dimension callouts): **18.1"L x 9.1"W x 4.3"H** (briefcase-style unit with a fold-down handle) — fits the tray with real margin: ~4.9" of length, ~6.6" of width, ~1.4" of height to spare. Two 6.3"-diameter induction zones, 1800W total, plus an included removable non-stick cast-iron griddle pan that lays across both zones — a direct match for "two burner or griddle" in one unit.
- **Price: \$189.99** (confirmed current listing price).
- Requires magnetic (induction-compatible) cookware — test with a magnet before buying pots/pans if you don't already own compatible ones (the included griddle pan is already induction-ready). See the cookware recommendation below.

Power check — confirmed: the front console AC outlet is rated at **1500W (verified against the real van, Appendix A)**, and a second AC outlet exists in the **back passenger area** (the cooktop + Power strip 1 plug in there — Section 5). Both outlets are **CONFIRMED** to share the van's single 1500W inverter — treat 1500W as the TOTAL AC budget across everything at once.

Cookware: UMETRE 9-Piece Detachable-Handle Induction Set (PURCHASED July 2026)

Bought in place of the planned Cook N Home stainless set. The detachable-handle design is what won: the whole set nests flat and ships in an **11" x 11" box that drops straight into the rear pantry's ~13" pot crate** (Component 1) — the Cook N Home set's fixed handles never would have. Induction-compatible (verified against the COOKTRON), dishwasher-safe. Trade-off, noted honestly: it's a nonstick coating rather than bare stainless, so pack silicone/wood utensils and expect to re-buy pans in a few seasons of camp abuse.

Kitchen unit: bought product, no slide-out tray to build

Unlike the earlier design (a plywood box with a built-in drawer slide for the cooktop), the JAGAHABA kitchen unit is a standalone product with its **own** slide-out mechanism and stove space already built in — there's no cooktop tray to build. Set the COOKTRON cooktop into that built-in space and run its cord through a grommet at the back of the unit, then a **SHORT** hop to the **REAR** outlet right beside the tailgate (Section 5 — no seam crossings; the front console outlet is the DELTA 3's alone). No plumbing, no water lines, no sink.

Multiport AC power strips (2 total)

Location	Purpose
Power strip 1 — mounted on the rear pantry's deck edge (relocated from Panel A — Component 1/6)	Phone/device charging, reading light, and the Windmill Smart fan — easy reach from the sleeping area, right by where your head/torso would be
Power strip 2 — mounted ON the slide-out kitchen unit (travels to the cook position; cord has a slack loop for the slide)	Powers the induction cooktop and any other small kitchen appliance (kettle, blender)

Each strip runs on its own short cord to the REAR outlet's 2-way tap (Section 5) — Power strip 1 no longer taps onto the cooktop's line, and neither runs forward anymore. They're still not unlimited: a 6-8 outlet power strip is for convenient access to the inverter's one 1500W budget, not for adding capacity — don't run the induction cooktop and multiple other high-draw devices at the same time, and remember the DELTA 3's charging cord (front console) draws from the **SAME** inverter pool.

Recommended power strip: a compact 6-outlet strip with individual switches (e.g., a basic surge-protected strip, ~\$15-20 each) — the individual switches let you kill power to specific outlets without unplugging everything, useful for lights left on overnight.

All cooktop and power strip hardware costs are already folded into the master BOM in Section 4.

8. Modular Lift-Out Design & Weight

Panel A, Panel B, and Panel C each rest unbolted on the van floor — nothing is fastened down, and no module shares a frame rail with its neighbor. Pull the bumper strips free, lift any of the three out by their exposed 2x2 top rails (no tops or skirts in the way — the old routed hand-holds are gone), and the sleeping deck is gone for normal minivan use; drop them back in and the alignment pins register everything in the same spot every time. The rear pantry is not one of these three lift-out modules — its prefab drawer units just sit on Panel C's deck behind the cleat pocket and strap (loosen the one cam strap, no tools, Component 1) — lift the drawer units out first and its deck is clear. **The fridge and kitchen unit are the one deliberate exception to "nothing is fastened down"** — see below.

Securing heavy components (fridge + kitchen unit)

The "rests unbolted, lifts out" design that works fine for the lightweight sleeping panels is not safe for the fridge (up to ~90lb loaded) or the kitchen unit (45lb). In a hard stop or collision, an unsecured item that heavy becomes a projectile — this is the one place in the whole build where "modular" has to give way to "actually tied to the vehicle's crash-rated steel." (Non-invasive, though, survives: see the no-drill note just below.)

The Sienna's factory cargo hooks are not adequate for this. Toyota's own documentation is explicit that they're for hanging a cargo net, not for restraining loose cargo — confirmed by checking the owner's manual guidance, not an assumption. Do not use them to anchor the fridge or kitchen unit.

Owner decision (July 2026): no drilling, no vehicle modification. An earlier revision of this section through-bolted 8 Snap-Loc E-track anchors into the floor pan — 16 drilled holes, with a full drill-spec / rustproofing / corrosion-warranty procedure to go with them. The owner rejected that approach outright: nothing in this build may put a hole in the van. What follows is the replacement — a **no-drill design** where every restraint lands on factory steel Toyota already rates for crash loads, using hardware that comes back out without a trace. Dropping the drill also dissolves the old plan's worst liabilities for free: no corrosion-warranty exposure, no drilling blind above the fuel tank and brake lines, no working on your back under the van.

What carries the load instead — two kinds of factory hardpoint, both crash-rated:

- **The 3rd-row strikers.** At the front edge of the folded-3rd-row area sit **3 crash-rated latch loops**: the strikers the 3rd-row seat legs lock into when the seats are deployed (photographed and confirmed present, July 2026). They're engineered to hold an occupied bench seat in a collision — FMVSS 207-class loads — which makes them the strongest exposed steel in that zone, and with the 3rd row folded they sit empty. The fridge and kitchen live at the tailgate end, directly over the 3rd-row stowage well, so these loops are the **nearest** rated hardpoints to the mass they restrain.
- **The 2nd-row long-slide floor rails.** The Sienna's 2nd-row seats ride on long seat tracks bolted through the floor — and the owner's camper-mode plan parks the seat carriages fully forward behind the front seats, which leaves the **rear ends of both floor rails exposed and unused**. Those rails are the vehicle's own seat anchorage (the strongest bolted-to-floor steel in the cabin), and their rear ends are assumed to reach the striker-row area under Panel B's rear edge — close enough for the anchor board's forward members to grab — **MEASURED Aug 1 2026 (F8a): the rail rear ends sit 42" from the closed hatch, 2.5" nearer than the striker row**, so the tongues reach comfortably and get fabricated to 42". The seat *mount points* themselves sit further forward under Panels A/B and stay in reserve for the panel train, but the **rails' rear ends are the anchor board's forward connection**.

The anchor board. Three striker loops have to restrain two appliances plus a slide, so everything first unifies onto one chassis: a comb-shaped **3/4" ply anchor board** lying on the floor under Panel C, cut so nothing changes height where height matters. It is **ONE piece of plywood** — a single 46" x 33" comb, bridge and strips continuous — so there is no ply-to-ply joinery anywhere in it (drawings: the *Anchor Board — Assembly & Connection Details* sheet in Renders — top view, the riser stack section, the kitchen strap elevation, the tongue/strap side view, and the cut plan):

- **A full-width bridge** (~46" x 6") across the front of the appliance zone (Y ≈ 29–35 in Panel C coordinates — forward of the closed fridge tray and the kitchen unit, still under Panel C).
- **Three strips running rearward from the bridge:** a narrow one under the driver-side fridge rail line, a wide center strip that serves double duty (the passenger fridge rail's riser AND the kitchen's bay-side L-track), and a 1.5" strip in the band at the panel's passenger edge (the kitchen's outer L-track). The strips stop ~2" short of the tailgate edge so Panel C's rear corner legs keep standing on the bare van floor — **Panel C's lift-out design is untouched, and it still never structurally touches the fridge or its slide**.
- **No wood joints at all — cut the whole comb as ONE piece (assembly sheet, V5).** The bridge and all three strips come out of a single 46" x 33" blank, so **no glue and no lap screws appear anywhere on this board**; every fastener left on it holds *hardware* to the ply, not ply to ply. It fits the stock you already buy: the 3/4" sheet carries only Panel C's deck plus some ripped cleats, leaving a contiguous spare of at least 48" x 60" — the comb drops in with ~26" of sheet length still left over for the control-panel backer board, the rear-pantry hold-down cleats and the spare-tire skid battens. Three things to get right when you cut it:
- **Face grain runs along the strips** (the 33" direction, which is the sheet's 96" length). That's the axis a rearward pull loads in tension, and the strips are the narrow members — the panel-edge one is only 1.5" wide, so it wants its face plies running lengthwise. The bridge takes its load in-plane over a 6" depth, where cross-grain costs nothing that matters.
- **Fillet every inside corner, ~1/2" radius** — drill a 1/2" hole centered on the corner first, then saw into it. A square re-entrant corner is where a comb starts a tear, and it is the only place this one-piece board is weaker than three separate strips would have been.
- **The falloff isn't waste.** The two gaps come out as clean rectangles (16.85" x 27" and 20.5" x 27") — keep both as 3/4" offcut stock. Cut them by ripping with a track saw or a circular saw on a straightedge (both cuts run the full 27" and *stop* at the bridge line), then crosscut the strips free.

Why one piece is worth the sheet space: the striker straps hold the *bridge*, and everything bolted to the strips hangs off it in a rearward pull. As one piece that load path is continuous plywood instead of three glue lines working in tension. (The steel tongues are not wood joints either — they bolt through the bridge with the 1/4-20 T-nuts.) - **Nothing under the fridge tray or under the kitchen unit — this is a skeleton, NOT a full false floor.** The tray keeps hanging between the slide rails at its 0.5" floor clearance and the kitchen sits on the bare van floor as before — the board occupies only the bridge and the strips beside them, so *leg_height*, the 0.33" end-rail clearance, and the kitchen-drawer stack are all unchanged. - **The fridge riser angles bolt on top of the rail-line strips** (1/4-20 machine screws into T-nuts from below). The risers — and with them the rails — ride ~0.85" higher on the mat + board stack, but nothing else moves: the moving member simply screws to the 1x3 apron lower down in their shared overlap (~1.9" of the 3" rail still faces the 2.5" apron — enough screw band, asserted in *params.scad*), so the tray keeps its 0.5" floor hang and the fridge stack height is untouched. - **The kitchen straps to the board:** short lengths of **aluminum L-track** (rated ≥2,000 lb/ft) screw to the two kitchen-side strips, and the 4 **Erickson ratchet straps** (400 lb WLL / 1,200 lb break each) criss-cross over the unit into **L-track stud fittings with D-rings**

(WLL 1,333 lb each) — the same criss-cross as the old plan, with the board's track replacing the 4 floor anchors. - A **non-slip rubber mat** goes under every strip — friction, trim protection, and rattle control in one.

Three load paths, each matched to its direction:

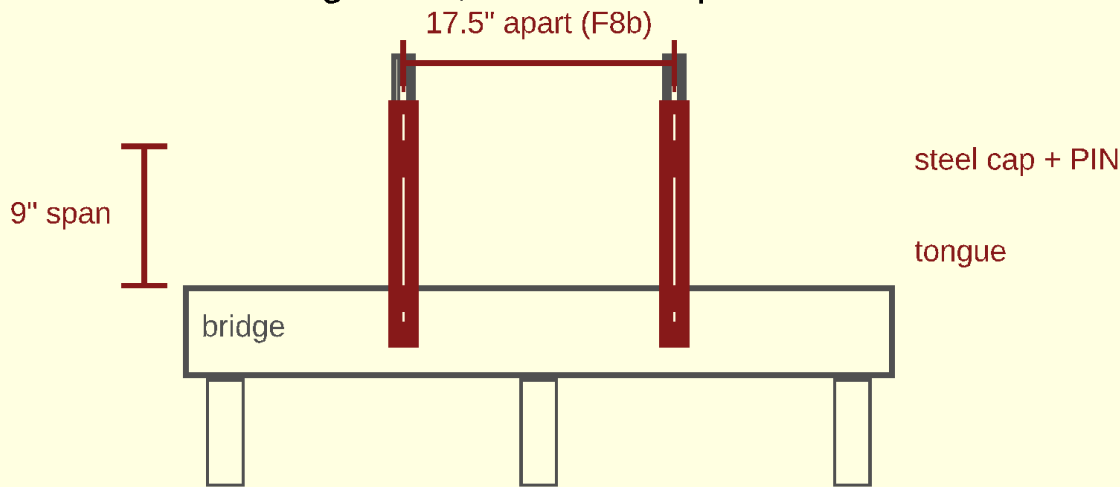
1. **Forward (the panic-stop direction, and the big one)** — a **bolted link to the floor rails**: two **2" x 3/16" steel flat-bar rail tongues** bolt onto the bridge and run forward along the floor to the **rear ends of the 2nd-row long-slide floor rails**, where a **vertical ¼" pin** drops into the ¼" hole in each rail's steel end cap — **no new holes in the vehicle**. That makes the forward restraint a positive steel-to-steel connection to the van's own seat anchorage, and it backstops rearward and lift too. (A tension strap to a *forward* anchor goes slack the moment the load moves forward — that's why this path is a hard link, not webbing.)
The old compression fallback is gone: F7 found no square hard face at the striker row to butt the bars against, just carpet over a soft ~1" step, so this connection is now load-bearing rather than one of two options. The full detail is below.

How the tongues connect to the rails

SECTION 8 — TONGUE TO 2ND-ROW RAIL CONNECTION

All measured Aug 1 2026. Governing limit is hole-edge bearing in the cap's top face, on an ASSUMED 0.08" thickness (~1,400 lb/tongue) — measure it.

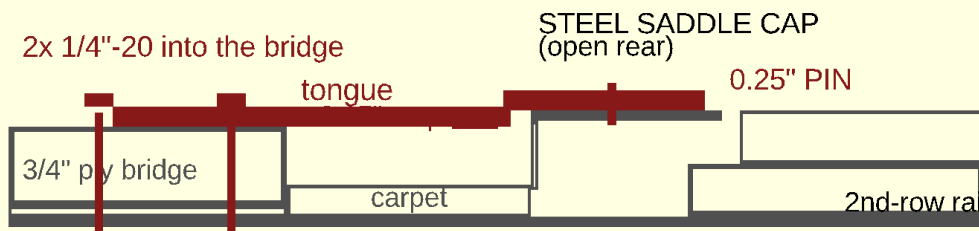
1 PLAN — looking down, forward is up



Rail ends 42" from the closed hatch (F8a).

Tongues 16" long: 4" on the bridge + 9" span + 3" on the cap.

2 SIDE SECTION — vertical scale exaggerated



The tongue rides ON the cap's top face (1") and the pin drops into its 0.25" hole. The cap's rear is OPEN, so nothing butts: forward load is pure SHEAR on the pin. Shim the step; a hold-down keeps the pin seated.

3 THE ONE CHOICE LEFT, AND THE DEAD ENDS

HOLD-DOWN (choose one)

ESSENTIAL, not optional: a vertical pin lifts straight out. Either

- a clip hooked over the cap's side skirts (the open rear helps), or
- a saddle clamp on the rail just forward of the cap.

RULED OUT — do not revisit

- fastener in the track slot: the slot runs FORE-AFT, so friction only, ~430 lb
- nut bar from the end: cap + pin block it
- butting the rail's end face: no such face, the cap's rear is open
- bolt through the hole: no nut access

Tongue to 2nd-row rail connection detail

Plan, side section, and the three ways to engage the rail. The tongue and its fasteners are drawn in red; the van, the rails and the board are context.

What the survey settled. F8a measured the rail rear ends at **42" from the closed hatch**. The board is 46" x 33", so its bridge ends at 33" and each tongue has only **9" of open floor to span** — much shorter than the 10–14" originally assumed. **Cut both tongues 16" long:** 4" lapped onto the bridge, 9" of span, 3" overlapping the rail.

What is at the rail end (MEASURED Aug 2026): each rail terminates in a **moulded plastic end cap** held by a single flush, slotted (flat-blade) fastener in a round recess — try turning it 90° first, since a fastener that shape is often a quarter-turn catch rather than a screw. **Keep both caps** for passenger mode. Under that plastic cap is a **steel saddle cap** closing the rail's end, described below. Rail centres measure **17.5" apart** (F8b), and the track's own slot is ¾" wide over a 1"-deep channel — recorded because it rules the in-slot options in or out, but the final design doesn't use it.

The forward load is carried by a VERTICAL PIN in shear (owner decision, Aug 2026). Getting to that took five revisions, so the reasoning is worth stating once and the dead ends are listed at the end of this section rather than re-argued.

The rail's slot **runs fore-aft — the same direction as the load**, because a seat track exists for a carriage to slide along it. So anything sitting *in* the slot resists forward load only by **friction**: a ¼"-20 torqued sensibly gives ~1,200 lb of clamp, steel-on-steel $\mu \approx 0.18$, so **~215 lb per bolt** — ~430 lb across two tongues against a ~155 lb loaded board. That was the original plan's forward path and its weakest link.

The cap is a SADDLE, and its rear is open (photos, Aug 2026). It's a U-shaped steel stamping straddling the rail's end: a **flat top face at 1.0"** carrying a single ¼" **round hole** (with two *closed* oblong stiffening dimples either side — not usable), **two vertical side skirts** hanging down, and an **open rear** between them. That geometry decides everything:

- **The pin is VERTICAL.** It drops down through the tongue into the hole in the cap's top face.
- **There is no bearing face.** The rear is open and its edge is a rolled hem, so nothing butts against anything. **Forward load is carried entirely by the pin, in shear.** The butt-joint idea is dead — see below.
- **The tongue rides ON the cap's top face** (1.0"), which is 0.15" above the board's own plane (0.85"). **Shim that step** under the tongue's lap on the bridge — washers are enough.
- **Drill one ¼" hole in the tongue** and fit a ¼" bolt as the pin, nipped on the tongue's own accessible face, plain end down. No lug, no bend, no fabricated part, and no nut anywhere you can't reach.

⚠ **The hold-down is now essential, not optional.** A vertical pin lifts straight out, so something has to keep the tongue seated. The cap's **open rear makes this easy** — a clip hooked over the side skirts, or a saddle clamp on the rail just forward of the cap. **This is the one part of the joint left for whoever builds it to choose.**

Capacity, and where the limit actually sits

¼" mild-steel pin, single shear	~0.049 sq in × ~36 ksi ≈ 1,760 lb per pin , ~3,500 lb for the pair
Hole-edge bearing in the cap's top face ← governing	at an assumed 0.08" thickness: 0.25 × 0.08 = 0.02 sq in → ~1,400 lb per tongue , ~2,800 lb for the pair

That's ~18 g on a ~155 lb loaded board, and ~6× the friction-only option — but note the limit moved off the pin and onto the **cap's sheet thickness**. 🛠 **Measure it** to replace that assumption with a real number. Nothing in this design is crash-rated.

Four things that were tried and dropped

Kept short, so they don't get re-proposed:

Rejected	Why
Fastener in the track slot (T-bolt, channel nut)	The slot runs fore-aft — the same direction as the load — so it resists forward motion by friction only : ~1,200 lb clamp × μ 0.18 ≈ 215 lb per bolt , ~430 lb for the pair. This was the original plan's forward path and its weakest link
Nut bar slid in from the end	Impossible — the steel cap closes the channel and a pin behind it blocks entry from the rear
Butting the tongue on the rail's end face	No such face exists: the cap's rear is open, its edge a rolled hem
A bolt through the cap's hole	Possible but awkward — no access behind the top face to start a nut. A pin needs no nut , because the load is shear, not clamp

(F7 had already ruled out butting the **striker-row step** — carpet over soft trim, not a hard face.)

- 🛠 One minor item remains, and it only sets the pin's **length**: how deep the hole is, and whether it breaks through into the channel.
2. **Rearward + lift — 3 ratchet straps** (400 lb WLL each) from stud-fitting D-rings on the bridge, forward and down into the 3 striker loops, tensioned. These also pin the board against the slide's cantilever moment when the loaded fridge is pulled to full extension.
 3. **Lateral** — the kitchen's criss-crossed straps, the striker straps' splay, the well and van walls ~1" away, and the mat's friction.

The straps and tongues cross under **Panel B's rear bottom rail** at the B/C seam — cut shallow relief notches in that rail's underside where they cross (3/16" deep for the tongues, a saw kerf for the webbing). That's a modification to build lumber, not to the van.

The anchor board secures the tray to the van — it doesn't secure the fridge to the tray. As built, the fridge just sits on the plywood tray behind its 1x3 side aprons' lip; nothing stops it lifting or shifting off the tray itself in a hard stop, the way the kitchen unit's own top straps stop it lifting off its footprint. The BougeRV Rocky 40 has a carry handle at each end (visible in the manual photos) that doubles as a tie-down point: add **1 low-profile cam strap over the top, hooked to the fridge's 2 end handles and down to 2 small D-rings screwed into the tray's side apron** (same pattern as the DELTA 3 drawer's own hold-down, Component 8). Snug it enough to stop vertical lift, but loose enough that it doesn't bind the reversible lid when the tray's pulled to its fully-open position — check that clearance once installed (Component 7, step 3).

Why this margin is enough — same DOT yardstick the old plan used: commercial cargo securement rules (49 CFR 393, Subpart I) size tie-downs for 0.435g forward / 0.5g rearward / 0.25g lateral deceleration, with aggregate working load at least 50% of the cargo's weight. Cargo here is ~135 lb loaded (fridge ~90 lb + kitchen 45 lb) plus ~20 lb of board — the 50% rule asks for ~80 lb of aggregate WLL. The 3 striker straps alone provide **1,200 lb, 15x that**, before counting the rail-

tongue link, which isn't strap-limited at all. Every link in the chain is rated: straps 400 lb WLL each, stud fittings 1,333 lb, L-track ≥2,000 lb/ft, and the striker loops and seat rails themselves are seat-anchorage hardware rated far beyond any of it — the straps govern, exactly where overbuilding is cheapest. (A DIY conversion isn't held to DOT's certification process; the baseline is used as an engineering floor, not a legal one.)

Gate: the Appendix A floor survey (F1–F8) is COMPLETE (Aug 1 2026) — its findings are folded into this section below. Nothing gets drilled, and the board's outline is now cuttable. For reference, what the survey settled: where the load surface is solid vs folded seatback vs well void (F1, F4 — a board strip bearing on springy seatback needs filler blocking down to the well's structural bottom, per F3's depth), the strikers' actual fore-aft position and whether the folded seatbacks leave all 3 loops exposed and hookable (F4), where the 2nd-row floor rails actually end and what their end hardware offers the tongues to bolt or clamp to (F8 — this decides the bracket detail), and whether the step at the striker row gives the fallback a square face to bear on (F7). The strikers are confirmed to EXIST (photo, July 2026); their position — assumed ~46–50" forward of the hatch, which puts the tongue run at ~10–14" — is not yet measured, and the rail ends are assumed to reach the same zone. The install renders draw the board, straps, rails, and tongues at this ASSUMED geometry — re-run `./render.sh` once the survey pins the real numbers. **And keep the passenger rear corner clear: the hybrid's 12V battery access panel lives there (directly under the rear AC outlet) — the board's kitchen-side strip stops short of it, and battery service stays "unstrap + slide the kitchen out."**

Install and upkeep — this one block replaces the old plan's entire drill-spec / rustproofing / warranty / under-the-van sequence. Total install is about an hour with a drill, driver, and wrench, all of it inside the van, all of it reversible:

- 1. Bench-build the board first:** cut the comb outline as ONE piece from the 3/4" sheet's spare (final dimensions only after the F1–F8 survey; grain along the strips, ~1/2" fillet at each inside corner — assembly sheet, V5), then counterbore the underside and seat the 1/4-20 T-nuts flush, then bolt on the riser angles, the L-track lengths, the bridge's stud-fitting D-rings, and the two steel rail tongues (their forward ends drilled/bent to match what F8 found at the rail ends). A drop of **blue medium-strength threadlocker (Loctite 242 or equivalent)** on each machine screw — these fasteners hide under the appliances, so they should never need re-checking. Let it cure per the label (~24h) before loading.
- 2. Set the board in the van** on its non-slip mat, bolt/clamp the tongues to the rear ends of the 2nd-row floor rails (or butt them against the striker-row step, if that's where the survey landed), and notch Panel B's rear bottom rail where the tongues and straps cross under it.
- 3. Hook and tension the 3 striker straps** (bridge D-rings → striker loops), then install the slide, tray, fridge, and kitchen per Component 7 and set the kitchen's 4 criss-cross straps.
- 4. Re-tension every strap after the first drive**, then check at the start of each trip — the striker-strap ratchets live under Panel B's open bay, so re-tensioning means lifting the bed platform, not removing any panel.
- 5. Nothing to rustproof, seal, or crawl under the van for** — no holes were cut, so the old yearly rust inspection is gone too. To return the van to stock: release 7 straps, undo the 2 rail-tongue connections (any factory hardware goes back exactly as found), and lift the board out with everything on it. Zero evidence it was ever there — and the 2nd-row seats bolt back onto their untouched rails for passenger mode (Section 9).

Approximate weight per module (plywood + frame + legs only, rough estimates — excludes mattress and small hardware):

Module	Est. weight
Rear pantry — 4× prefab drawer units + pot bin (each unit lifts out on its own at ~3.2 lb)	~15 lb total
Panel A (incl. divider + 1 drawer box/slide + corner braces + 2 bottom rails + glide strips + WAVE 3 overhead shelf, no top)	~40 lb (pull the DELTA 3 drawer box off its slides first and it drops to ~18 lb)
Panel B (bare cube frame — bottom rails on all 4 faces + corner braces; no top, no divider, no drawers)	~15 lb
Panel C (frame + front bottom rail + deck + front wall + kitchen drawer & cheeks + control enclosure — the fridge TRAY + slides stay bolted to the anchor board (strapped to the van, no holes), and the fridge/kitchen units are tracked separately below)	~61 lb — the heavy one (unclip the kitchen drawer box from its slides first: ~48 lb)
BougeRV Rocky 40 fridge (empty weight, manual spec)	40.6 lb / 18.4 kg (realistically 60-90 lb loaded — why it's on a 379 lb locking slide, not a lift-out module)
JAGAHABA kitchen unit (real product weight)	45 lb
EcoFlow DELTA 3 Plus + Smart Extra Battery (Panel A right drawer, included in that drawer's own weight above)	~48 lb
EcoFlow WAVE 3 (Panel A left bay, stored in place — unlike the old tailgate tray, it doesn't need removing for driving)	33.7 lb

Panel A (~40 lb) and Panel B (~15 lb) are comfortable one-person lifts; Panel C (~61 lb) is the outlier — unclip its kitchen drawer box from the slides first (2 minutes, drops it to ~48 lb) or make it a two-person lift. Losing the tops — and, for Panel B, all its drawers and divider — still puts A and B well under the ~50 lb rule of thumb for a solo carry (weights in the table above): **Panel B is a bare cube frame at ~33 lb, Panel A is ~43 lb** (one right-side drawer + the WAVE 3 open bay + the overhead shelf), and **Panel C is ~49 lb** (frame, deck, front wall, fridge tray, kitchen drawer, and the small control enclosure). The heaviest single lift is Panel C at ~49 lb, still a solo carry. The rear pantry's **~15 lb** of prefab drawer units ride on top of Panel C when installed but lift out one at a time (~3.2 lb each — loosen the strap); each is a trivial one-hand lift even loaded.

Fit and quiet-in-transit checklist:

- Each panel is lifted by gripping its exposed 2x2 top rails (no tops or skirts in the way) — the old routed hand-holds were removed; a two-hand, centered grip keeps the lift balanced.
- Bumper strips + alignment pins + seam draw-latches (hardware in Section 3, installed in Section 6, Component 5) go at the Panel A/B and Panel B/C seams. The bumpers stop the panels rubbing/rattling, the pins register them in the same spot each time, and the 4 over-center draw latches (both sides of each seam, low on the bottom-rail band) clamp the three modules into one rigid beam so they can't creep or rock relative to each other — with the bed platform tying the A/B tops, the low latches complete a top-and-bottom couple. All hand-released, so any panel still lifts out in seconds.
- Do the full lift-out/reinstall dry run in Section 6, Component 10 before you consider the build done — that's the real test of whether the modular fit is tight enough to stay quiet but loose enough to lift easily (with a second person, given the weight note above).

Test the well depth, fridge dimensions, and drawer weight against your actual van and materials before cutting — the numbers above are solid starting estimates but real-world results vary by trim year and exactly what plywood you source.

9. Seat Removal & Reinstall (Camper ↔ Passenger Mode)

The 2nd-row seats come out entirely for camper mode (Section 1) and go back in for passenger use. The Sienna's 2nd-row seats carry SRS (airbag/occupancy) wiring, so pulling them without the right procedure leaves airbag warning lights on the dash — this build uses **SRS airbag emulators (0651 Round x2 + 5522 "Square" x2 — purchased July 2026; VERIFY the 5522 mates with the rectangular plug)** plugged into the vacated van-side floor connectors so the airbag system reads normally with the seats out.

The 30-minute battery wait is a safety step, not a convenience step — the airbag computer holds backup electrical power in capacitors that must fully drain before any SRS connector is touched. Do not skip or shorten it in either direction of the transition.

Transitioning to Camper Mode (Removing Seats)

1. **Turn off the vehicle** completely and open the rear trunk hatch.
2. Locate the 12V battery in the right rear trunk wall panel and **disconnect the negative terminal**.
3. **Wait exactly 30 minutes**. Do not skip this; the airbag computer holds backup electrical power in capacitors that must drain completely.
4. Unbolt the seats and **unplug the floor harnesses**.
5. Firmly click the **0651 Round** and **5522 "Square"** emulators (purchased July 2026 — verify the 5522 mates with the rectangular plug; the original spec said 6134 Rectangle) into the van-side yellow floor plugs. Give them a light tug to verify they are locked.
6. Reconnect the negative 12V battery terminal and tighten it. Start the van; the dashboard lights will clear normally.

Transitioning to Passenger Mode (Reinstalling Seats)

1. Turn off the vehicle and **disconnect the negative 12V battery terminal** again.
2. **Wait 30 minutes** to de-energize the SRS system.
3. Unplug the Ling Labs dongles and store them safely in your glovebox or center console for the next trip.
4. Bolt the captain's chairs back down securely and **plug the factory seat harnesses back into the floor**.
5. Reconnect the negative 12V battery terminal.

Storage Tip

When in camper mode, the vehicle-side plugs sit flat on the floor. Use a small piece of plastic wrap or a ziplock bag secured with a rubber band around the emulators to prevent dirt, dust, or spilled camping liquids from getting into the electrical contact points.

Appendices — Working Sheets

These references consolidate the plan into sheets you actually work from. Four are **also a live web page** (interactive check-off boxes, or a calculator; all printable); the weight budget is also a spreadsheet file:

- **Shopping List** — <https://claude.ai/code/artifact/24d09bf4-2f51-42d5-879d-003ab1c65326>
- **Cut List** — <https://claude.ai/code/artifact/fc8fa178-40f9-40dc-9ecf-f8dc181ad479>
- **Build Sequence** — <https://claude.ai/code/artifact/eff00cab-1e69-4a54-8eec-2d9346072e92>
- **Leveling Block Calculator** — <https://claude.ai/code/artifact/149333c6-8f02-47a2-915f-52d26d9059d9>
- **Weight Budget** — spreadsheet in the repo as `weight_budget.csv`

Appendices B and C consolidate Sections 4 and 3; Appendix B (build order), Appendix E (leveling calculator), and Appendix F (weight budget) are new here.

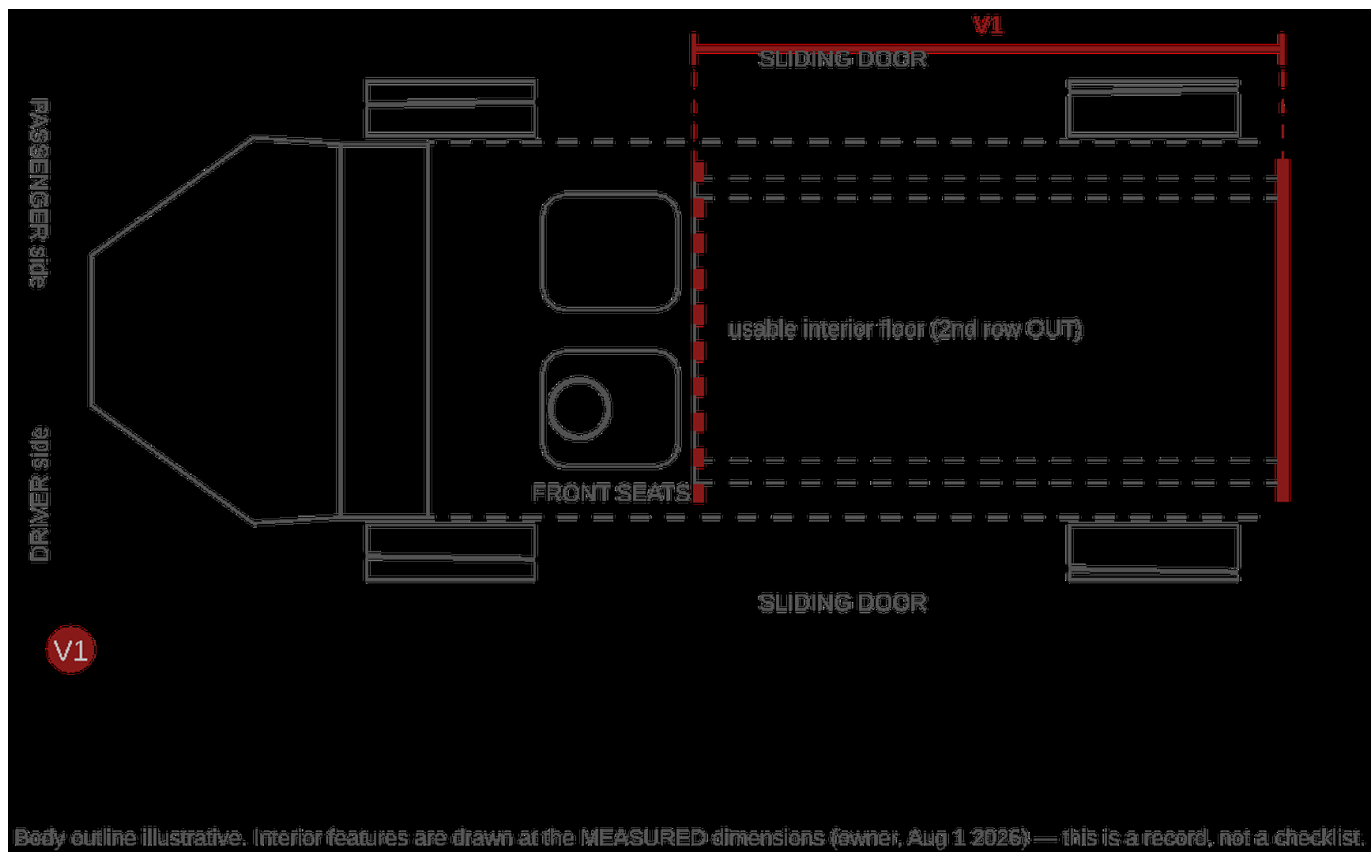
Appendix A — Vehicle Measurements & Rear-Floor Survey (COMPLETE — measured Aug 1 2026)

DONE for the van — surveyed Aug 1 2026 (owner), 2nd row out and 3rd row folded as it sits in camper mode. Every V and F line below now carries a measured finding, and `params.scad` has been updated to match. The fridge/kitchen table further down is still open (those are bought products, measured on arrival). **Read the four findings that moved the design before building anything: V1 (interior 2¼" shorter), V3 (2" lower), V4 (vents 1" deeper per side) and V7 — the walls do NOT flare out at the sleeping plane, which cut the bed platform from 52" to 49".** Every number below either said UNVERIFIED somewhere in this plan (an estimate that drives real cut dimensions) or came from an online listing/photo rather than a hands-on measurement. All of them feed directly into `params.scad`, and all of them have already been applied there — every diagram, cut list, and BOM total in this plan is drawn from the measured numbers, not the old estimates. This appendix is kept as **the record of what was measured, what it changed, and what is still open**, so any figure in the plan can be traced back to a tape reading. Where a measurement came in smaller than the estimate, the `assert()` guards in `params.scad` are what forced the design to adjust rather than silently overflow the van.

The van (once the 2nd row is physically out)

Reading these drawings. Same format as the rear-floor survey below: each measurement gets its own plan of the whole Sienna — **front at the left, tailgate at the right**, driver side along the bottom — with **only that measurement drawn in red** and tagged (V6a, V6b, ...). Those tags match the blank lines under each drawing one-for-one. Several of these are not plan dimensions at all — interior height, the gate opening, wall-to-wall width up at platform height, floor slope — so those sections carry an **elevation or section inset**, and the plan just shows where it is taken. Fore-aft distances all work from the **closed hatch**, the same datum the survey uses.

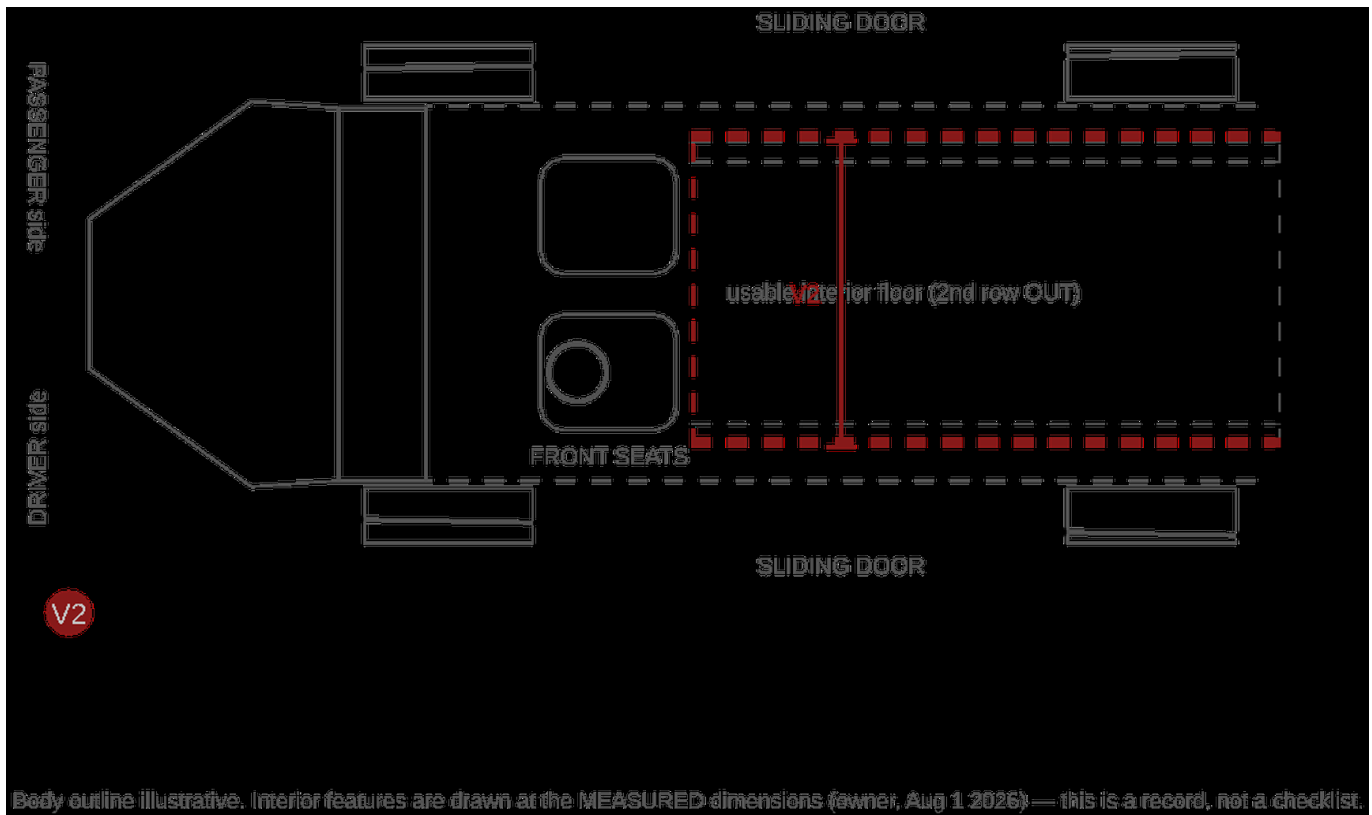
V1 — Interior length — closed hatch to front seatbacks (was row #1)



Van measurement V1 — interior length

- **Measure:** The interior length along the floor, from the closed hatch to the front seatbacks.
- **Feeds:** `van_interior_length` — the number the whole panel train is sized against: Panel A + B + C fill it exactly, so if it comes back short, every panel shrinks.
- **Previous estimate:** 96" — UNVERIFIED
- **V1** — hatch to seatbacks: **93¾"** (MEASURED — owner, Aug 1 2026) — **2¼" shorter than the estimate.** `van_interior_length` = 93.75. Because V5 also freed the 2" that was reserved for hatch curvature, the panel train only had to give up ¼" net: **Panel C went 36" → 35¾"**. It has to be Panel C — Panels A and B are asserted to stay the *same* length (they share one set of step diagrams and one cut-list entry), and their combined length is the bed platform's 58" side rails, so shaving either one would ripple through the lumber list for nothing. Panel C has ~7" of fore-aft slack around the 28.74" fridge.

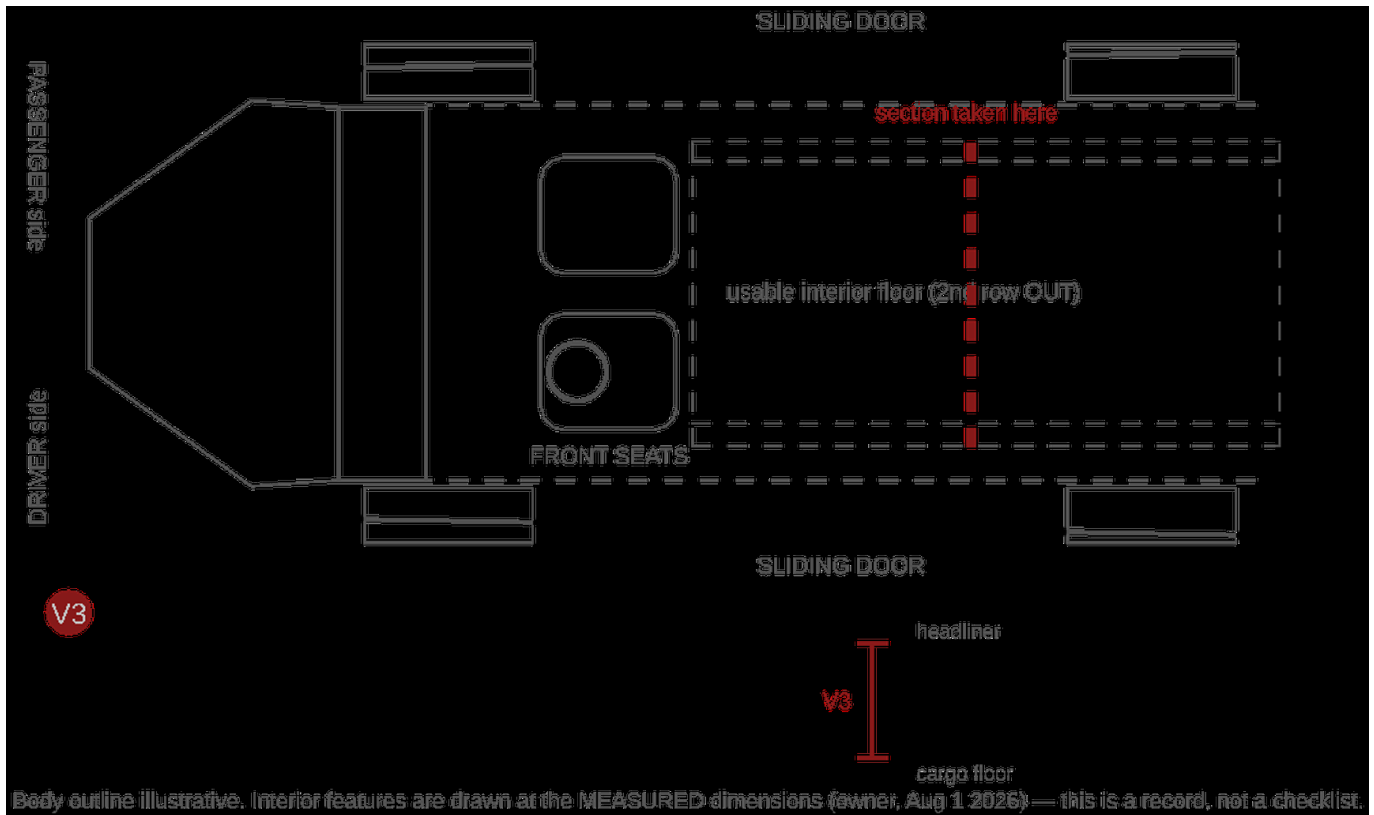
V2 — Interior width — between the wheel wells (was row #2)



Van measurement V2 — interior width

- **Measure:** The width at the floor, at its narrowest point between the two wheel wells — not at a wider spot fore or aft of them.
- **Feeds:** `van_interior_width` — every 46"-wide panel and the floor envelope come off this.
- **Previous estimate:** 48.5" — already verified (reconfirm)
- **V2** — floor pinch, wheel well to wheel well: **49"** (MEASURED — owner, Aug 1 2026) — $\frac{1}{2}$ " *better* than the figure this was designed against, so nothing downstream changes. The floor opens out to **54" forward of the wheel wells**, but the pinch is what gates the 46" boxes. `van_interior_width` = 49.

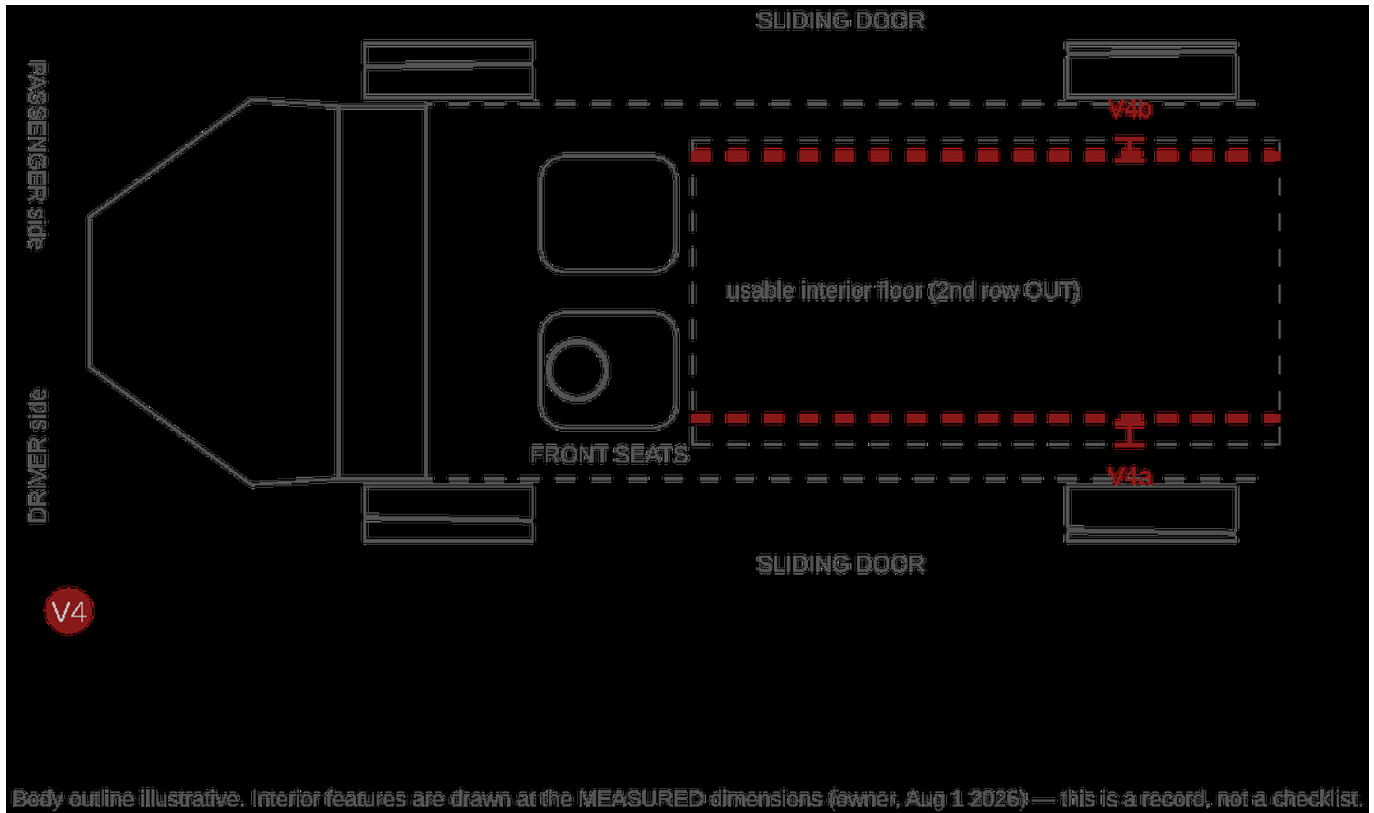
V3 — Interior height — floor to headliner (*was row #3*)



Van measurement V3 — interior height

- **Measure:** Cargo floor up to the headliner, taken at the sleeping run rather than at the tailgate where the roof drops away.
- **Feeds:** `van_interior_height` — caps the whole stack: deck + mattress + sitting headroom.
- **Previous estimate:** 44" — already verified (reconfirm)
- **V3** — floor to headliner: **42"** mid-van (MEASURED — owner, Aug 1 2026) — **2" lower than assumed**, and only **37" at the gate** where the roof drops (that is the V6 gate height). `van_interior_height` = 42. Knock-on: sitting headroom over the mattress drops 21.5" → **19.5"**, and the rear-pantry cluster's roof clearance drops to ~6.7" — a third drawer tier would now foul the headliner by 1.7", so the 2×2 cluster is the ceiling.

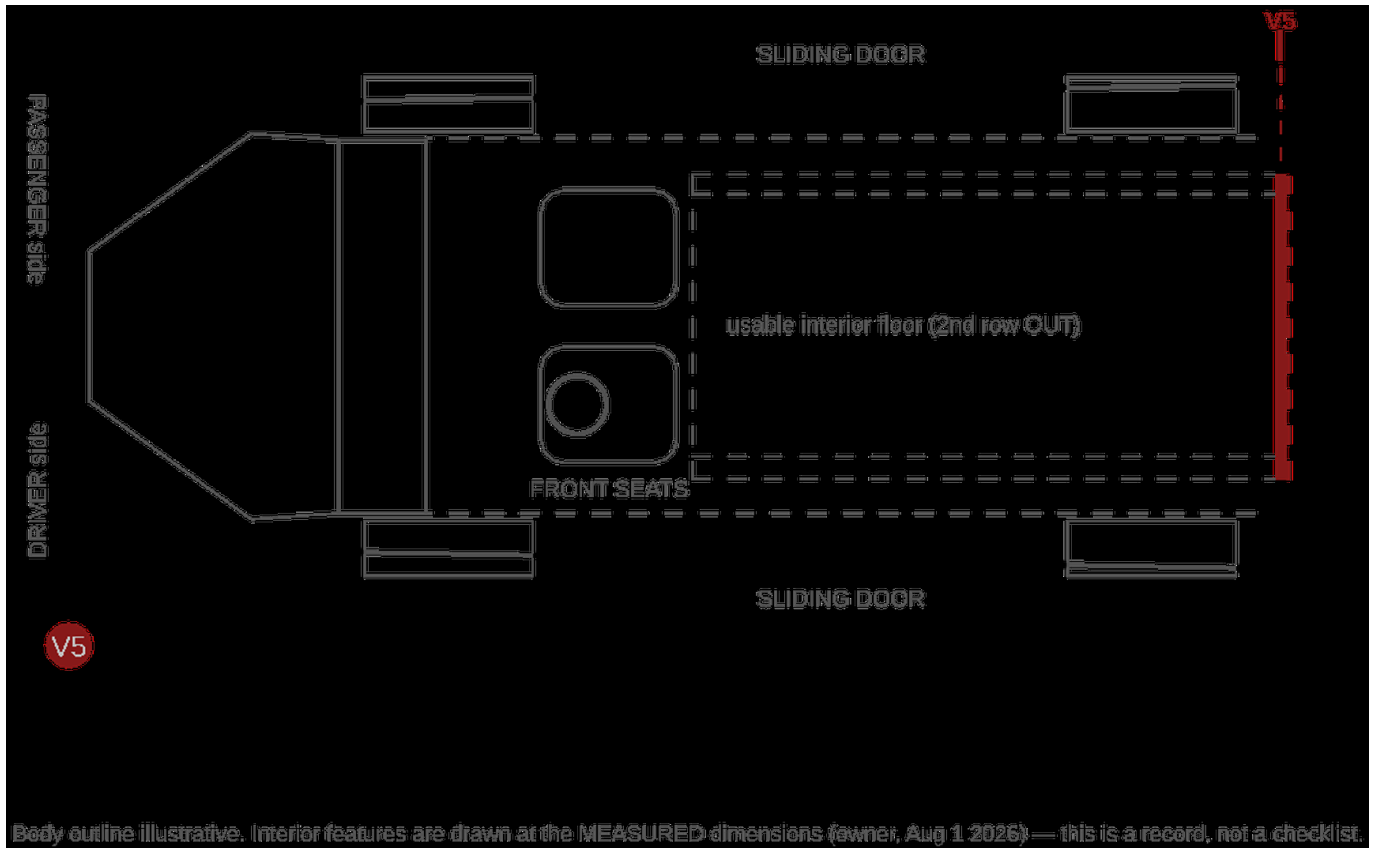
V4 — Vent intrusion width, each side, at floor level (was row #4)



Van measurement V4 — vent intrusion

- **Measure:** How far the floor vent / trim kick eats into the usable width down at floor level. One measurement per side. The deck may overhang it, but a **leg may not stand on it**.
- **Feeds:** `vent_intrusion_width` — drives `leg_inset`.
- **Previous estimate:** 2.5" per side — already verified
- **V4a** — driver side: **3.5"** (MEASURED — owner, Aug 1 2026)
- **V4b** — passenger side: **3.5"** (MEASURED — owner, Aug 1 2026)
- **1" per side deeper than assumed.** `vent_intrusion_width` = 3.5, so `leg_inset` grows to 3.5" — the legs step further in from the deck edge. Sketched on the sheet as ~3¼" over a ~15" fore-aft run; **3.5" is the number to build to**. Floor-level usable width is now 42" (49 – 2×3.5), and the leg line at 39" still sits inside it.

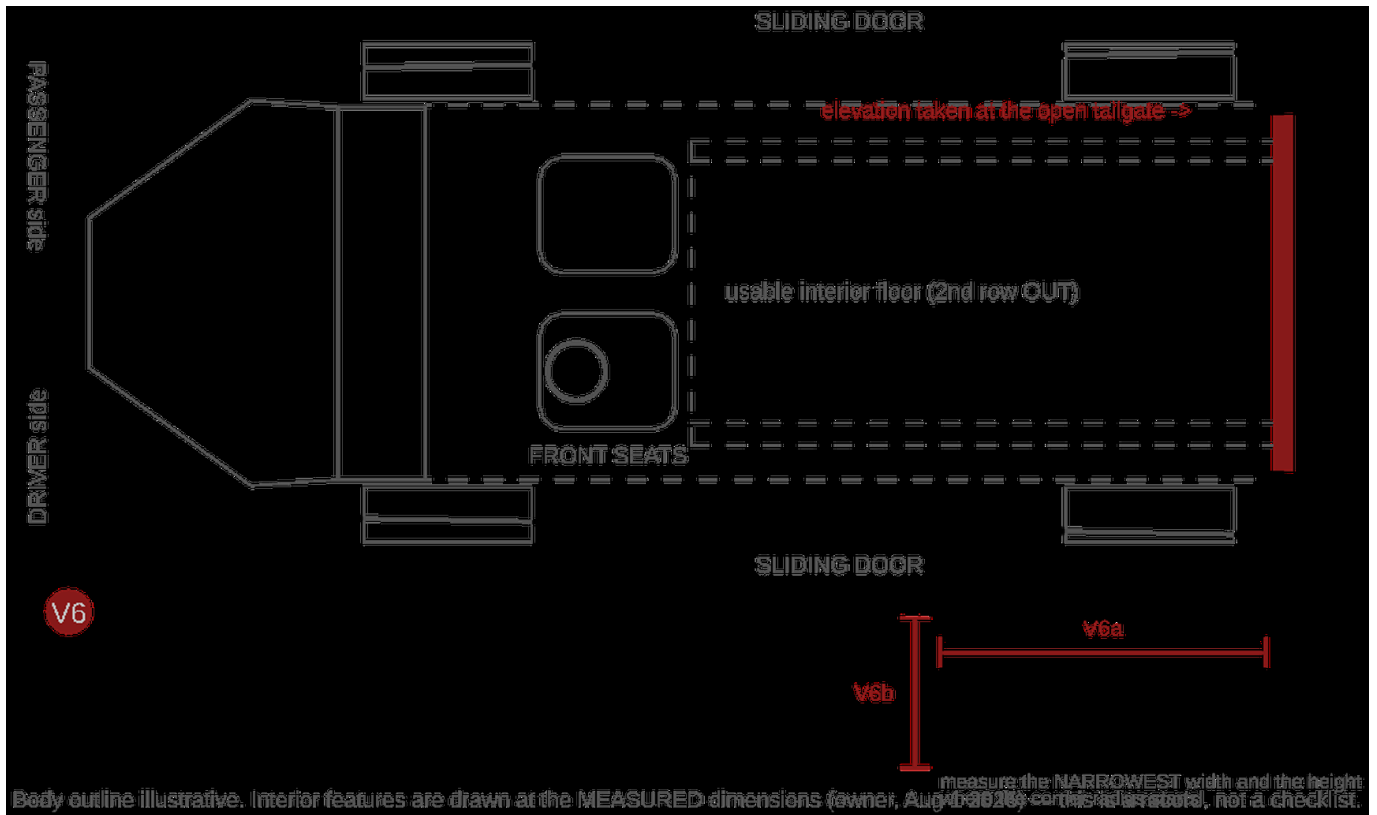
V5 — Hatch curvature clearance (was row #5)



Van measurement V5 — hatch curvature

- **Measure:** How far forward of the closed hatch you have to stop building, because the glass and trim curve inward above the floor.
- **Feeds:** `hatch_curvature_clearance` — it is why the panels total 94", not 96".
- **Previous estimate:** 2" — already verified
- **V5** — hatch to the usable boundary: 0" (MEASURED — owner, Aug 1 2026) — *"the build isn't high enough to hit curvature."* Nothing has to be reserved at the tailgate end: `hatch_curvature_clearance` = 0, which is what gave back most of what V1 took away. **One thing did break:** the rear-pantry drawer units (14.3" deep) used to be allowed to sit 0.3" proud of the deck edge *into* that 2" reserve. There is no reserve to hang into now, so **pantry_len went 14"** → **14.5"** and the deck carries the whole drawer (costs ½" of sleeping run).

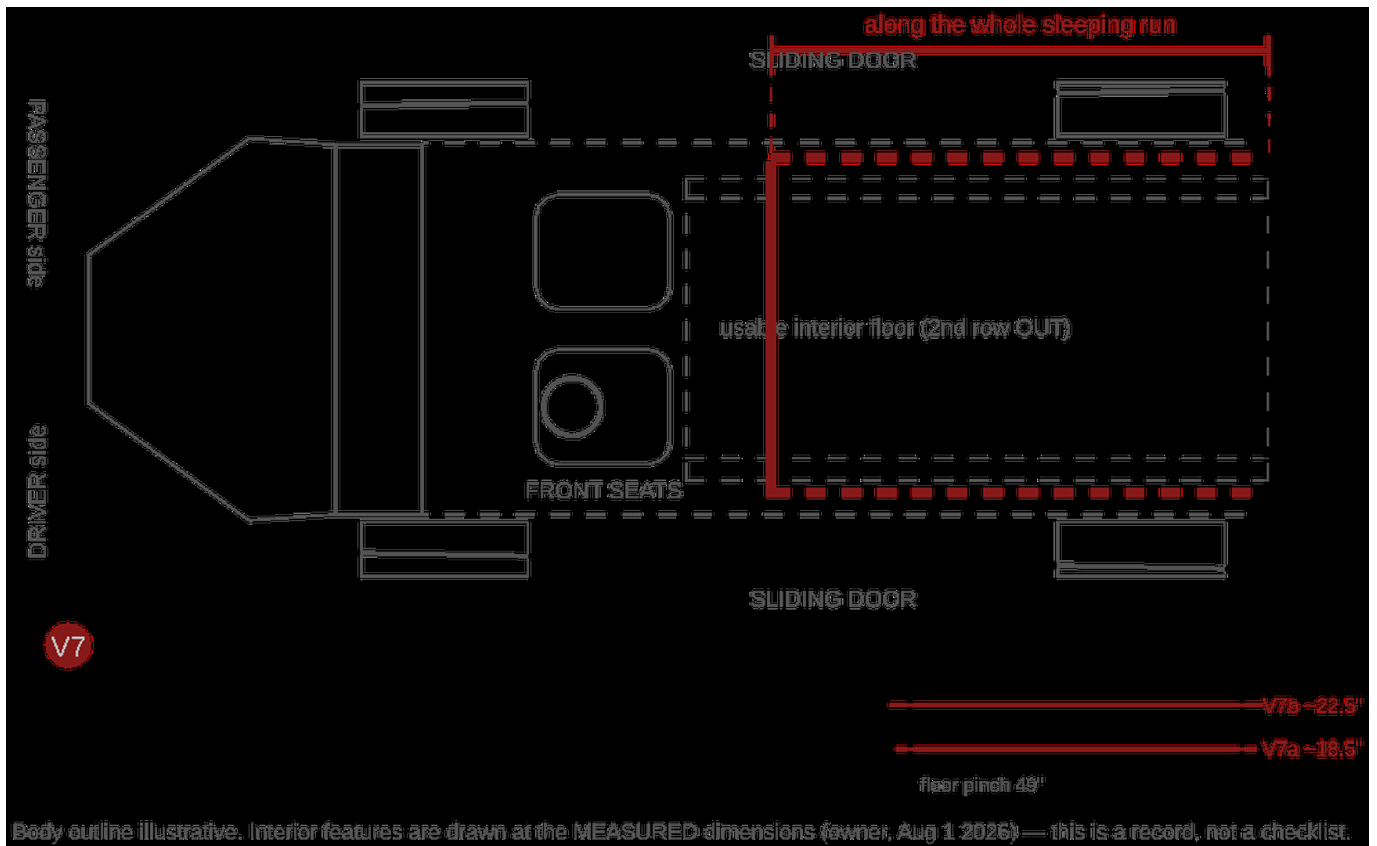
V6 — The gate (tailgate) opening — width and height (was row #6)



Van measurement V6 — gate opening

- **Measure:** Standing at the open hatch: the opening's width at its **narrowest** point, and its height **where the rounded corners start cutting in** — not the widest or tallest point of either.
- **Feeds:** `gate_opening_width` / `gate_opening_height` — these gate what can physically pass into the van at all.
- **Previous estimate:** 48" / 36" — both UNVERIFIED
- **V6a** — narrowest width: **50"** (MEASURED — owner, Aug 1 2026) — 2" better than the estimate; the 46" boxes now have **2" of clearance per side** instead of 1".
- **V6b** — clear height: **37"** at the centre of the opening (MEASURED — owner, Aug 1 2026), but only **20.5" out where the corner radius starts cutting in**. `gate_opening_width` = 50, `gate_opening_height` = 37. **Carry the 18.5"-tall boxes through the middle of the gate** — a module walked in flat against one side will foul the rounded corner well before it runs out of height in the centre.

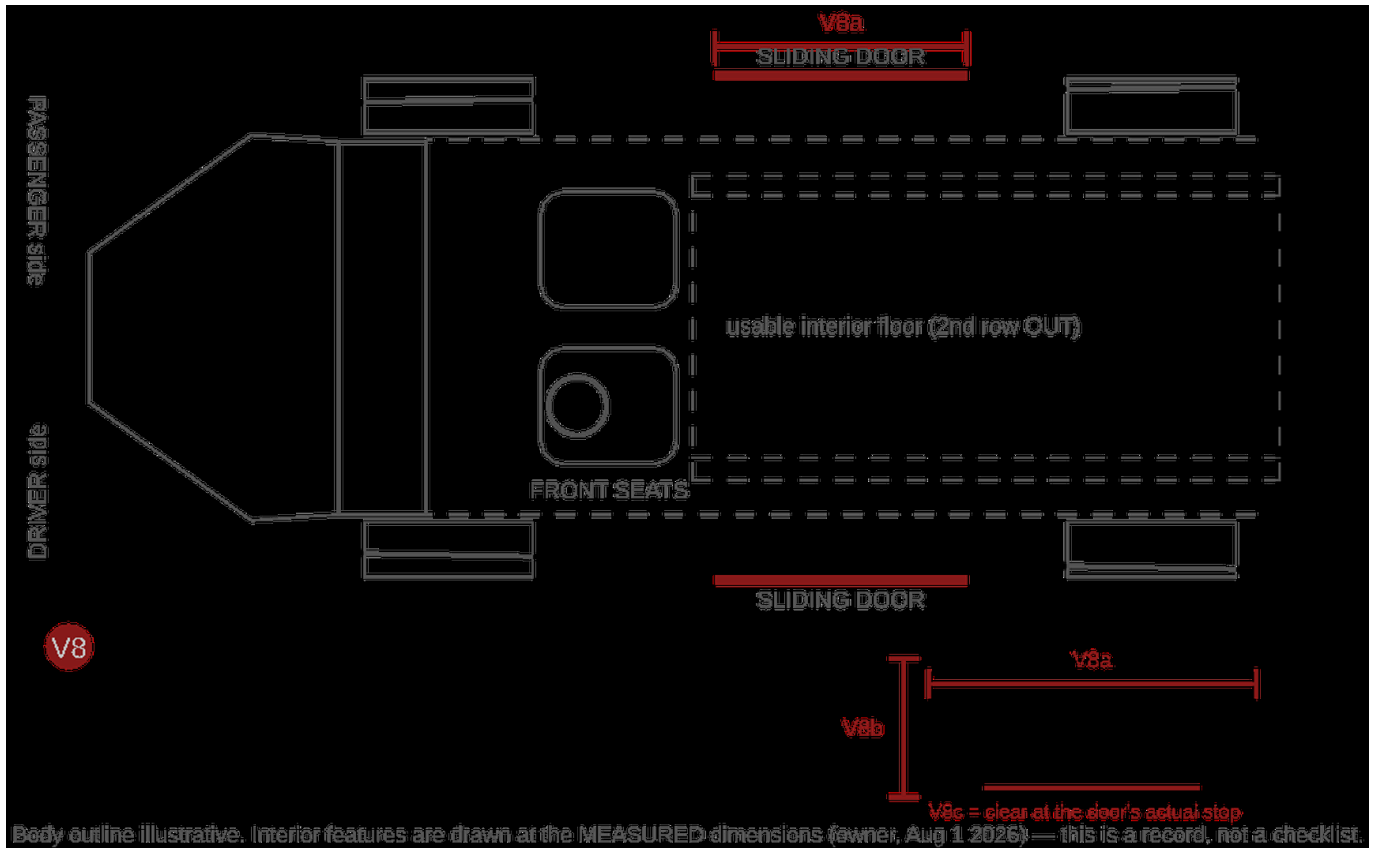
V7 — Wall-to-wall width up at platform height



Van measurement V7 — wall-to-wall at platform height

- **Measure:** Interior wall-to-wall width at ~18.5" and ~22.5" above the floor, taken at several points along the sleeping run. The floor pinch between the wheel wells is one number; the assumption was that the walls flare wider up here, which is what would let a 52" platform fit at all.
- **Feeds:** Gates `bed_frame_width` — the 52" platform needed ≥ 53 " at platform height.
- **Previous estimate:** — UNVERIFIED, and the platform-width gate
- **V7a** — at ~18.5" above the floor: **50"** (MEASURED — owner, Aug 1 2026)
- **V7b** — at ~22.5" above the floor: **49.5"** (MEASURED — owner, Aug 1 2026)
- **⚠ THIS ONE FAILED ITS GATE — the biggest finding of the survey.** The walls barely flare over the wheel wells: **50" is the usable width** at the sleeping plane (owner), against the ≥ 53 " the design assumed. So the cantilevered platform came down from **52" → 49"** (`bed_frame_width`), i.e. **1½" of overhang per side instead of 3"**, keeping ~½" of insertion clearance per side. The slats shorten with it: **42" instead of 45"**. New param `van_platform_width` = 50 records the limit and asserts against it.
- **✅ Resolved by dry fit (owner, Aug 2026): the 50" HEST goes in, with a little play to spare.** It is still the widest thing in the van — the usable width is also 50", and the frame under it is 49", so the foam overhangs its own frame by ~½" per side. The walls **narrow** with height here (50" at 18.5", 49.5" at 22.5"), so what little pinch there is lands on the mattress's top outer edges rather than its base, and solid foam absorbs it (no coils to distort). No trimming, chamfering or replacement needed — build the platform at 49" as specified.

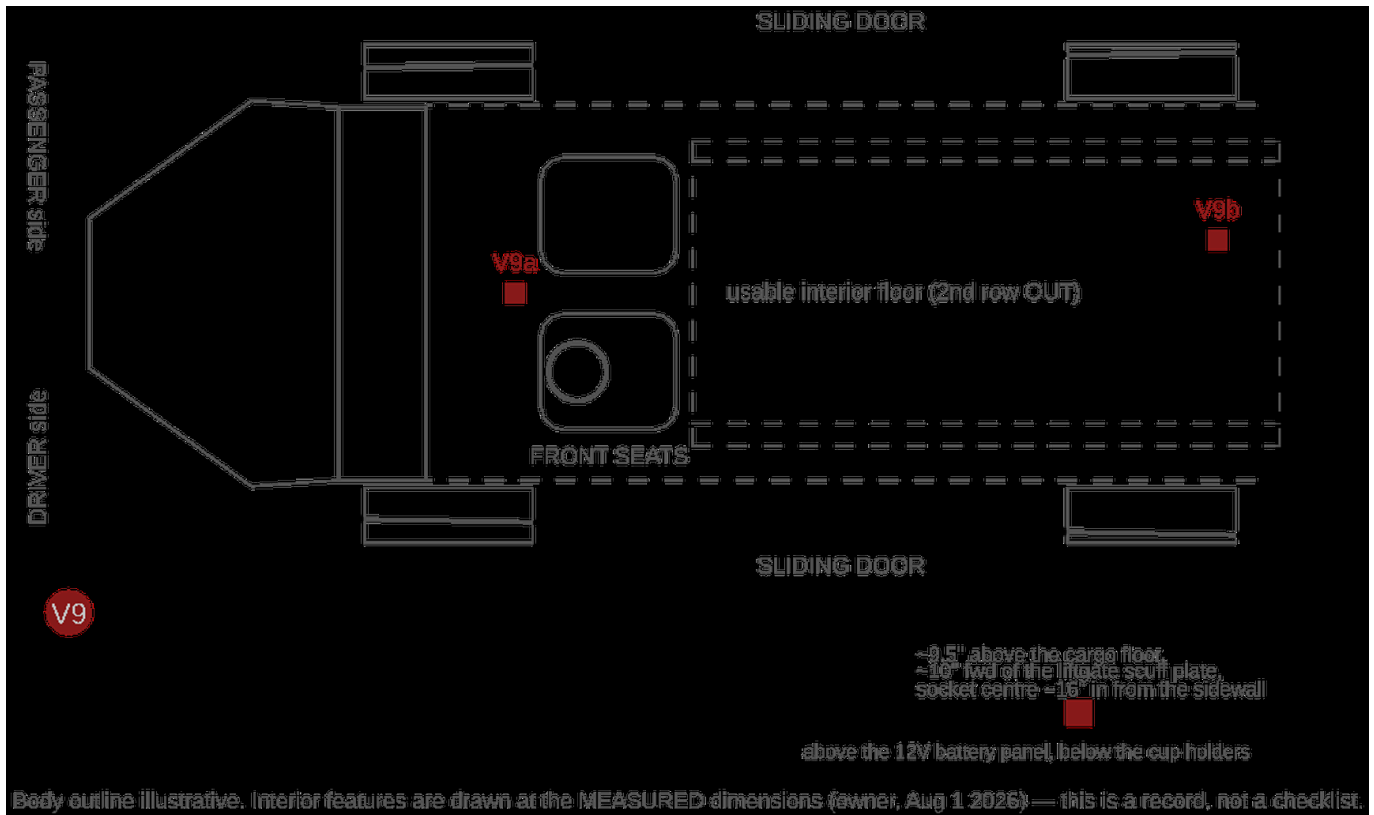
V8 — The side sliding door opening (was row #7)



Van measurement V8 — side door opening

- **Measure:** The door opening's fore-aft width and its height — and the one that actually matters, the **usable clear** width and height at the door's real stopping point, not the rough opening.
- **Feeds:** `side_door_opening_width` / `side_door_opening_height` — decides whether the DELTA 3 stack stays in Panel A or has to move (see the reachability note in Section 1).
- **Previous estimate:** 40" / 40" — both UNVERIFIED
- **V8a** — fore-aft width: **35"** (MEASURED — owner, Aug 2026). The **50"** first written on the sheet turned out to be the **door panel's own external width**, not the aperture.
- **V8b** — height: **45"** (MEASURED — owner, Aug 2026). The **58"** first written on the sheet was likewise the **door panel's external height** (it could not be a cabin dimension — the cabin is 42" tall).
- **V8c** — usable clear at the door's actual stop: **29"** (MEASURED — owner, Aug 1 2026) — **this is the number that governs, and it is 11" tighter than the 40" assumed**. The sliding door parks 6" short of the aperture's forward edge, so 6" of the 35" is never open. `side_door_opening_width` = 35 / `_height` = 45 record the aperture; `side_door_clear_width` = 29 is what the reach check runs off. Knock-on: with the door's fore-aft position still a placeholder, the check reports Panel A reachable and **Panel B blocked outright** — fine, because Panel B was already a top-loaded bay with no drawers, and it is why its bed top is now two lift-out halves (Component 2). Re-run the check once `side_door_y0` is measured.

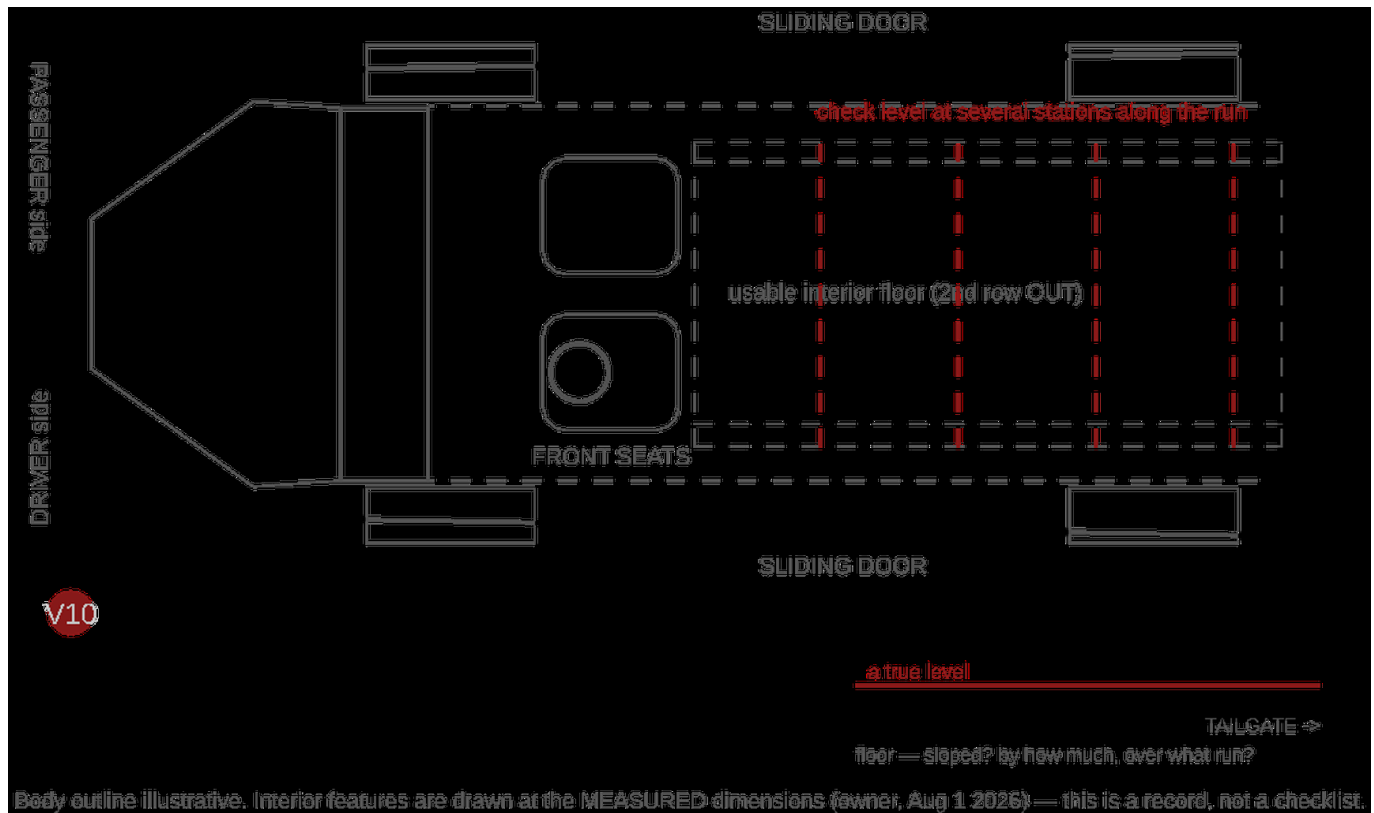
V9 — The two AC outlets



Van measurement V9 — AC outlets

- **Measure: VERIFIED: 2 outlets total, both fed by the one 1500W inverter**, so they share a single power budget. Front: centre console. Rear: passenger-side rear quarter trim — **22.5" above the cargo floor, ~10" forward of the liftgate scuff plate, socket centre 10" laterally in from the right sidewall** (MEASURED — owner, Aug 1 2026), directly above the 12V battery access panel and below the rear cup holders. *(This corrects the ~9.5" / ~16" figures carried in earlier drafts.)*
- **Feeds:** Section 5/7 — the cooktop and Power strip 1 tap the rear outlet; DELTA 3 charging has the front one. Section 1/7 uses the 1500W rating for the DELTA 3's charge rate and the cooktop check.
- **Previous estimate:** front 1500W — VERIFIED; rear position — VERIFIED
- **V9a** — front console outlet — wattage rating: **1500W, unchanged** (the line was struck through on the sheet as already verified).
- **V9b** — rear outlet position: **22.5" up, 10" inboard of the right sidewall** (MEASURED — owner, Aug 1 2026). **This is better news than the plan assumed:** at 22.5" the socket sits **4" ABOVE the 18.5" deck plane**, not buried below it — the cooktop and Power strip 1 cords reach it from deck level instead of having to drop down the gap between the kitchen unit and the van wall. Section 5's routing is easier, not harder; see the corrected note there.
- **NEW check** — does the quarter trim / cup holder intrude inboard of the 46" deck width at deck height near that corner?: **the trim sits ~10" in from the sidewall at outlet height**, so the corner is where the deck and the trim come closest — keep the anchor board's kitchen-side strip clear of it as Section 8 already requires.

V10 — Cargo floor level — flat, or sloped front-to-back? (was row #9)



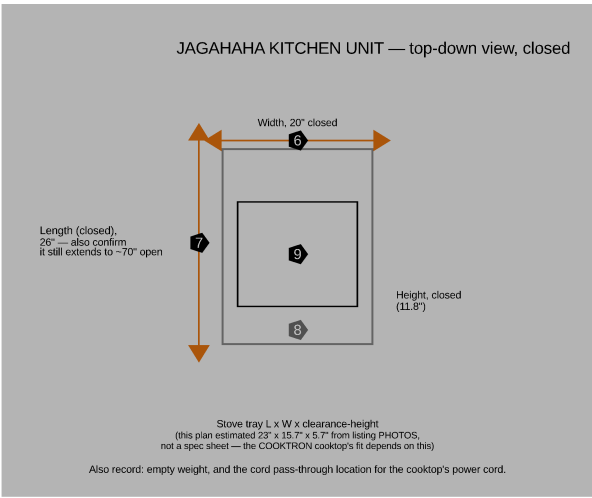
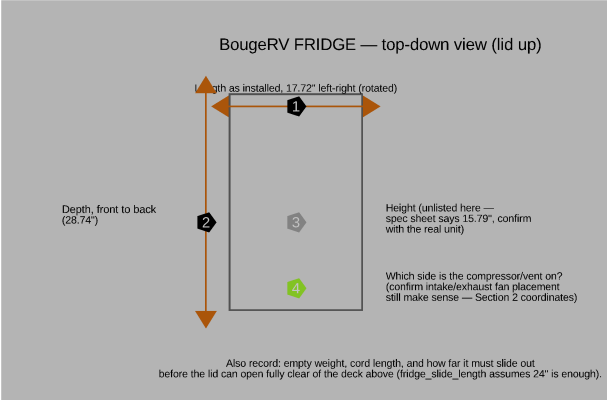
Van measurement V10 — cargo floor level

- **Measure:** Lay a long level along the run and check at several stations. Nothing in the model accounts for a slope: every module's four legs are one length, which only gives a level deck if the floor itself is level.
- **Feeds:** Not currently a param — a slope means per-leg shimming rather than one `leg_height`.
- **Previous estimate:** assumed flat — UNVERIFIED
- **V10** — level, or slope over what run?: **mixed — mostly level; level at the ends, with minor change in between** (MEASURED — owner, Aug 1 2026). No slope big enough to force per-leg shimming, so the one-length-legs assumption holds. The leveling feet already on every leg absorb the "minor change" — set them once on a flat surface and adjust only if the deck rocks.

Also worth noting while you are in there — no drawing needed

- **Folded 3rd-row well depth.** Already covered by **F3b** in the rear-floor survey below, which measures the same well. It confirms `leg_height` (17" on Panel C, fridge-driven; Panel A/B legs are 16.25" with the deck recess — that clears it).
- **Rear 12V accessory outlet — exact position.** Informational only, **not a blocker**: this design no longer uses it. The fridge and fan system run off the DELTA 3 stack instead (Section 1).
- **Factory sunroof — PRESENT** (MEASURED — owner, Aug 1 2026), opening size not measured. Only relevant if you later pursue a roof vent — measure it then.

The fridge (BougeRV) and kitchen unit (JAGAHABA), once purchased



Measurement guide: fridge and kitchen

#	Measure	Goes in params.scad as	Listing spec (unverified)	Your measurement
1	Fridge length as installed (left-right, the 450mm side)	fridge_ext_length	17.72"	17¾" (owner, July 2026 — matches)
2	Fridge depth, front to back incl. handles	fridge_ext_width	28.74" (body 28.03")	27¾" (owner, July 2026 — ~1" under; hard to tape a rounded-corner case square, and easy to undercount how far the handle protrudes. Spec used, since the tray is already cut to it.)
3	Fridge height	fridge_ext_height	15.79"	15⅝" (owner, July 2026 — matches within rounded-corner tolerance)
4	Which side is the compressor/vent on?	(confirms the intake/exhaust fan placement in Section 2 still makes sense)	Manual (CR04001): compressor, louvers, control panel & power port ALL at one end; free-air clearances 200mm that end / 100mm sides	Manual answers the "which end" question — that end faces the tailgate (Component 7). Still confirm on the physical unit, and see Section 2's clearance note for how the fan system substitutes for the manual's free-air numbers
5	Fridge empty weight	(Section 8 weight table)	40.6 lb (18.4 kg, manual)	
5b	Floor vents at Panel C's REAR corners	Panel C's rear legs sit at the TRUE corners now	must be vent-free for ~4"	✅ CLEAR — the rear-corner floor vents do not reach the leg area (owner, Aug 2026). Panel C's rear legs land on solid floor
—	Fridge cord length	(confirms it reaches the new DC line's connection point inside Panel C's void — Section 5)	—	
—	Fridge slide clearance — how far it must come out before the lid opens fully	fridge_slide_length (confirms 24" is enough)	24" slide	
6	Kitchen unit width, closed	kitchen_box_width	20"	
7	Kitchen unit length, closed	kitchen_box_length	26"	
8	Kitchen unit height, closed	kitchen_box_height	11.8"	
9	Stove tray length x width x clearance height	(Section 7 — the COOKTRON cooktop's fit depends on this; the current numbers came from listing photos, not a spec sheet)	23" x 15.7" x 5.7"	
10	Kitchen unit empty weight	(Section 8 weight table)	~45 lb	
—	Cord pass-through location for the cooktop's power cord	(Section 5/6 cord routing)	—	
—	Confirm it still extends to ~70" open	(sanity check — this happens outside the vehicle, doesn't affect interior length)	~70"	

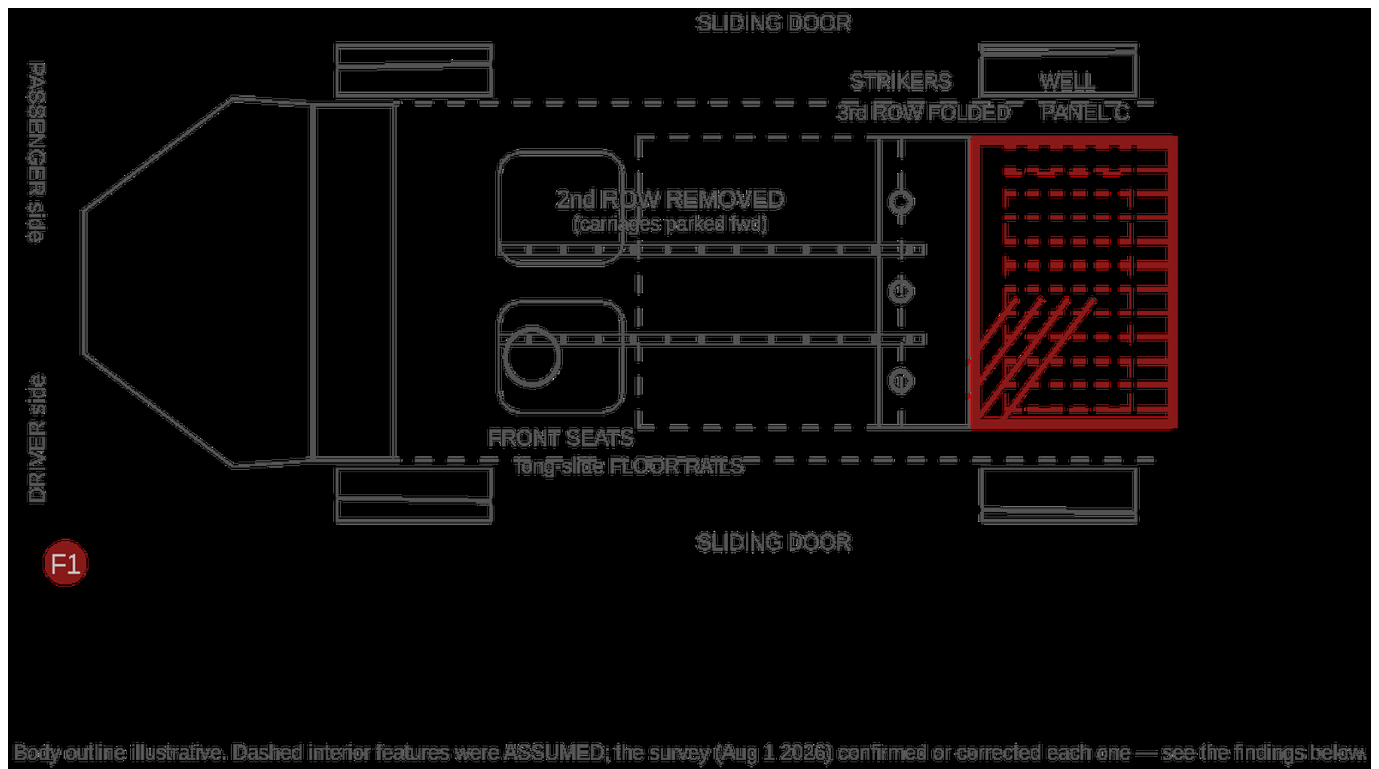
The rear floor under Panel C — what does the anchor board actually sit on and strap to? (SURVEYED Aug 1 2026 — findings below)

HEADLINE FINDING (owner, Aug 1 2026): the feared geography mostly is not there. **The stowage well is full of folded seats, not open**, and the folded seatbacks make a load surface solid enough to bear on across the whole Panel C footprint (F1/F2/F3) — so no filler blocking, and the well's edges stop being a constraint. Two fallbacks died with it, though: **there is no hard step face at the striker row** for the steel tongues to butt against (F7 — it is carpet over a soft ~1" step), and **the 3rd row's seat-mount bolts are occupied by the seats** and unavailable as hardpoints (F6). That leaves Section 8's load path as designed: **the 2nd-row rail rear ends in tension/compression, plus the 3 striker straps**. The rail ends measure **42" from the hatch** and the striker row **44.5"** — both nearer than the 46–50" assumed, so the steel tongues come out at or just under the short end of the 10–14" range. Fabricate them to the measured 42".

The original problem statement (owner, July 2026, from a photo of the actual cargo area): the rearmost stretch of the Sienna — exactly where Panel C, the fridge slide, and the kitchen unit live — is not plain sheet-metal floor. It's the 3rd-row stowage area: a **deep cargo well** the folded seatbacks drop into, so much of the "floor" back there is actually **folded seatback over a void**, plus the seats' own latch/hinge hardware and (passenger rear corner) the hybrid's 12V battery access panel. Section 8's no-drill securing design — an anchor board bearing on this surface, steel tongues bolted to the rear ends of the 2nd-row floor rails (fallback: butting the striker-row step), straps into the 3 striker loops — needs the same geography mapped before its outline can be cut: where the surface is solid, where the well and its edges are, exactly where the strikers and the rail ends sit, and what Panel C's legs land on. Do the whole survey with the 3rd row folded exactly as it will sit in camper mode and the 2nd-row carriages parked fully forward. (The old version of this checklist gated 16 drilled floor-pan holes; the drilling is gone, and with it the under-van fuel/brake-line clearance crawl.)

Reading these drawings. Each measurement below gets its own drawing of the whole vehicle in plan, so you can see where in the van you are working. All eight are the same base plan — **front at the left, tailgate at the right**, driver side along the bottom — showing the body outline, the front seats, the empty 2nd-row bay with its long-slide floor rails, the folded 3rd row over its stowage well, and Panel C's footprint. **Only that section's measurement is drawn in red**, tagged with lettered items (F4a, F4b, ...) that match the blank finding lines under each drawing one-for-one — so there is a line to write on for every thing you are asked to measure. F3 and F7 add a small section inset, because a depth and a step height cannot be shown in plan; F8 adds three detail bubbles for the rail-end cases you have to tell apart. The body outline is illustrative, and every dashed interior feature is drawn at its ASSUMED position — confirming those is the entire point of this survey.

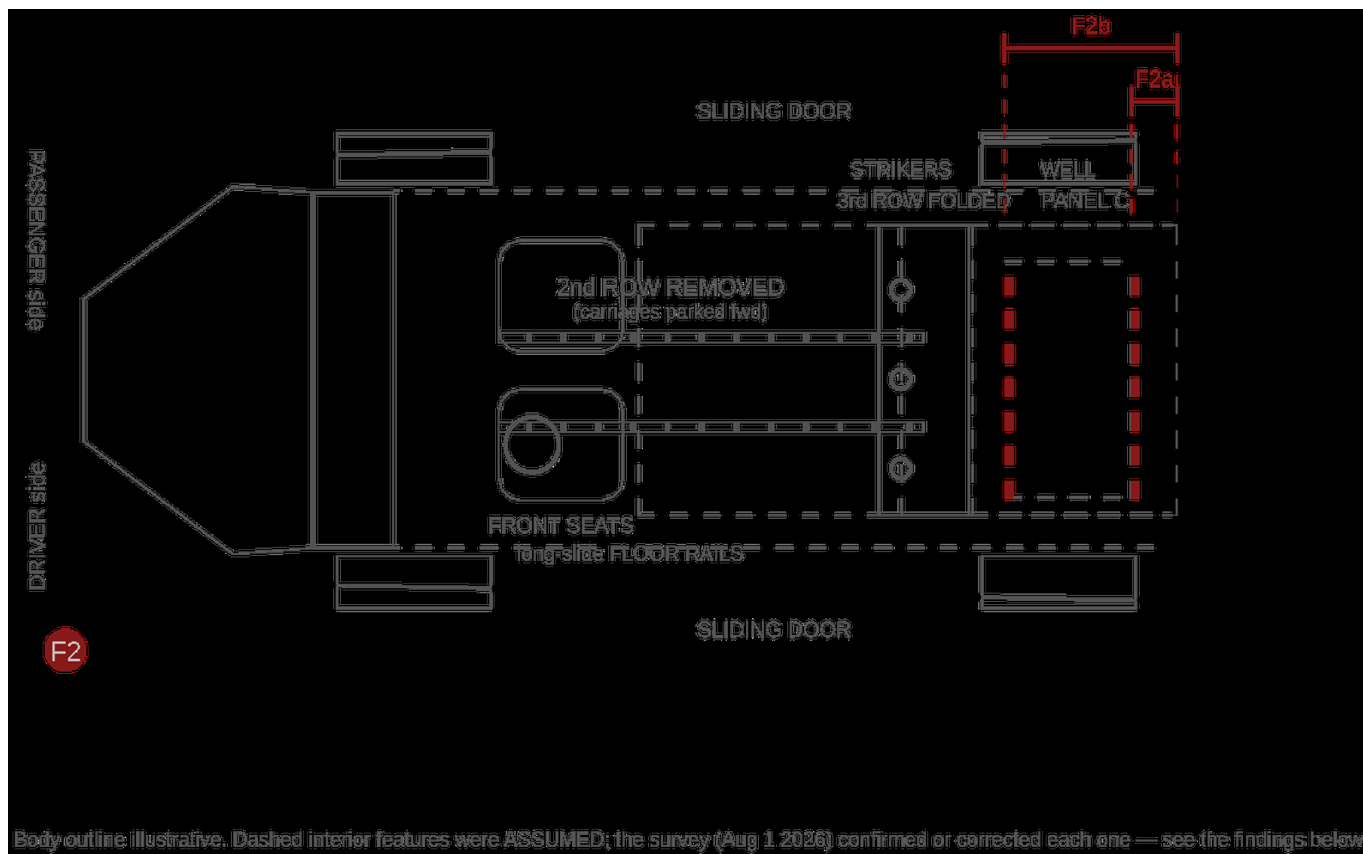
F1 — Map the load surface into its three zones



Survey F1 — load-surface map

- **Measure / check:** With the 3rd row folded flat, map the load surface: press and knock along the whole Panel C footprint and mark with masking tape where it is **folded SEATBACK**, where it is the **well/tub** (hollow underneath), and where it is **carpet over solid pan**.
- **Feeds:** Where the anchor board's strips can bear solidly, and where they would need filler blocking.
- **Previous assumption:** assumed usable as-is — UNVERIFIED
- **Finding (MEASURED — owner, Aug 1 2026):** "**Seats in all areas. Can handle load.**" There is no three-zone map to draw — the folded seatbacks cover the whole Panel C footprint and bear the load. **No filler blocking needed.** The trade-off is that the bearing surface is upholstered rather than steel, so the anchor board's rubber mat is doing real work; keep the straps tight.

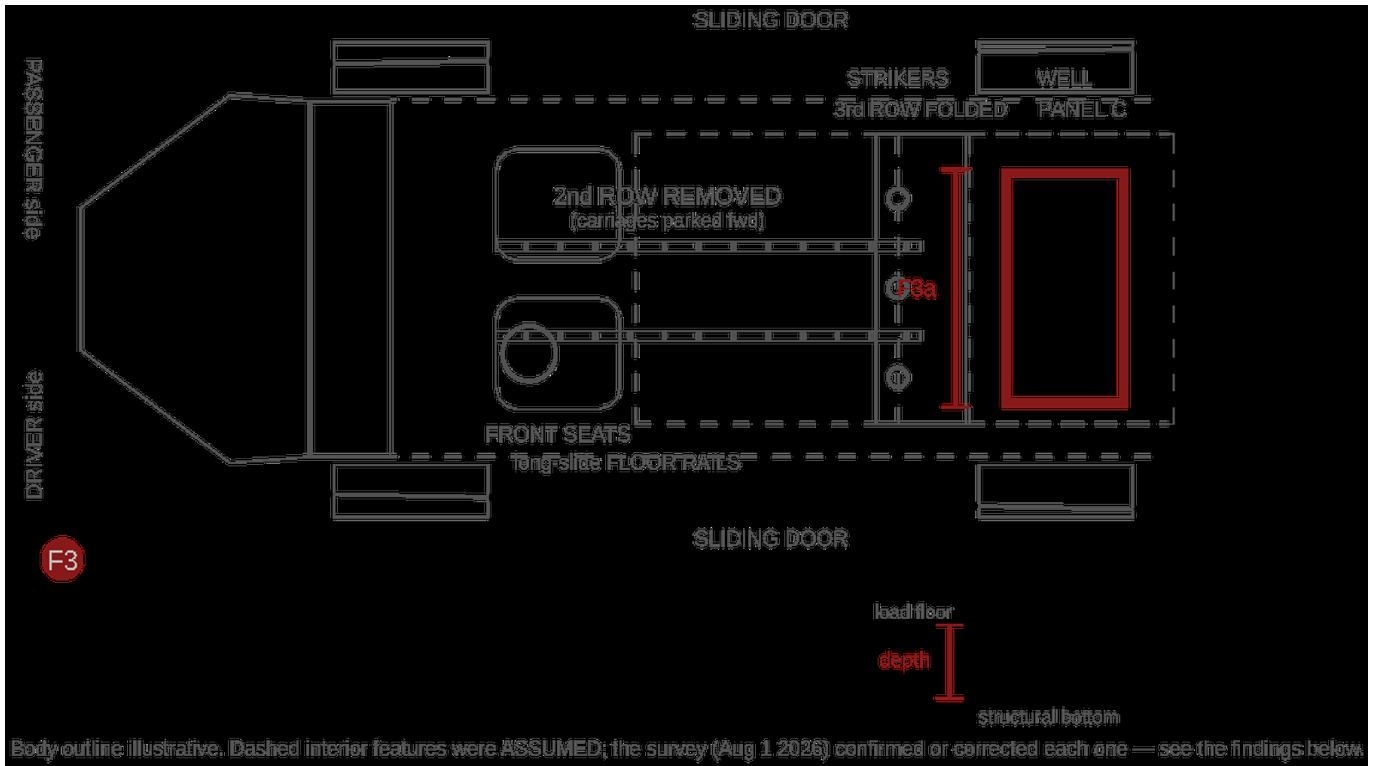
F2 — Fore-aft position of the stowage well



Survey F2 — well edges, fore-aft

- **Measure / check:** Where the stowage well's rear and forward edges sit, both measured from the closed hatch (the same fore-aft datum as V1).
- **Feeds:** The board's strip lengths, and where filler blocking starts and stops.
- **Previous assumption:** —
- **F2a / F2b** (MEASURED — owner, Aug 1 2026): **not applicable** — "doesn't matter, seats fill it in." With the 3rd row folded into it, the well is not an open cavity with edges to work around, so there is nothing for the strip lengths or filler blocking to be positioned against.

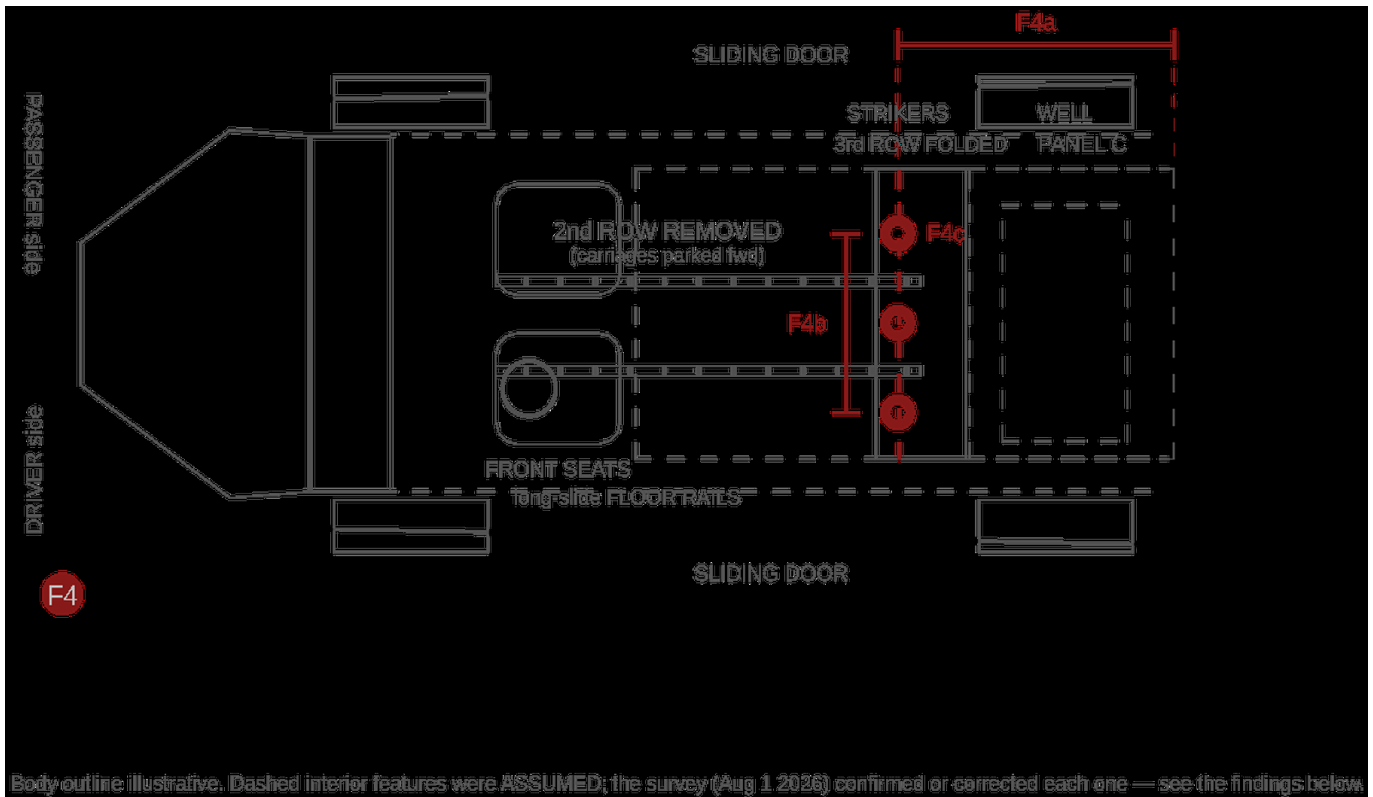
F3 — The well's width, and its depth



Survey F3 — well width and depth

- **Measure / check:** The well's width across the van, and its depth below the surrounding load floor.
- **Feeds:** Sizing the filler blocking that carries a board strip across the well, down to its structural bottom.
- **Previous assumption:** —
- **F3a / F3b** (MEASURED — owner, Aug 1 2026): **not applicable** — "well is filled with seats, doesn't matter." No depth to blocking-out, and nothing here challenges `leg_height` (17" on Panel C, 16.25" on A/B).

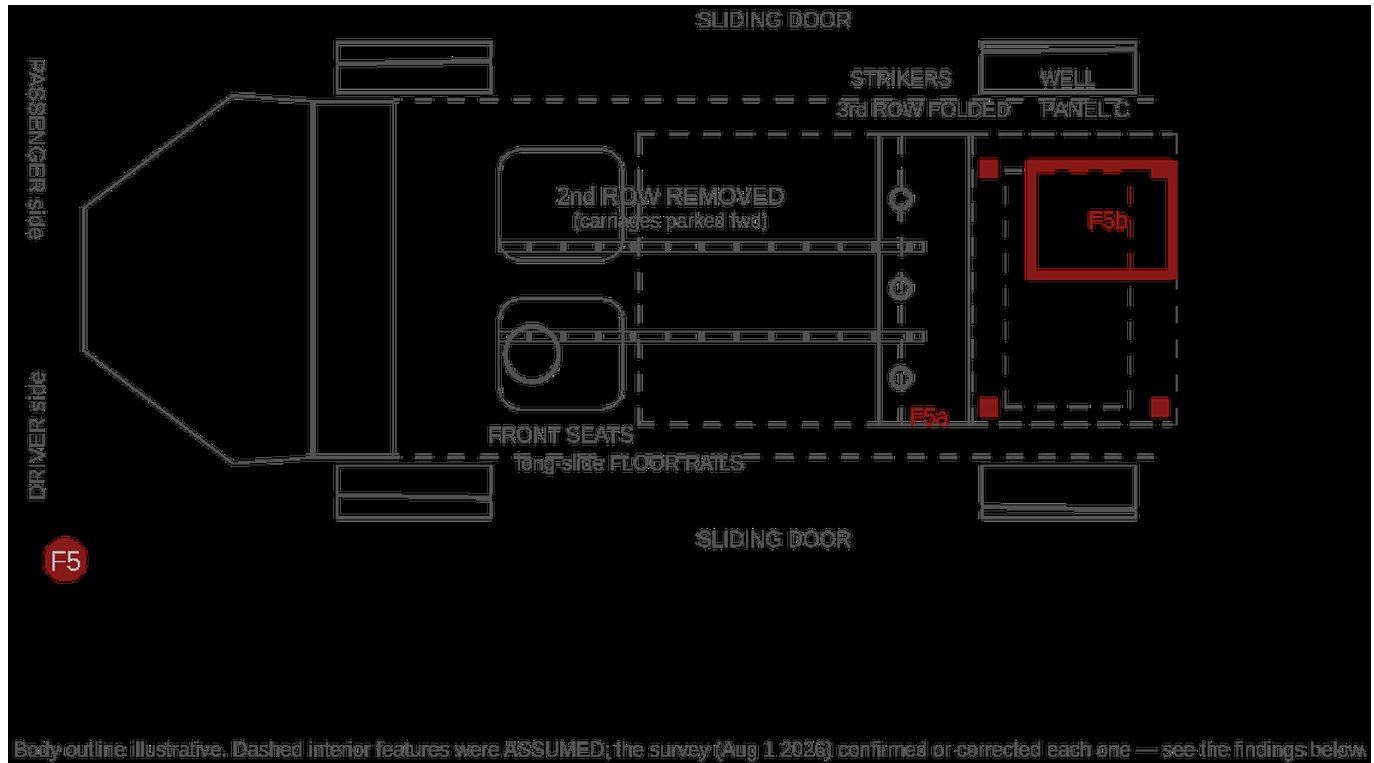
F4 — The 3rd-row striker loops



Survey F4 — striker loops

- **Measure / check:** For each of the 3 loops: its fore-aft distance from the hatch, its lateral position, and its inside clearance (a strap hook has to seat in it). Also whether the FOLDED seatbacks leave all 3 loops exposed and hookable.
- **Feeds:** Section 8's striker straps plus the steel tongues' length (assumed striker row ~46–50" from the hatch → ~10–14" tongues).
- **Previous assumption:** ~46–50" from hatch — UNVERIFIED. The strikers are confirmed PRESENT (photo, July 2026) but not yet measured.
- **F4a** — hatch → striker row: **44.5"** (MEASURED — owner, Aug 1 2026) — 1.5" nearer than the near end of the assumed range.
- **F4b** — loop-to-loop spacing: **~12.5–13"**, with the outer two loops **~11" in from each sidewall** (scaled off the tape in the Aug 2026 striker-row photos; the finding line itself says "see pictures"). Three loops across the row, roughly 11" / 24" / 37" on a tape zeroed at the driver sidewall.
- **F4c** — inside clearance / exposure: **loops are 1.5" tall, and all 3 are exposed and hookable with the seats folded** — "yes" (MEASURED — owner, Aug 1 2026). 1.5" of clear height is enough for a cam-strap S-hook or a small carabiner; check the hook you buy against it.

F5 — What Panel C's legs and the kitchen bear on



Survey F5 — leg and kitchen bearing points

- **Measure / check:** Where Panel C's 4 legs and the kitchen unit's footprint actually bear: solid floor, or folded seatback? A deck leg standing on a seat cushion is springy — it will not sit level or solid.
- **Feeds:** leg_height, deck levelness, and the kitchen unit's strap-down.
- **Previous assumption:** —
- **F5a** — what each of the 4 leg pads bears on (MEASURED — owner, Aug 1 2026): **the REAR legs land on folded seats; the FRONT legs land on the floor, but beside the vent intrusions** (see the Aug 2026 photos). So two legs bear on upholstery — springy, exactly the case this section warned about. **Set the leveling feet with the deck loaded**, not empty, so the foam-and-fabric compression is already taken up, and re-check after the first night out.
- **F5b** — what the kitchen unit's footprint bears on: same mixed surface as the rear legs (folded seat). It is strapped down rather than standing on levellers, so compression matters less — but keep the strap tight enough that settling doesn't loosen it.

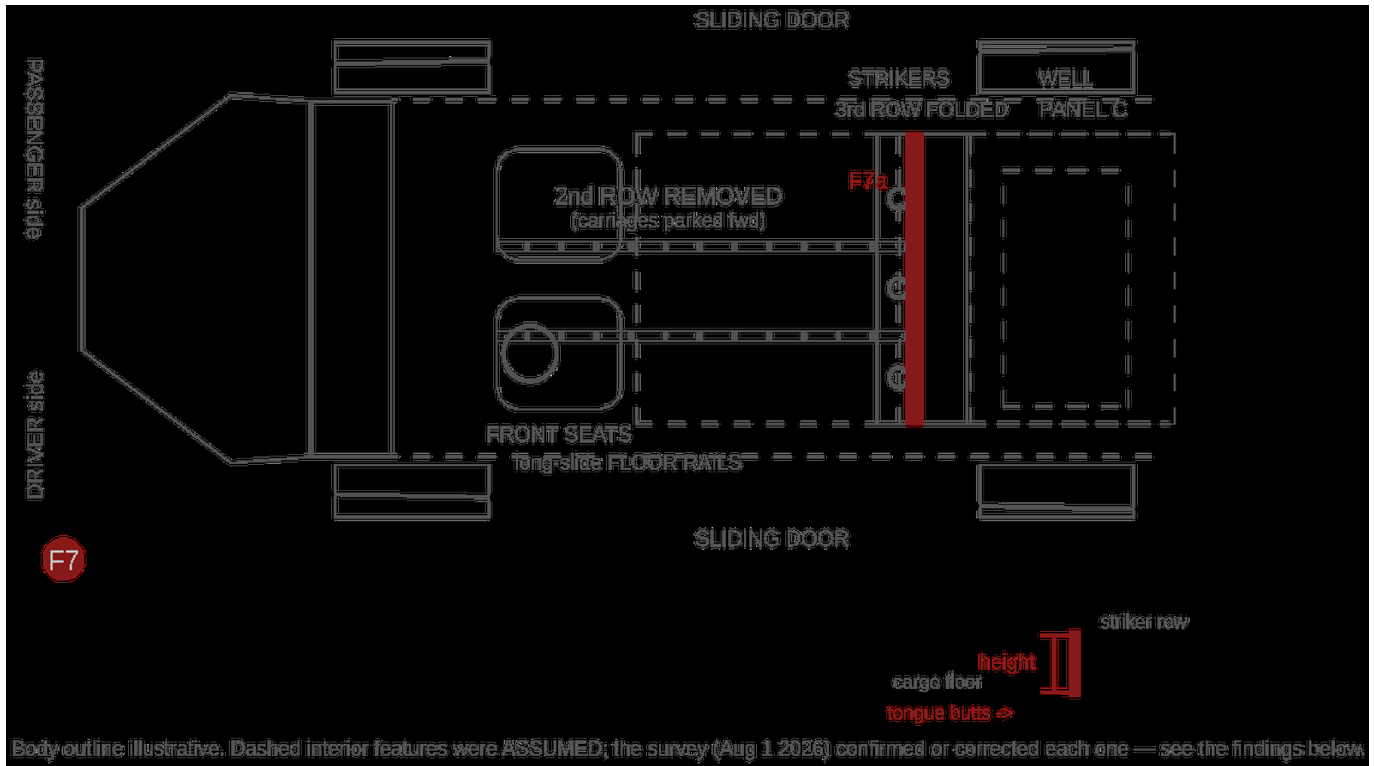
F6 — Fold the 3rd row, or remove it?



Survey F6 — fold vs remove the 3rd row

- **Measure / check:** Does a solid bearing surface require REMOVING the 3rd row rather than folding it? If so, does the 3rd row carry its own SRS / seatbelt-pretensioner wiring?
- **Feeds:** Section 9 covers 2nd-row removal only — a 3rd-row removal procedure (and possibly more emulators) would be NEW scope. Removal would also expose the 3rd row's own seat-mount bolts as bolt-in anchor-plate hardpoints (no new holes) — the strongest points in the zone if the striker plan needs help.
- **Previous assumption:** assumed fold-only — UNVERIFIED
- **F6a** — does the row have to come OUT? **No** — "folded is good." (MEASURED — owner, Aug 1 2026) **Section 9 stays as written: 2nd-row removal only.** No 3rd-row removal procedure, no extra SRS emulators, no new scope.
- **F6b** (MEASURED — owner, Aug 1 2026): **no SRS / pretensioner wiring on the 3rd row** — but the seat-mount bolts are in use by the seats and are **NOT available** as anchor-plate hardpoints. The "bolt to the 3rd row's own seat mounts" fallback is therefore off the table; the rail ends + striker straps carry the load.

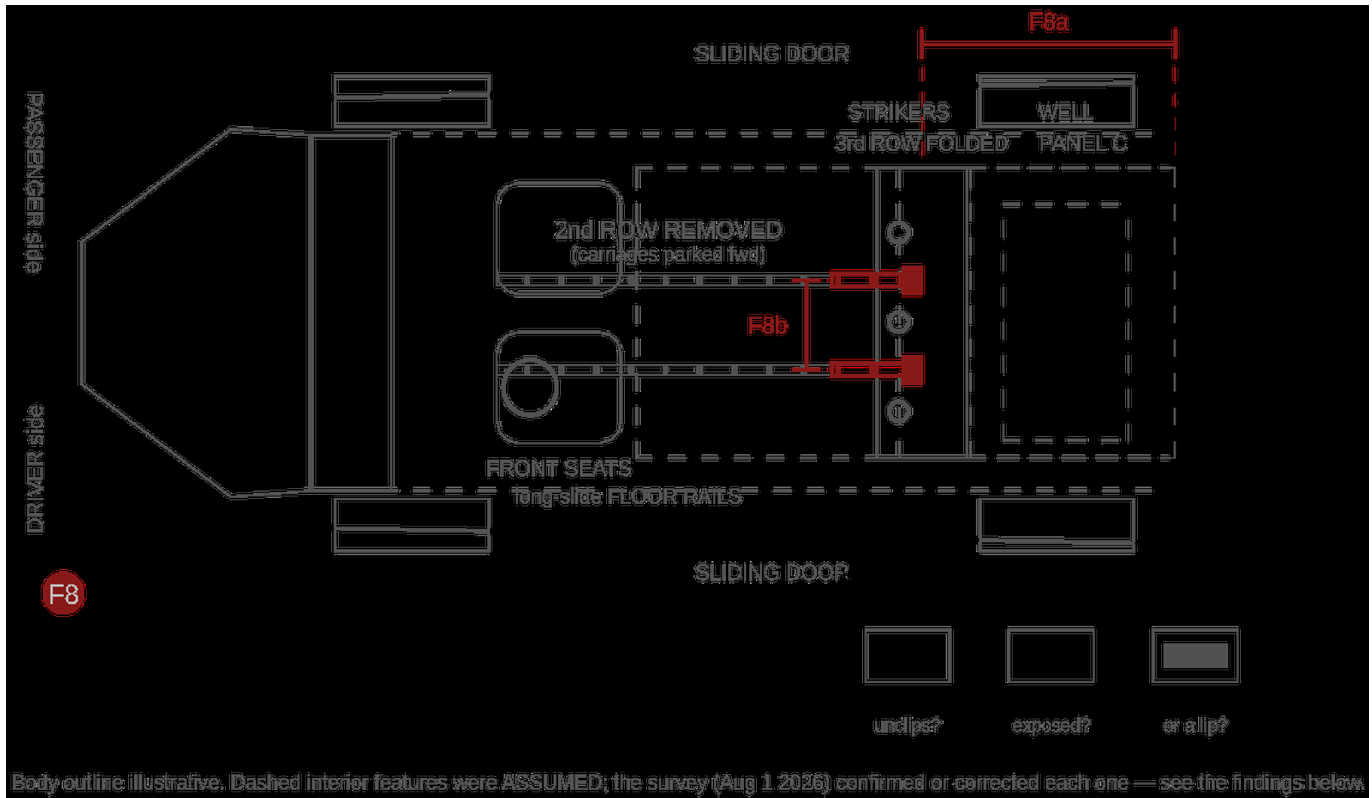
F7 — The step/riser at the striker row



Survey F7 — step/riser at the striker row

- **Measure / check:** Is there a square, hard vertical face at the step by the striker row (metal under the trim?) for Section 8's two steel tongues to butt against — and how tall is it?
- **Feeds:** The FALLBACK forward load path, in compression, if F8 rules the rail ends out.
- **Previous assumption:** assumed hard step at the striker row — UNVERIFIED
- **F7a** — is the step face square and hard? **No** — "**covered in carpet**" (MEASURED — owner, Aug 1 2026). Not a square metal face, so **the compression fallback is out**: do not size the steel tongues expecting to butt them against this step.
- **F7b** — step HEIGHT: **1"** (MEASURED — owner, Aug 1 2026) — low and soft. It will stop a board from creeping rearward under gentle load, but it is not a crash-path member.

F8 — The 2nd-row rails' rear ends



Survey F8 — 2nd-row rail rear ends

- **Measure / check:** With the seat carriages parked fully forward: where do the rails' REAR ends sit (distance from the hatch, lateral spacing), and what is at each end — an end cap that unclips, exposed track bolt heads, an open track slot, a lip a clamp plate could grab? Photograph the end hardware close-up.
- **Feeds:** Section 8's PRIMARY forward connection: the anchor board's 2 steel tongues bolt or clamp to these rail ends (no new holes). This measurement decides the bracket detail and the tongue length.
- **Previous assumption:** assumed the rail ends reach the striker-row zone — UNVERIFIED
- **F8a** — hatch → rail REAR ends: **42"** (MEASURED — owner, Aug 1 2026). **The rail ends reach — with room to spare.** They sit 2.5" NEARER the hatch than the striker row (44.5", F4a), so the tongues are shorter than the assumed 10–14": **fabricate to the measured 42"**. This is the PRIMARY forward connection and it survives the survey intact, which matters because F6 and F7 both eliminated fallbacks.
- **F8b** — rail lateral spacing: **17.5"** centre to centre (MEASURED — owner, Aug 2026). This is what the anchor board's bridge gets drilled to.
- **What is AT each end** (MEASURED Aug 2026): a **moulded plastic end cap** held by **one flush slotted fastener**, over an **open-top track channel with a 0.75" slot**. The metal track sits **0.5"** above the floor pan and the plastic housing over it reaches **1.0"**. Under the plastic cap is a **steel saddle cap** with a ¼" hole in its flat top face — see Section 8's connection detail (short version: a vertical pin into that hole carries the forward load).

Photos to take while you're in there (each one answers a specific question later, at the workbench, without crawling back in):

1. **Wide shot from the open tailgate**, 3rd row folded as it will sit in camper mode, with a tape measure laid fore-aft along the floor from the hatch — establishes the Y datum every other photo hangs off.
2. **Carpet/liner pulled back at the tailgate end** — what's actually under it: painted steel, a plastic tub, or the seat mechanism.
3. **Straight-down shot into the stowage well** (seats up if needed) with the tape showing its depth and its front edge's distance from the hatch.
4. **Close-ups of the seat latch/hinge hardware** anywhere it falls inside Panel C's 35.75" footprint — this is what a board strip or a deck leg would be fighting for space with.
5. **The passenger rear corner** showing the 12V battery access panel — the one spot the anchor board must stop clear of (Section 8).
6. **The striker row straight-on**, tape held across it — all 3 loops in frame, with the step's face visible below them. This pairs with F4 and F7.
7. **Each 2nd-row floor rail's rear end, close up** (carriages parked forward) — the end cap/bolts/slot the tongue bracket has to mate with. This pairs with F8 and is what the bracket gets fabricated against, at the workbench.

If the survey comes back ugly — rail ends short of the tongues' reach or with nothing to grab (F8), loops buried under the folded seatbacks, no hard face at the step, nothing solid for the strips to bear on — the Section 8 design flexes rather than dies: the striker-row step as the tongues' compression stop (F7), filler blocking down to the well's structural bottom (F3), the well's forward wall as a bearing target (F2/F3), or the 3rd row's own seat-mount bolts with anchor plates (F6 — likely means removing the 3rd row, which is new Section 9-style scope). Every one of those changes geometry that `params.scad` currently takes on faith, so settle this survey before cutting Panel C or the anchor board.

Appendix B — Build Sequence & Assembly Order

The order to build in — **dependency-first, not the Component numbering**. Work top to bottom; the "needs" is what must be finished before each phase.

Phase 0 — MEASURE & PREP, before any cutting (Appendix A, 9) — *critical* 1. ~~Measure the real van and update `params.scad` ~~ — **DONE Aug 1 2026** (Appendix A): interior 93.75", liftgate 50×37", side-door usable clear 29", wall-to-wall at platform height **50"** — which failed the ≥53" gate and cut the platform to 49". *Still open: the side-door aperture height and its fore-aft position (`side_door_y0`).* 2. ~~Confirm the rear-corner floor vents don't reach the last ~4" (Panel C's rear legs sit there).~~ **DONE Aug 2026 — they do not reach the leg area; Panel C's rear legs land on solid floor.** 3. Remove the 2nd-row seats; install the Ling Labs SRS emulators (Section 9). 4. Buy & verify materials (Appendix C); verify the Rocky 40 + JAGAHABA dimensions on arrival (both purchased July 2026).

Phase 1 — Cut all plywood & lumber (Section 3 / Appendix D) 1. Cut both sheets + all boards. The ½" sheet nests to ~95% — lay it out before cutting. 2. Drill the leg leveling-insert holes now (far easier before assembly). 3. Pre-cut the Panel C front-wall openings (fan hole, 2 grommets, louver — Front Wall render).

Phase 2 — Build the three frames (Components 2/3/4, step 1) 1. Panel A frame: rails + 4 legs + divider + 2 END-face bottom rails (underside 1") + 2 diagonal braces. Corners: 2× 2" screws + bracket + glue. 2. Panel B frame — the full cube: bottom rails on ALL 4 faces, no divider/drawers. 3. Panel C frame: front pair inset 3.5", REAR pair at the true corners; FRONT-face bottom rail only. 4. Install the 12 leveling feet + star knobs.

Phase 3 — Panel C build-out (fridge / kitchen / cooling) (Components 4 + 7) 1. Set the fixed top between the rails on its bearer cleats (flush with the rail tops — deck recess) and screw it down; add the front wall (mount the low intake louver). 2. Bench-build the no-drill anchor board (bridge + strips + risers + L-track + rail tongues, Section 8), then set it in the van on its mat, bolt/clamp the tongues to the 2nd-row floor rails' rear ends, and tension the 3 striker straps — **the board's final outline needs the Appendix A floor survey (F1–F8) done first**; nothing is drilled into the van. *Verify strap tension after the first drive.* **The hybrid's 12V battery access panel is at the passenger rear corner (under the rear outlet) — the kitchen-side strip stops short of it, and it stays reachable by unstrapping/sliding the kitchen out.** 3. Fridge on its tray + locking slide (driver corner); kitchen unit strapped down (passenger corner). Let the fridge stand 12h before first power-up. 4. Hang the kitchen drawer; velcro the 2 utility bins into the open bay (no door — it was cut). 5. Cooling + control cluster: intake fan (blows IN), exhaust fan + NTC on the fridge's kitchen-facing wall, W1209 + fuse block + switches on the backer board in the open utility bay.

Phase 4 — Rear pantry (prefab) (Component 1 — *needs Panel C's finished deck; nothing to build*) 1. Stack the 4 IRIS drawer units 2×2 on the deck's tailgate end (felt tape + Velcro ties between rows). 2. Screw the cleat pocket to the deck (cab side + both sides) + 2 flush D-rings. 3. Run the cam strap across the drawer fronts; set the pot bin + corner cleat + bungee in the open bay. 4. Mount Power strip 1 + the ROLL bubble level on the deck edge; route the cord.

Phase 5 — Panel A build-out (EcoFlow bays) (Components 2 + 8) 1. Right bay: R1-biscuited drawer box on a 20" slide + catch; DELTA 3 cleats + D-rings + a cable tray. 2. Left bay: 2 glide strips + an overhead shelf on 1×1 cleats at ~13.7" (WAVE 3 slides out beneath it).

Phase 6 — Bed platform (Component 2 — *needs Panel A & B built*) 1. Assemble the three pieces — Panel A: 2× 29" rails 49" apart + 5× 42" slats; each Panel B half: 2× 29" rails 24.5" apart + 5× 17.5" slats (pocket screws or R2 biscuits). Fit Panel B's 26" centre bearer first, so both halves have something to land on, and **screw Panel A's section down** — it is permanent. 2. Screw the PITCH bubble level to the driver-side rail edge.

Phase 7 — Seam hardware (Component 5): bumper strips + 2 alignment pins at the A/B and B/C seams.

Phase 8 — Cord runs (Component 6): cooktop + Power strip 1 → SHORT cords to the REAR AC outlet (2-way tap, no seams); DELTA 3 charging → front console outlet (no seams); fridge DC → the one long run, Panel A → C with 2 SAE seam disconnects.

Phase 9 — Install in the van (Components 10 + 8) 1. Set Panel A → B → C; register the alignment pins. 2. Connect power + stow the DELTA 3 stack and WAVE 3; load Panel B — RJ-MODINI spare flat on its skid (liner under), valve up, tool case in the wheel, 2 totes on top, cam strap to the footman loops. 3. Lay the platform → HEST mattress → tension-rod curtain.

Phase 10 — Commission & test (Component 10) — *critical* 1. Lift-out / reinstall dry run (each panel by its top rails; pantry drawer units by loosening the cam strap). 2. Level the van at the WHEELS (Appendix E) + set the interior feet once. 3. Power-up + cooling test (fridge stood 12h; fans ramp off the NTC); pull every drawer/slide. 4. Owner-place the low-level CO monitor + fire extinguisher.

Appendix C — Materials Shopping List

Consolidates Section 4's BOM into a shopping order with links. Prices are estimates; items marked **UNVERIFIED** need a live-listing check. **Purchased** = already bought (July 2026); **owned** = existing gear, not priced.

ASK THE STORE TO CUT THE SHEETS (owner, Aug 2026) — hand them the cutting layout in Section 3: trim $\sim\frac{1}{2}$ " off all four factory edges of every sheet, then the numbered crosscuts before the rips, exact-fit parts $\sim\frac{1}{8}$ " oversize. You come home with rectangles; only the anchor board's comb outline, the $\frac{3}{4}$ " $\times\frac{3}{4}$ " cleats and the holes are left for your saw.

Lumber & Plywood — $\frac{3}{4}$ " Baltic birch sheet (\$65), $\frac{1}{2}$ " Baltic birch sheet (\$50), **two $\frac{3}{8}$ " 4x4 handy panels (\$50 — corrected Aug 2026; a half-sheet was 2 sq ft short)**, weight swap), 12 $\times$ 2 $\times$ 2 pine 8ft (\$96–180), 6 $\times$ 1 $\times$ 4 pine 8ft (\$30).

Frame / fasteners — rear pantry: 2 $\times$ IRIS 3-packs, Home Depot #500163 (\$61) + cam strap/D-rings/felt/Velcro hold-down (\$26) + pot crate (\$15) + 2 $\times$ Sterilite 28-Qt Panel B totes (\$24) + spare hold-down hardware: footman loops + cam-strap 6-pack (\$24) + the RJ-MODINI spare kit (\$350 — stows in Panel B at the axle); corner brackets $\times$ 12 (\$12); diagonal corner braces $\times$ 8 (\$24); seam draw-latches $\times$ 4 (\$16); wood screws (\$20); wood glue (\$8); [Ryobi R-series biscuit assortment](#) (\$12); alignment dowel pins (\$3); anti-rattle bumper strip (\$10).

Drawers & slides — Panel A drawer slides (GlideRite 20" 100lb 5-pack, *purchased*, \$47); drawer catch (\$3); kitchen-drawer slide 24" (\$16) + catch (\$3); fridge slide 24" heavy-duty locking (VADANIA, *purchased*, \$79).

Leveling — leg leveling feet (3 four-packs, \$36); star-knob grips (3 four-packs, \$36); Larbeti stick-on bubble levels 2+2 (\$10, **PURCHASED**); [Lynx leveling blocks 10-pack](#) + chock (\$40).

Fridge & kitchen — [BougeRV Rocky 40](#) (\$400–500 less JULY7 coupon, **PURCHASED**); [JAGAHABA slide-out kitchen](#) (\$300–350, **PURCHASED**); [COOKTRON induction cooktop](#) (\$190); UMETRE 9-piece detachable-handle cookware set (purchased in its place) (\$65–90).

Electrical & cooling — 120mm 12V fans $\times$ 2 (\$18–20); W1209 controller + probe (\$10); Nilight fuse block (*purchased*, \$18); Ampper switches 10-pack (*purchased*, \$9); enclosure (\$12–15); snap-in louver vent $\times$ 1 (\$6 — the door louver went with the door); grommets (EASIEAH 20-pack, *purchased*, \$10); power strips $\times$ 2 (\$50); AC cords (\$24 — two short rear-outlet hops + the DELTA 3 charging line); fridge DC cord + SAE seam disconnects (one OYMSAE 2-pack does both, *purchased*, \$22); 2-way rear-outlet tap (\$5); cord clips (\$9); fused car-plug $\rightarrow$ SAE adapter + SAE $\rightarrow$ 12V-socket pigtail (\$18); 18AWG hookup wire, crimp assortment + WAGO lever nuts (\$30).

Anchoring (no-drill, Section 8) — L-track 48" 4-pack (\$95); stud fittings w/ D-ring 10-pack (\$45); Erickson ratchet straps, 2 4-packs (\$64); anchor-board ply + 1/4-20 screws/T-nuts (\$20–45); steel flat bar for the 2 rail tongues (\$15–25, + rail-end bracket bolts/clamps speced after F8); non-slip mat (\$25); fridge-slide steel riser angles $\times$ 2 (\$24) + 1 $\times$ 3 apron stock (\$6).

Comfort & interior — [HEST Dually Long mattress](#) (\$530–590); Windmill Smart fan (\$69, **PURCHASED**); tension rod + blackout curtain (\$40); [PeaceOut Sienna bug nets](#) ($\sim$ \$197); [EcoFlow WAVE vent kit](#) (\$39); non-slip mat (\$12); WAVE 3 hose hook (\$4); WAVE 3 glide strips (\$6); DELTA 3 drawer hardware (\$15).

Found storage — DELTA 3 drawer tray (\$8); WAVE 3 overhead shelf, from offcut (\$5); utility-bay bins $\times$ 2 + hook-and-loop tape (\$8).

Finishing / vehicle — utensil bins + non-slip liner (\$12); wood sealant (\$20); sandpaper/misc (\$15); SRS airbag emulators — 0651 Round $\times$ 2 + 5522 Square $\times$ 2 ($\sim$ \$60, **PURCHASED**; verify the 5522 fits the rectangular plug).

Already owned (not priced): EcoFlow DELTA 3 Plus + Smart Extra Battery, EcoFlow WAVE 3.

Total: $\sim$ \$3,370–3,740 with the fridge + kitchen, or $\sim$ \$2,670–2,890 if you already own equivalents (see Section 4 for the full itemized BOM and every purchase link).

Appendix D — Cut List (consolidated)

Finished sizes; add saw kerf. Full context in Section 3.

3/4" Baltic birch plywood — 1 sheet (4×8): Panel C deck 1× 36×46"; rear-pantry hold-down cleats ~4× 1×1×12–14" (ripped from the offcut) — the rest is spare stock.

1/2" Baltic birch plywood — 1 sheet (4×8): kitchen-drawer box (bottom 1× 16×26"; sides 2× 26×4"; front/back 2× 15×4") + hanging cheeks 2× 26×6.2"; battery-drawer bottom 1× 20×25". **3/8" half-sheet:** battery-drawer walls (2× 25×14.5", 2× 20×14.5"); fridge tray 1× 17.72×28.74"; Panel C front wall 1× 46×17".

2×2 pine — 12 boards (8ft), 1015.5" total: Panel A long rails 2× 29"; Panel B long rails 2× 29"; Panel C long rails 2× 35.75"; end rails 6× 46"; center divider (Panel A) 1× 26"; legs 12× 16" (cut 1" short); bottom rails 5× 46" + 2× 26"; Panel B centre bearer 2× 26".

1×4 pine — 7 boards (8ft), bed platform: side rails 6× 29"; slats 5× 42" + 10× 17.5".

Trim / cleats / scrap: WAVE 3 overhead shelf 1× ~20.75×14" (½" ply offcut); WAVE 3 glide strips 2× 20×1" (UHMW/laminate); DELTA 3 locating cleats 4× 1×1 blocks.

Appendix E — Leveling: Block Calculator

Per-site leveling is done at the **wheels** with the [Lynx blocks](#), driven by the phone [Sienna Block Calculator](#) (open it in Chrome and *Add to Home screen*, or use the offline copy [sienna_block_calculator.html](#)). The interior leg feet are then a **one-time** set against the van's own floor.

Using it: park (engine off, brake on) → read the two bed-mounted bubble levels (pitch on the platform rail, roll on the rear-pantry deck edge) in degrees → enter both readings + which end/side is low → it returns blocks-per-tire. Set the one-time van settings once (wheelbase 120.5", track ~68", and *your measured* block height, Lynx ≈ 1.5"). Stack ahead of the low tires, drive up slowly, chock, re-check; if any tire needs >4 blocks, re-park.

The math (so it's not a black box): a low END needs $\tan(\text{pitch}^\circ) \times \text{wheelbase}$ of lift; a low SIDE needs $\tan(\text{roll}^\circ) \times \text{track}$. Blocks per tire = $\text{round}(\text{that deficit} \div \text{block height})$. One 10-pack (1.5"/block over the 120.5" wheelbase) covers about 2.9° of pitch correction.

Appendix F — Weight Budget & Weight-Distribution Analysis

Every component's weight, computed from the real material dimensions (Baltic birch at 42.5 lb/ft³, pine at 28 lb/ft³) plus manufacturer specs for the appliances/power gear. The live spreadsheet is in the repo as `weight_budget.csv` (open it in any spreadsheet app; regenerate with `weight_budget.py` if you change materials). Figures marked *est.* are estimates to confirm.

Component	Category	Weight (lb)	Zone	Notes
Panel A frame (2×2 pine)	Structure	12.0	front	rails+legs+divider+bottom rails (16.25" legs — deck recess)
Panel A drawer box + slides (¾" birch, ½" bottom)	Structure	22.2	front	holds the DELTA 3 stack
Panel B frame (2×2 pine, full cube)	Structure	12.9	mid	bare-frame deep-storage box (16.25" legs — deck recess)
Panel C frame (2×2 pine)	Structure	10.0	rear	rails+legs+front bottom rail
Panel C deck (¾" birch, 33×43 recessed + cleats)	Structure	27.2	rear	drops between the rails onto bearer cleats, flush (deck recess)
Panel C front wall (¾" birch)	Structure	7.2	mid	intake fan + grommets
Fridge tray (¾" birch + 2 1×3 side aprons)	Structure	6.4	rear	hangs between the side-mount slide rails
Kitchen drawer + ½" cheeks (birch)	Structure	13.9	rear	hung over the kitchen unit; 18" box (Aug 2026)
Rear pantry: 4× IRIS 12"W drawer cluster + hold-down	Structure	15.0	rear	prefab 2×2 drawers (est.) + cleats/strap — replaces the ~42 lb plywood pantry
Panel B: 2× Sterilite 28-Qt under-bed totes	Hardware/misc	4.0	mid	restacked ON the spare (2 of the original 4)
RJ-MODINI spare kit + jack (Panel B, flat at the axle)	Hardware/misc	40.0	mid	T155/85R18 steel-wheel kit + cleats/strap — VERIFY kit weight
Bed platform (1×4 pine, 6 rails + 15 slats, 3 lift-out pieces)	Structure	23.8	front	Panel A section + Panel B's 2 centreline halves (Aug 2026)
Panel B centre bearer (2× 2×2 pine, 26")	Structure	1.9	mid	halves the deck's unsupported span over Panel B, 46" → ~22" (Aug 2026)
Fridge (BougeRV Rocky 40, empty)	Appliances	40.6	rear	manual spec; +20–40 lb loaded
Kitchen unit (JAGAHABA)	Appliances	45.0	rear	listing spec
EcoFlow DELTA 3 Plus + Extra Battery	Power/climate	48.0	front	Panel A passenger drawer
EcoFlow WAVE 3 A/C	Power/climate	33.7	front	Panel A driver bay
Mattress (HEST Dually Long)	Bedding	35.0 <i>est.</i>	mid	solid foam + waterproof cover
Fridge + kitchen-drawer slides (heavy)	Hardware/misc	8.0	rear	24" heavy-duty pairs
Fridge-slide steel riser angles (2)	Hardware/misc	9.8	rear	2×2×3/16" × 24" steel, fixed rails → anchor board (aluminum angle saves ~6 lb)
12 leveling feet + star knobs + inserts	Hardware/misc	5.0	mid	4 per panel
Anchor board (¾" ply + 2 steel tongues) + L-track + 7 straps + mat	Hardware/misc	22.8	rear	no-drill striker tie-down chassis at Panel C (Section 8)
Corner brackets, braces, screws, glue	Hardware/misc	6.0	mid	spread across all modules
Seam draw-latches (4) + pins + bumpers	Hardware/misc	1.5	mid	at the A/B + B/C seams
Electrical (2 fans, controller, fuse, 2 strips, cords)	Hardware/misc	6.0	mid	cooling + power runs
Cooktop + cookware (stowed)	Hardware/misc	12.0	rear	induction top + pots in the kitchen

Subtotals: Structure **152.5** · Appliances **85.6** · Power/climate **81.7** · Bedding **35.0** · Hardware/misc **115.1** **Build total (added): ≈ 470 lb** — after the lighter-wood swaps, the deck recess and the prefab-drawer pantry swap, plus Panel B's 2 storage totes (~4 lb), the **RJ-MODINI spare kit stowed at the axle (~40 lb)** — carried inside deliberately instead of on a hitch basket, which would have added its own 30–60 lb of carrier plus a ~1.4× lever on the rear axle — and the no-drill anchor board (~23 lb, +15 over the drilled E-track hardware it replaced).

But the 2nd-row seats come OUT (Section 9), so the conversion *removes* weight too:

	Weight
Build (added)	+470 lb
2nd-row seats removed (2 × ~48–70 lb) <i>est.</i>	~120 lb
NET permanent weight added vs. stock	≈ 350 lb

So the van, converted and empty, weighs only about **350 lb more than a stock seatless curb** — and because the seats were part of the factory curb weight, pulling them **frees ~120 lb of payload** for your gear. **Weigh your actual seats to confirm** — captain's chairs vary; 120 lb is an estimate.

Loaded scenarios (payload = everything you add to the van — build + people + cargo):

Scenario	Weight
Net build (added – seats removed)	≈ 350 lb
+ provisions (food, water, cookware, bedding ~120 lb)	≈ 470 lb
+ 2 occupants (~340 lb, in the front seats while driving)	≈ 810 lb

Weight distribution

- **Fore-aft:** the build's own center of mass sits ≈ **51.9" back** from the front seatbacks — aft of the 47" geometric center (the prefab pantry and the axle-stowed spare pull it forward; the anchor board pulls it back). Breaking it into thirds: **front 30%** (140 lb — Panel A + the DELTA 3/WAVE 3 stack + the bed platform), **middle 25%** (118 lb — Panel B now carries the spare + totes at the axle), **rear 45%** (210 lb). That rear share is the **tailgate cluster** — the prefab drawer pantry (~15 lb), the fridge (41 lb), the kitchen unit (45 lb), the anchor board (~23 lb), and Panel C's own structure — most of which has to live at the back for slide-out access.
- **Lateral:** ≈ **0.7" toward the passenger side** — trivial (the fridge-slide steel risers on the driver side pulled it in a bit more). This is by design: the DELTA 3 stack (48 lb, passenger) and the WAVE 3 (34 lb, driver) sit in opposite bays of Panel A, and the fridge (driver) and kitchen (passenger) also cross-balance in Panel C. The reviewer's "heavy battery off to one side" concern is effectively cancelled by the WAVE 3 on the other side.
- **Height:** everything is low (floor-level bays) except the 16.8" pantry cluster at the rear and, when parked, the sleeping load on the ~20–27" platform. Overall CG stays low — no rollover concern beyond a stock minivan.

Effects on drivability

- **Payload headroom:** loaded for travel (net build + provisions + 2 people, with the seats out) is ≈ **810 lb** — and that already includes the spare tire + jack. A 4th-gen Sienna's payload is roughly **1,100–1,300 lb** (the AWD Woodland is at the lower end), so there's a **~290–490 lb margin** — the removed seats buy back ~120 lb of that. Comfortable, but not unlimited once you add lots of water and gear. **VERIFY the exact figure on the driver door-jamb sticker** — that number, and the **rear GAWR**, are the hard limits.
- **Rear bias:** with 45% of the build mass in the rear third (much of it at/behind the rear axle), the fixed build alone loads the rear axle and slightly lightens the front — which reads as lighter steering, a bit more rear squat, and a raised headlight aim. **But while driving, the two occupants (340 lb) sit up front and more than offset that**, so the *loaded* vehicle is reasonably balanced. The rear bias matters most when driving **solo with a full rear and no front passenger**.
- **Lateral / height:** negligible effect — the ~0.7" lateral offset is unnoticeable, and the low CG keeps handling stock-like.

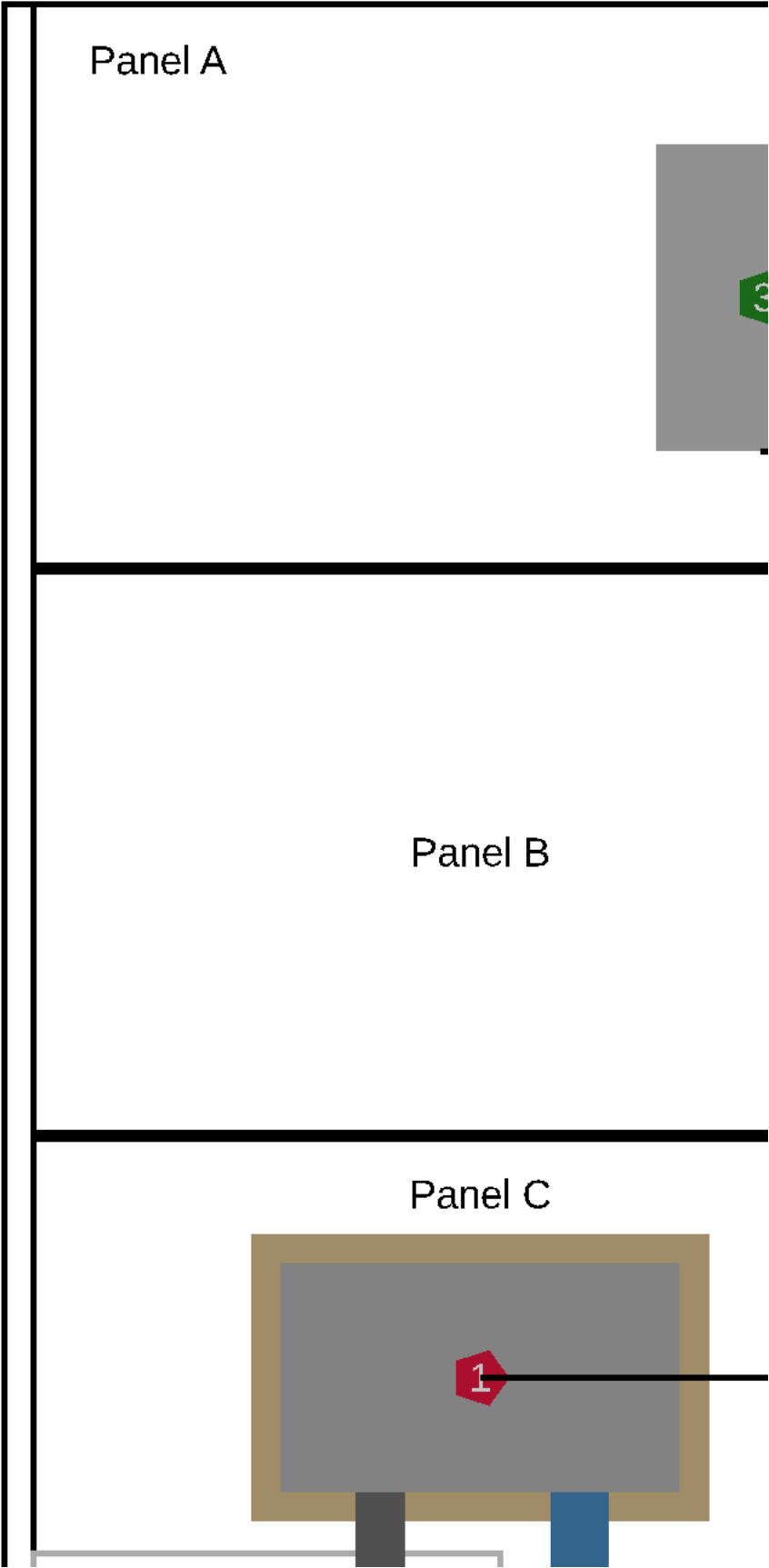
Recommendations

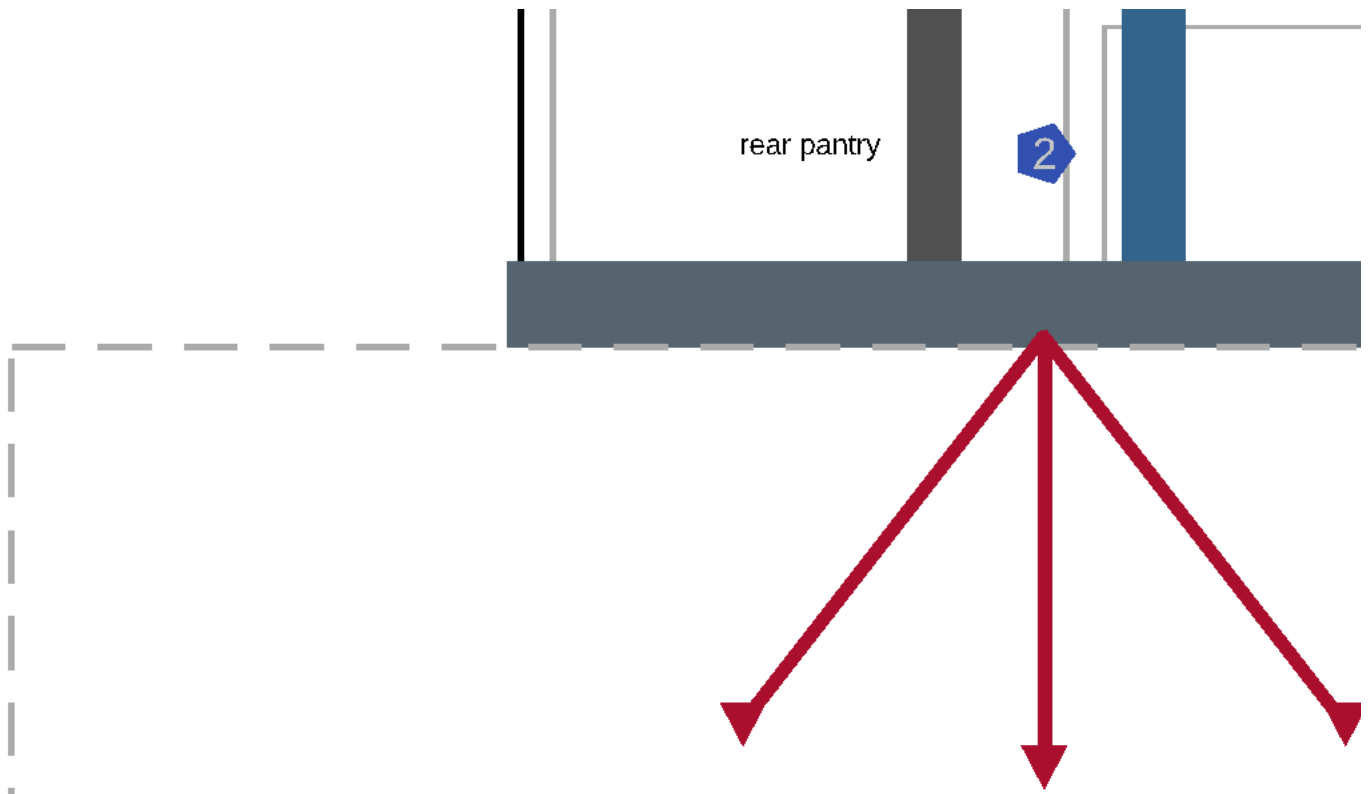
1. **Verify payload + rear GAWR** on the door-jamb sticker before loading, and stay under both. The Woodland AWD has the least payload — treat ~1,100 lb as the working assumption until you read the sticker.
2. **Stow heavy movable cargo forward/mid, not at the tailgate.** Water, canned food, and tools belong in **Panel B** (middle, near the axle) — that offsets the fixed rear cluster. Water is the big one: **8.3 lb/gal**, so a 6-gal jug is ~50 lb — keep it low and mid/forward.
3. **Re-aim the headlights** after the build is in and loaded — rear squat raises the beam.
4. **Set tire pressures to the door placard** (consider the upper end for the rear given the bias) and re-check with the van loaded.
5. **Driving solo?** Expect a slightly lighter front end with a full rear — nothing unsafe at these weights, just drive to it.
6. **Leg-foot load is not a concern:** ~441 lb + occupants + cargo (~1,000 lb worst case) over 12 feet is well under the feet's 330 lb-each rating (~85 lb/foot worst case, a quarter of capacity).
7. **If you add heavy items later** (second battery, water tank, awning), re-run `weight_budget.py` and re-check the rear axle before a trip.

Appendix G — Nighttime Van + Tent Cooling Setup

A **night-only** cooling arrangement for hot-weather campsites (worked out for a Texas-summer trip, but applies anywhere overnight lows still run warm): condition the van and an attached tailgate tent as one shared air volume with the WAVE 3 alone, and skip the factory AC overnight entirely. This builds directly on the WAVE 3 "with a tent" configuration already described in Section 1 — this appendix just adds the setup diagram and the night-specific hints.

Sienna interior envelope
FRONT





5

TENT — EXAMPLE FOOTPRINT (swap for your
127" x 96" — VEVOR SUV tailgate tent



Night-only setup: skip the factory AC here — it needs the van (doors/tailgate shut) to run efficiently, which conflicts with the tent sleeve needing the tailgate open. Seal the tent's own m vents shut and add a small circulation fan inside the tent to the WAVE 3's output to the tent's far end — see Section 1.

Why no factory AC at night: the Sienna Hybrid's electric AC compressor cools the cabin well, but it needs the van **sealed** (tailgate/doors shut) to work efficiently, and it cycles the gas engine on and off to keep the hybrid battery fed — noisy, intermittent, and it burns fuel right through the park's quiet hours. A tailgate-attached tent needs the opposite: the tailgate **open** so the connection sleeve has something to wrap around. Trying to run both at once means either the AC fights a leaky open tailgate (worse cooling, more engine cycling) or the tent's connection collapses — there's no clean way to combine them. The WAVE 3 has neither problem: it's silent-ish, battery-only, and its whole design point (Section 1) is blowing straight through the open tailgate into the shared van+tent space.

Setup steps (matches the numbered callouts in the diagram):

1. Carry the WAVE 3 from its Panel A storage bay back to Panel C's deck, just forward of the rear-pantry cluster, on a non-slip mat — facing the open tailgate.
2. Route its intake/exhaust hoses past the pantry and out through the open tailgate gap to true outside air. No window seal needed — the open tailgate is the vent path.
3. Leave the DELTA 3 Plus + Smart Extra Battery in Panel A's right (passenger) drawer — that's the WAVE 3's power source. Run its cord the length of the van (Panels A → B → C) to reach the unit.
4. Attach the tent's elastic connection sleeve around the open tailgate/liftgate opening.

5. Pitch the tent in its normal footprint just beyond the tailgate (swap in your own tent's dimensions — the diagram uses an example ~127" x 96" SUV tent as a stand-in).
6. Zip every one of the tent's own mesh windows/vents shut except the one facing the van — every extra opening is cool air leaking out into the Texas night air.
7. Turn the WAVE 3 on and let it condition the shared van+tent volume; hoses keep dumping heat/moisture out the tailgate gap the whole time.

Hints:

- **Add one small low-power circulation fan inside the tent**, aimed to push the WAVE 3's output toward the tent's far end — its own airflow may not reach that far unassisted, and a clip fan costs only a few watts against the 2048Wh bank.
- **If you want the factory AC's extra cooling power too**, use it for a short **pre-bed knockdown** only: seal the van fully (tailgate shut, tent not yet attached) and run the AC 10-15 minutes before quiet hours to clear the day's heat soak, then switch to the WAVE 3 + tent setup above for the actual overnight run.
- **Battery budget**: expect roughly 7-10 hours of overnight runtime from a full DELTA 3 stack in this configuration (nighttime ambient is far easier to hold than a daytime load) — plenty for one night; plan a solar/shore-power top-up if running it multiple nights back to back.
- **If you need two independently-cooled sleeping zones instead of one shared space** (e.g., people sleeping in both the van and the tent), the WAVE 3 can't do both — the alternative is to seal the van and run the factory AC there while pitching the tent as a **fully stand-alone** shelter (its own door, sleeve not attached to the tailgate) with its own separate cooling. That trades the shared-air convenience for two independently sealed zones.
- **Safety note carries over from Section 1/4**: this configuration doesn't run the engine overnight, so it doesn't add to the CO-monitor concern that a factory-AC-all-night approach would — but keep the low-level CO monitor active regardless if you ever do run the engine near the sleeping area.

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